# Principles of Instrumentation 2025–26

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**Abstract**

This Year-3 elective course provides an introduction to the principles and practice of instrument science. This is a "directed study" course with significant academic engagement through regular contact sessions and a substantial 'hands-on' element in the lab. You will be introduced to concepts in the electronics of scientific instruments through a combination of course notes and problem sheets released weekly, Q&A sessions and student-led seminars, and practical laboratory work. In lab, you will use the National Instruments LabVIEW$^{\mathrm{TM}}$ ELVIS prototyping system to build and characterise key instrument components such as input/output transducers, amplifiers and filters, and to gain a practical understanding of some of the theory underpinning electronic signal processing.

These notes constitute the core material that will guide you through the course. They were developed by a number of colleagues that taught Principles of Instrumentation going back many years. I formatted them anew and extended them in places to make them more 'self-contained' in nature, as previous materials were complemented by traditional lectures – which we will not have. A few sections will be released every week, amounting to around 10 pages at a time, which you should read carefully in advance of the Q&A session.

You may want to consult the recommended textbooks that cover our syllabus, whenever you want more detail or a deeper (or different!) explanation. Although I expect that the content of these notes will suffice in the majority of the cases, I cannot stress enough that you will need to "take control of your own learning", since you will not hear the same material covered and repeated various times in lectures and slide-sets as you may be used to. In addition to the recommended textbooks, there are many good materials online that cover our syllabus and you should consult them until you are happy that you have a good understanding of each topic. In many cases you are advised to try out the derivations covered or suggested here, and the problem sheets will be immensely helpful in bringing these topics to life – as will the lab sessions later on.

Finally, as this is still a relatively new course format for the department, I will appreciate your feedback at any time!

## Contents

# 1. Instrumentation: characterising performance

## *1.1. Measuring physical parameters: true value and error*

The *error* in a measurement is the difference between the value recorded by an instrument and the *true value*. The true value is a rather difficult concept, since it is in general both unknown and unknowable. Take for example the measurement of mass, for which the international standard was (until very recently!) a piece of platinum-iridium alloy held at the International Bureau of Weights and Measures in France. This individual weight was – for over a century – *the* kilogram.[1] There are various copies of this, which are held by national standards labs around the world, the accuracies of each are estimated by analysis and occasional comparison. The problem is that the actual mass of the standard kilogram can never be pinned-down to an arbitrarily high level of accuracy. For example, as soon as it is removed to be compared against something it must be touched, cleaned, exposed to air and so on, which inevitably results in some material being either lost or gained. Even if we could, at some instant in time, know exactly how many atoms it contains, what is the exact mass of each atom? As physicists, we understand that there is no such thing as an exact value for any physical parameter, which is why we always have an error in any good measurement. The important thing is to know how good our estimate of the error is. This is often harder to do than measuring the parameter itself.

## *1.2. Calibration*

So, until very recently the *primary standard* for the kilogram was the single mass held in France; nationally held copies are the *secondary standards*; and there are still a whole chain of copies of copies down to the weights used in laboratories in industry and academia, which can be traced back to the primary standard with some well established level of accuracy. *Calibration* is the process of comparing our instrument with some reference measurement which can be traced back to one of these standards, thereby allowing us to quantify the error. Whilst the true value of any measurement is inherently unknown, it is generally taken to be the value which we would measure if we used the best possible instrument available in the field, i.e. one which is best by both design and calibration. So, in a calibration laboratory, a set of scales could be calibrated by comparing the value of a test-mass on the 'device under test' with a reference instrument with a known and traceable calibration.

In general, the *reference measurement* is the best estimate we can make of the true value. We compare our instrument with the reference measurement, and can hence make some assessment of the *accuracy*, as illustrated in Figure 1.1. Usually, calibration also implies adjustment, whereby the output is tuned to minimise the error. This may be by some kind of electronic or mechanical compensation, with the aim being to release the instrument with a calibration certificate which states the estimated *accuracy* of the device.

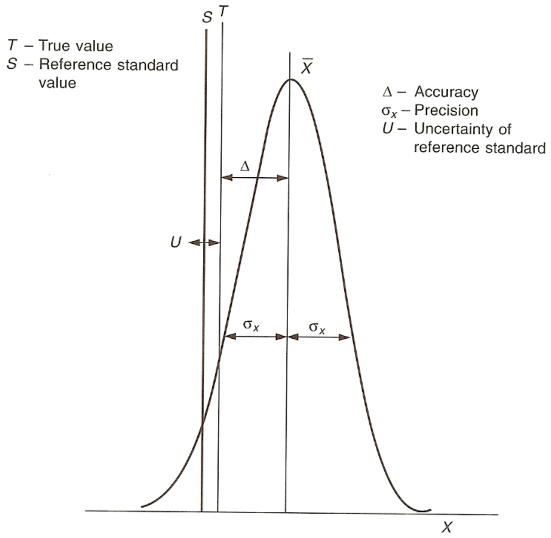


Figure 1.1: Precision, accuracy and related terms.

## *1.3. Precision is not accuracy*

The error described above is a *systematic error*. We also anticipate *random error* in a sequence of measurements due to natural stochastic variability (noise) in the underlying measurable, plus any additional noise added by the instrument or measurement process. Consider, for example, 10 trainee doctors taking a patient's temperature with a clinical thermometer which has as the smallest increment 0.1°C. This is the *resolution* of the sensor. We may expect to get a range of values with a mean of 37°C. We can plot a histogram of the results and, as we increase the number of measurements to, say, 1000, the histogram should tend to a normal distribution. If the thermometer has been calibrated (had its scale adjusted) then the best estimate of the reference measurement ("true value") is the mean of that histogram. The width of the distribution (its standard deviation) is the *precision* of the measurement. High-precision equates to high-repeatability. *Accuracy* tells us how far the mean is from the reference measurement. So, if we have an instrument with good resolution and repeatability, but poor calibration, then the result will be precise but not accurate. Figure 1.2 illustrates this point.

## *1.4. Characterisation*

In order to fully characterise the behaviour of an instrument we need to consider how it responds to both a static

[1] In 2019 the kilogram was redefined by taking the fixed numerical value of the Planck constant $h$ to be $6.62607015\times10^{-34}$ kg m$^2$/s, where the metre and the second are defined in terms of the speed of light $c$ and the transition between the two hyper-fine levels of the ground-state of the caesium-133 atom, $\Delta\nu_{Cs}$.

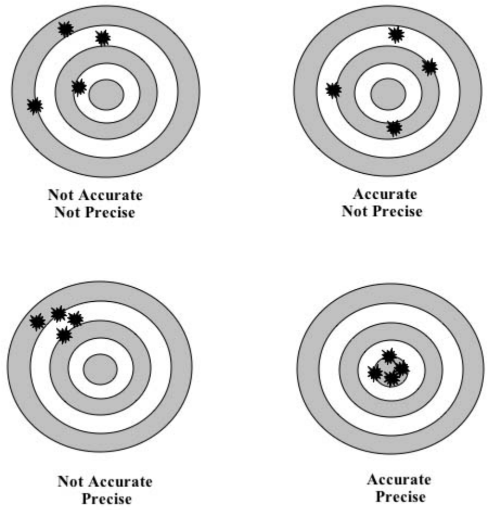


Figure 1.2: A precise measurement is not necessarily accurate.

input and under changing input conditions. How this is done and the relevance depends on the type of instrument being characterised but, in general, there are three types of measurement which can be made:

- Static response;
- Frequency response;
- Transient response.

The static response is what is generally considered to be the calibration of the instrument; the other two will give us further information about the time and frequency-domain behaviour of the device.

*Static response*

In this case the input is a set of static values which covers the *range* of the instrument, and the output is plotted for these – see Figure 1.3. We will usually want to be able to plot a straight line through the results which yields the *static sensitivity* (gradient) of the device and the *zero offset* (intercept). The input must be constant during the measurement and the output must be allowed to settle into the steady-state. If there is any non-linearity, then this can be expressed as the *non-linearity error*, usually taken to be the maximum deviation from the straight-line fit, divided by the full-range value, and expressed as a percentage. The input measurement is the reference value, so the static-response graph gives us our calibration; we can use it to correct the output of our instrument.

This approach is perfect for a set of scales or many other static parameters such as length. These are often referred to as *DC parameters*; however, many things we would like to measure are inherently variable *AC parameters* such as light intensity, electric/magnetic fields etc.

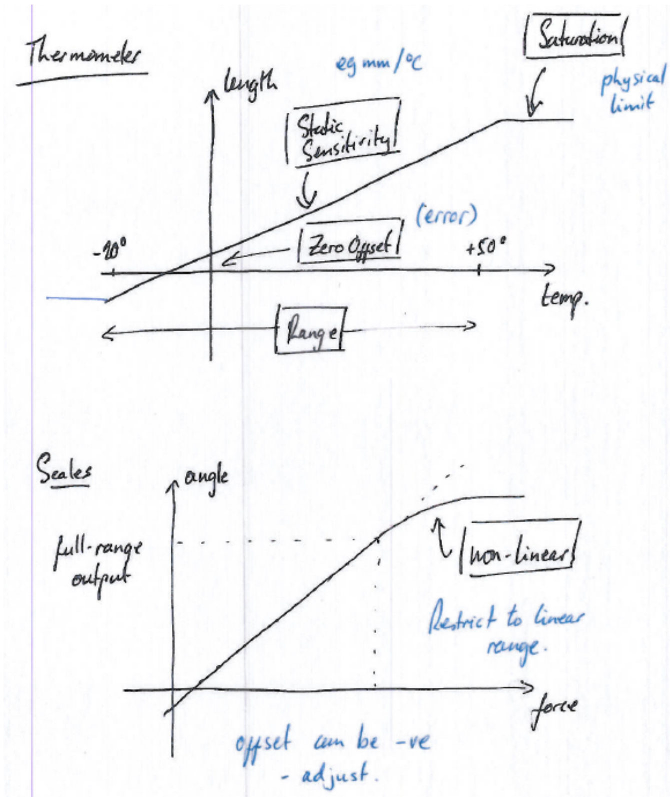


Figure 1.3: Examples of static response: thermometer and scales.

*Frequency response*

The *frequency response* is usually tested by applying a sinusoidal input signal which is stepped over the frequency range of the instrument. At each frequency step, the amplitude and phase of the output are measured with respect to the input. The result is graphed on a *Bode plot*, which is actually two plots: the first expresses the relative amplitude of output/input; the second gives the phase difference, with output leading input representing positive phase by convention. The frequency axis is usually logarithmic, and by convention the relative amplitude is given in dB, the phase in degrees. An example is shown in Figure 1.4.

The frequency response is the direct equivalent in the frequency domain of the static response. For any given input frequency we can predict both the amplitude and phase of the output. Since we know any arbitrary input signal can be decomposed into sinusoids, we can use the Bode plot to find the instrument's response to each, and hence predict what the output will be by summing the responses. This only applies if the instrument shows a nicely linear static response, which is why it is so important for instruments to be 'linear'. We will study this in much more detail later on; however, for now, it is important to understand that an instrument with *linear* characteristics can be analysed in the frequency-domain using Fourier techniques.

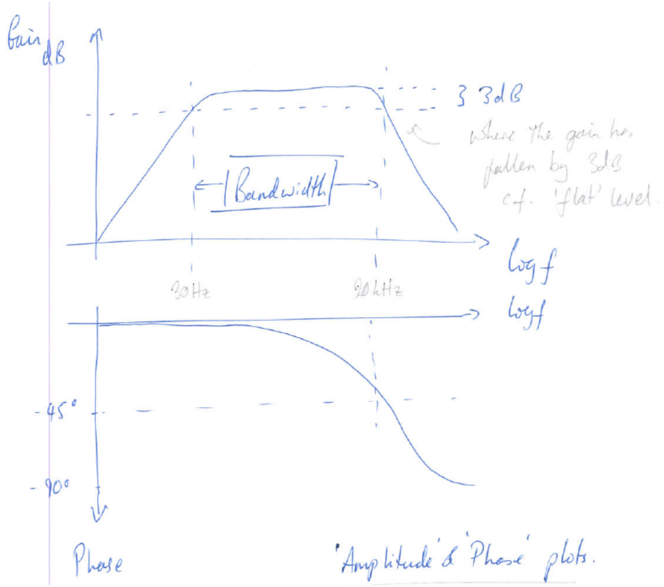


Figure 1.4: Bode plot for microphone.

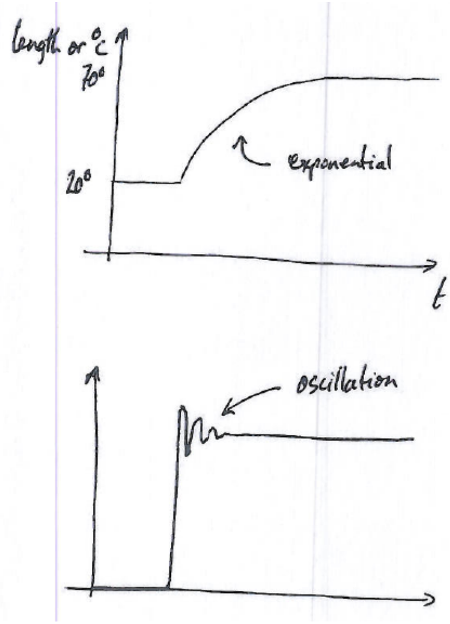


Figure 1.5: Transient responses for thermometer (upper panel) and scales (lower panel).

*Transient response*

We often wish to know the response to a non-sinusoidal transient change at the input such as when we place a mass on the scales (mathematically, a *step function*) – see Figure 1.5. Another useful test-input is the *delta function*. These can provide information on the settling-time for the output, as well as being used to test for system stability. Again, we will cover this in much more detail later on.

### *1.5. The decibel*

The dB is a way of expressing ratios on a logarithmic scale. It is ubiquitous in engineering and takes some practice to get used to. By definition the decibel expresses the ratios of two power levels

$$n\,\mathrm{dB} = 10\,\log_{10}\frac{P_2}{P_1}\,.$$

If we *double* the power output of an amplifier then we say the power has "gone up by 3 dB". Four times the power is 6 dB, which is the same as doubling the voltage (amplitude) output of the amp, since power is proportional to amplitude squared:

$$n\,\mathrm{dB} = 20\,\log_{10}\frac{A_2}{A_1}\,.$$

Further confusion can arise from the fact that dB are sometimes used to express absolute amplitude quantities, such as a voltage relative to 1 $\mu$V: 1 volt in dB$\mu$V is expressed as $20\,\log_{10}10^6/1$ dB. Another common quantity is the 'dBm' (decibel-milliwatt), used to indicate a power level with reference to 1 mW. This can be confusing and it is worth spending some time to become familiar with the usage.

### *1.6. Linearisation*

As mentioned above, a linear instrument is important to allow Fourier analysis. In a future lecture we shall define mathematically what is meant by the term 'linear system', and then we will return to this later in the course to see how the Fourier and Laplace transforms can be brought to bear to understand the behaviour of really quite complicated systems involving feedback, amplification and other linear operations. For now, it is important to understand that this is the reason that instrument designers pay so much attention to ensuring that the static response of the instrument is a straight line through the origin. If there is an offset then this can be compensated for electronically. If the output deviates from a straight line at high inputs then we can artificially limit the input. Both are examples of *linearisation*. Consider also the thermistor (Section 2), which has a roughly exponential temperature response. This can be coupled with a logarithmic amplifier to yield a linear response.

### *1.7. Loading*

As a final note on the subject of measurement errors, we must always be aware that the very act of measurement can disturb the measurement we are trying to make. Two simple examples. First, when we put a thermometer in a cup of tea, we cool the tea slightly. Secondly, imagine a pressure sensor which works by moving a diaphragm against a spring in response to pressure. In a constrained space, the volume of the sensor will change slightly and could influence the pressure measurement. These are both examples of the sensor *loading* the measurement. We will encounter many more examples of of loading in electronic systems.

## 2. Sensors and transducers

Generally, a *transducer* is any device which converts some physical parameter from one form into another, usually in a proportional and predictable way. All sensors are transducers, specifically they are *input transducers*, in that they convert some physical observable into a more useful form, usually – but not always – an electrical signal. The mercury thermometer, for example, is an input transducer which converts thermodynamic temperature into length. Typically though, we want the output of the transducer to be an electrical signal so that we can do further manipulation and processing using electronic circuits, and ultimately obtain a representation of the signal into a computer-readable form. Also, we may want to go in the other direction, i.e. converting an electrical signal into some kind of physical observable (such as light, heat, sound, movement etc.) for which we need an *output transducer*. Figure 2.1 illustrates the idea with a simple "public address" system. The microphone is an input transducer which converts speech (sound pressure variations) into a voltage signal. Then we have an amplifier, where we can also do more fancy things such as adjusting the tone (frequency content) of the signal. The amplifier drives a speaker which is the output transducer, whose job is to convert the amplified electrical signal back into a mechanical movement (of the speaker diaphragm) which in turn gives us sound. Note that we will be studying a more complicated system of sound/electrical transducers in the lab sessions.

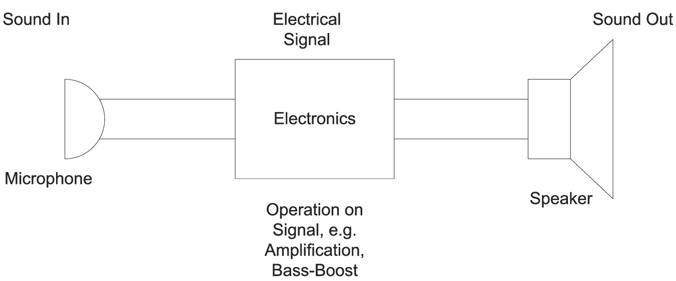


Figure 2.1: Input/output transducers.

There are thousands of different types of transducers and we will only look at a few. This is not a course on detector physics, and we do not have the time to study the details of many individual transducers. However, we will see that there are many similarities in their behaviour, so we will study them in a more general (that is to say device-independent) way. In this handout we will look briefly at two different types of input transducer: temperature and photon sensors. Note that, according to general convention, we will usually use 'sensor' and 'input transducer' interchangeably during this course.

### *2.1. Temperature transducers*

*Thermocouple.* This is the simplest temperature transducer/sensor since it directly generates a voltage. A thermoelectric voltage is generated across any metal when the two ends are at different temperatures. Using two junctions between dissimilar metals – one held at a known temperature, the other at the point we wish to measure – the voltage generated is proportional to the temperature difference between junctions. The voltage is small – of the order a few $\mu$V/°C – so precision voltage measurement is required. The technique requires some care; however, the thermocouple is robust, small, and operates over a very wide temperature range ($-200$°C to $+1500$°C).

*Platinum Resistance Temperature Device.* This is the standard for high-accuracy measurement over the temperature range $-200$°C to $+600$°C. The PRTD shows a reasonably linear variation in resistance as a function of temperature. Consequently, we just need to measure the resistance (e.g. with a multimeter) then apply a conversion factor to get the temperature. They typically have a rather slow response time, and can be reasonably expensive.

*Thermistor.* This is a semiconductor device where resistance decreases rapidly with temperature. The temperature coefficient of resistance is of the order 4%/°C, which is an order of magnitude larger than that of the PRTD. The thermistor is also small, cheap and robust, but not as accurate as the PRTD. It is a good choice for temperature measurement over the range $-75$°C to $+150$°C. They are available in all sorts of packages, for example the glass-encapsulated ones are good for harsh or chemical environments (though since glass is an insulator this slows down the response time to temperature changes). One drawback is that the temperature-resistance relationship looks like:

$$\frac{1}{T} = A + B \ln R + C(\ln R)^3 \,.$$

To go from $R$ to $T$ we need either a calculation or a look-up table, both of which are possible with electronics and/or a microprocessor, but this does mean that we cannot simply hook the thermistor up to a meter in the lab and take a quick reading of the temperature.

The point of this discussion is that none of the above are perfect and we have to make a choice according to: *i*) the temperature range we need to cover; *ii*) the accuracy required; *iii*) the harshness of the environment and *iv*) the ease of conversion into a temperature value. This is a general rule for transducers: the perfect device does not exist and we often need to compromise.

*AD590.* Where such a situation exists, the semiconductor industry is always ready to sell us a solution. The AD590 from Analog Devices is an integrated circuit device which produces a highly linear current output of 1$\mu$A/K. It works over a reasonable temperature range of $-55$°C to $+150$°C. It comes in a range of packages (Figure 2.2) and can be bought for under \$10 (more expensive than a thermistor but cheaper than a good PRTD). A simple circuit of resistors converts the current output to a voltage, then a multimeter gives us a very quick and easy direct reading of the temperature. The complexity of linearising the output as a function of temperature is all handled within the circuitry of the device (this is what we are paying for). We do not generally need to be concerned with what goes on inside the device as long as we have a good datasheet to describe its behaviour. The front-page of the AD590 is reproduced in Figure 2.3. It is useful to browse the complete datasheet and you are recommended to download a copy from the manufacturer's website at http://www.analog.com. Look through it to see the level of detail that the manufacturer thinks is important for the user to know. You also might want to identify the ways in which the performance of this transducer departs from 'ideal' behaviour.

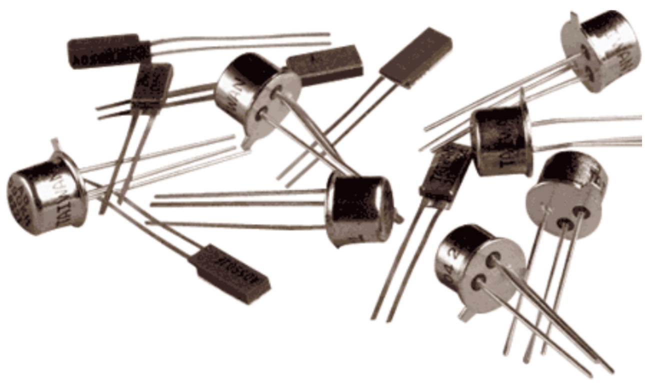

Figure 2.2: IC temperature transducer: Analog Devices AD590.

### *2.2. Photon sensors*

For the temperature transducers we are not, generally, expecting to operate at the limits of physics (unless we are trying to measure mK temperatures, but then we would use a different technique anyway). We often need to measure the temperature of our experiment or equipment and the above techniques will usually suffice. Next we will look at something more sophisticated: photon detection.

Photomultipliers tubes (PMTs) have been around for many decades and are still the sensor of choice for many photon detection applications. With the modern trend of implementing sensors in silicon, benefiting from mass production, robustness, etc., silicon photomultipliers (SiPMs) have become very popular in recent times. Also in this case sensor selection involves trade-offs across *many* parameters, and we shall discuss some examples here along with their operating principles.

#### *2.2.1. Photomultiplier tubes*

The photomultiplier tube (PMT) is effectively a type of light intensity transducer. It is based on vacuum-tube (i.e. 'valve') technology. It is capable of efficient photon detection at extremely low-light levels (down to the single photon). Manufacturers such as Hamamatsu Photonics or ET Enterprises Ltd (ETEL) have developed devices specifically for the physics community. At the extremes of measurement science we need to understand the physics of how the detector works in order to get the best out of it, which is why we will look into the principle of the PMT. Some models are shown in Figure 2.4.

The operation of the device is illustrated in Figure 2.5. A photon striking the photocathode liberates a photoelectron. The first "dynode" is held more positive than the cathode so the liberated electron is accelerated towards it and, on striking the dynode, several electrons are released – assume only 2 in this example. Each subsequent dynode is progressively more positive so, in this case, we get an electron multiplication of $2^n$ where $n$ is the number of dynodes. Now, in real devices we can get a gain in excess of 10 at each dynode. At the end of the tube the electrons are collected at a final anode and, flowing back to the power supply through a resistor, generate a voltage signal, the amplitude of which is proportional to the number of photons detected.

For a single photon detected (one photoelectron emitted from the photocathode) we can easily achieve a *gain* of 1–10 million (electrons at the anode per photoelectron emitted from the cathode), producing a few mV across a load resistor – which is an easily detectable voltage.

The PMT electrodes (photocathode, dynodes, anode) are typically biased using an external voltage divider circuit (see Figure 2.6) connected to a high voltage (HV) power supply. It is important to understand that the source of all the electrons which flow inside the tube is ultimately the power supply. In "dark" operation, a current flows around the loop which consists of the power supply and the resistors in the voltage divider alone. In the presence of light, some of these electrons are diverted into the tube via the dynodes, such that part of this current flows out of the tube via the anode. Ultimately, the tube will be saturated when all of the electrons from the HV supply are diverted into the tube. The entire current from the supply then flows through the anode and the voltage signal readout from the tube will be the maximum possible. This understanding is quite subtle and requires a working knowledge of basic DC circuits. We will cover this ground again in more detail in future lectures.

PMTs are not without their difficulties. A large (and somewhat dangerous) HV supply is needed (1–2 kV is usually applied across the tube). Even when there is no incident light, thermal emission of electrons from the photocathode means that a current always flows in the tube. This called the *dark current* and we want to minimise it since it adds noise and reduces the dynamic range of our measurement (these are both topics which will be discussed in more detail later on). One way to do this is to cool down the PMT to reduce the thermal emission. Note that the dark current appears as an *offset*, since the sensor shows an output even with zero light input.

ANALOG DEVICES

# 2-Terminal IC Temperature Transducer

# AD590

## FEATURES

**Linear current output: 1 μA/K**
**Wide temperature range: −55°C to +150°C**
**Probe-compatible ceramic sensor package**
**2-terminal device: voltage in/current out**
**Laser trimmed to ±0.5°C calibration accuracy (AD590M)**
**Excellent linearity: ±0.3°C over full range (AD590M)**
**Wide power supply range: 4 V to 30 V**
**Sensor isolation from case**
**Low cost**

## GENERAL DESCRIPTION

The AD590 is a 2-terminal integrated circuit temperature transducer that produces an output current proportional to absolute temperature. For supply voltages between 4 V and 30 V, the device acts as a high impedance, constant current regulator passing 1 μA/K. Laser trimming of the chip's thin-film resistors is used to calibrate the device to 298.2 μA output at 298.2 K (25°C).

The AD590 should be used in any temperature-sensing application below 150°C in which conventional electrical temperature sensors are currently employed. The inherent low cost of a monolithic integrated circuit combined with the elimination of support circuitry makes the AD590 an attractive alternative for many temperature measurement situations. Linearization circuitry, precision voltage amplifiers, resistance measuring circuitry, and cold junction compensation are not needed in applying the AD590.

In addition to temperature measurement, applications include temperature compensation or correction of discrete components, biasing proportional to absolute temperature, flow rate measurement, level detection of fluids and anemometry. The AD590 is available in chip form, making it suitable for hybrid circuits and fast temperature measurements in protected environments.

The AD590 is particularly useful in remote sensing applications. The device is insensitive to voltage drops over long lines due to its high impedance current output. Any well-insulated twisted pair is sufficient for operation at hundreds of feet from the receiving circuitry. The output characteristics also make the AD590 easy to multiplex: the current can be switched by a CMOS multiplexer, or the supply voltage can be switched by a logic gate output.

## PIN CONFIGURATIONS

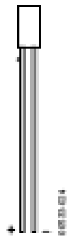


*Figure 1. 2-Lead CQFP*


*Figure 2. 8-Lead SOIC*

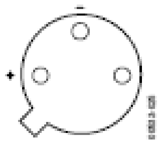


*Figure 3. 3-Pin TO-52*

## PRODUCT HIGHLIGHTS

1. The AD590 is a calibrated, 2-terminal temperature sensor requiring only a dc voltage supply (4 V to 30 V). Costly transmitters, filters, lead wire compensation, and linearization circuits are all unnecessary in applying the device.
2. State-of-the-art laser trimming at the wafer level in conjunction with extensive final testing ensures that AD590 units are easily interchangeable.
3. Superior interface rejection occurs because the output is a current rather than a voltage. In addition, power requirements are low (1.5 mW @ 5 V @ 25°C). These features make the AD590 easy to apply as a remote sensor.
4. The high output impedance (>10 MΩ) provides excellent rejection of supply voltage drift and ripple. For instance, changing the power supply from 5 V to 10 V results in only a 1 μA maximum current change, or 1°C equivalent error.
5. The AD590 is electrically durable: it withstands a forward voltage of up to 44 V and a reverse voltage of 20 V. Therefore, supply irregularities or pin reversal does not damage the device.




One Technology Way, P.O. Box 9106, Norwood, MA 02062-9106, U.S.A.
Tel: 781.329.4700 www.analog.com
Fax: 781.461.3113



Figure 2.3: AD590 datasheet (link).

Figure 2.4: Various PMT models from Hamamatsu Photonics.

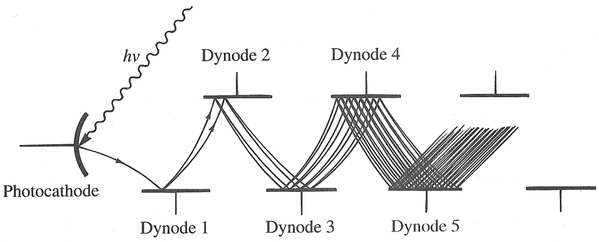


Figure 2.5: PMT operating principle (from Diefenderfer & Holton). This example shows a side-window PMT; most devices are front-illuminated PMTs using a semi-transparent photocathode.

The *quantum efficiency* (QE) of the photocathode is a very important parameter. Not all photons incident on the photosensitive layer lead to the mission of a photoelectron – since photons can simply be transmitted through, or a photoelectron may be created but never be emitted into the PMT vacuum. The QE is the fraction of incident photons which generate detectable electron emission. A QE of 25% is a typical minimum number, often as high as 40% in some PMT models.

*Further Reading.* Diefenderfer & Holton chapter 7 is good on the general topic of transducers and has more details of temperature transducers and PMTs. Note, however, that their figure 7.18 is wrong! (it is a useful exercise to

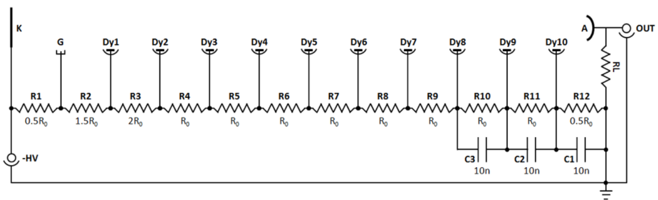


Figure 2.6: PMT voltage divider circuit. A single HV (here −1.0 kV) is applied to the cathode, and the circuit generates the voltages needed to bias the dynodes. The signal is collected at the anode and the current develops a voltage across the load resistor, $R_L$. The three capacitors ensure that the dynode electrodes can provide the required number of electrons without a significant voltage drop.

work out why). Horowitz & Hill chapter 15 is also very good, though it will make more sense once we have covered some of the electronics involved. For PMTs very good background and technical information can be found on the Hamamatsu and ETEL websites at www.hamamatsu.com and www.et-enterprises.com.

*2.2.2. Silicon photomultipliers*

SiPMs are silicon-based photon sensors consisting of an array of very small single-photon avalanche devices (SPADs) operating in Geiger mode, i.e. a few volts above breakdown. A small sensor 5 mm across may contain some 10,000 SPAD micro-cells (each typically 10–100 $\mu$m across). SiPMs are robust devices, requiring low bias voltages (tens of V), and offer extremely good characteristics for low-light (photon counting) applications. Some examples are shown in Figure 2.7.

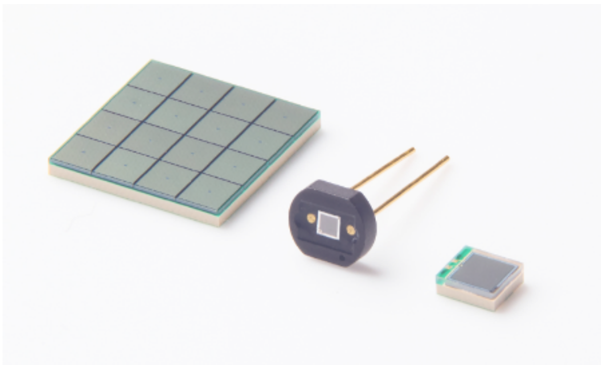

Figure 2.7: Hamamatsu SiPM devices and small arrays.

The silicon implementation of a SiPM is complex and varies depending on the wavelength of light and other factors – but the basic operating principle is relatively simple. A photon absorbed in a SPAD pixel produces a photoelectron which gets 'multiplied' in an avalanche process in the silicon. Each SPAD has a quenching resistor in series that limits ('quenches') the response of the pixel, i.e. the output signal of a cell is independent of the number of electrons which initiated the avalanche. Every signal in a single pixel produces the same-size response, typically $\sim$1M electrons – i.e. a gain comparable to a PMT. The outputs from all SPADs in the SiPM are connected in parallel, as illustrated in Figure 2.8, and summed across a common load resistor. At low light levels the output is highly linear and is proportional to the number of SPADs that 'fired' in response to the optical stimulus.

The probability that an incident photon is detected is not controlled by a quantum efficiency alone, as in the case of a traditional PMT, but rather by a *photon detection efficiency* (PDE). In addition to the quantum efficiency of the silicon, this parameter must take into account the probability that a photoelectron actually initiates an avalanche and, significantly, the 'fill factor' of the device, i.e. the ratio of active to total area of a pixel as a result of the dead space between SPADs. Nevertheless, SiPM PDEs

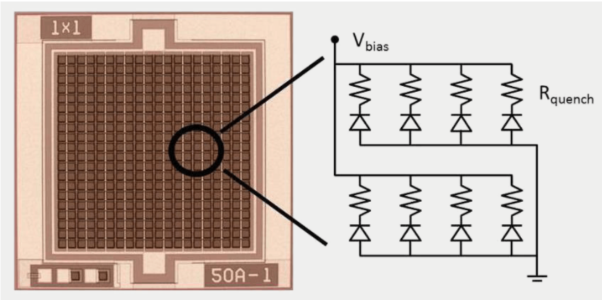


Figure 2.8: SiPM principle of operation (from Ketek GmbH).

can match PMT QEs in most wavelength regions, and we saw above that the gain is also comparable.

SiPMs have a number of advantages over traditional vacuum PMTs, as well as some disadvantages. Their ability to ‘count photons’ with extremely high resolution is perhaps their main advantage. On the other hand, they have extremely high *dark count rates* at room temperature, as well as *correlated noise* (e.g. an avalanche in one SPAD can trigger one in an adjacent cell).

In conclusion, photon detection is a good example of where a long and very successful technology is being slowly replaced by a more modern, silicon-based one, but even in this case no device is perfect: in any application it is critical to understand the detailed technical requirements and how this maps to the specification offered by each technology or device. The solution is often not obvious.

## 3. Input and output impedance

*Thevenin and Norton equivalent circuits.* We can consider any device such as a sensor to be modelled as a Thevenin or Norton source. Usually, we model a device which is inherently voltage-producing (e.g. LM35 temperature sensor) as a Thevenin source, and a device which is inherently current-producing (e.g. PMT) as a Norton source – but in practice the two models are interchangeable (you can convert a Thevenin model to a Norton model and vice-versa if you wish). Both these models give us the idea that the source is always associated with some kind of impedance known as the source or *output impedance*.

*Output impedance.* This is the impedance which we would measure 'looking into' the terminals of a source. This we can do by varying the load resistance which we connect it to, and seeing how the output signal (voltage or current) varies as a function of load, then using this information to calculate the effective impedance of the device. This is how it is done in first-year lab, in the experiment to calculate the output impedance of a signal generator.

We hope that the output impedance of a voltage source is low, and for a current source that it is high, and in both cases that it is real (resistive). This is generally true; however, any length of wire will have an inductance, and consequently there will always be some small reactive component of impedance too. This is a function of the design of the sensor, so we can try to keep the inductance low by minimising cable length and (especially) avoiding loops and coils of wire. In practice, sensor inductances are usually small and, since the reactance $X_L = \omega L$, the contribution to total impedance is negligible for all but the highest frequencies. There is also some capacitance in any sensor, and this causes more problems, typically. Any wire running next to another wire or the case of the sensor will result in there being some capacitance between the two conductors. This is called *stray capacitance*. There may also be some capacitance due to the physics of the sensor, e.g. the PMT where the anode and dynodes are metal plates. Overall, these capacitances can add up to some tens of pF, resulting in reactances of some kΩ at MHz frequencies. Stray capacitance is usually modelled as being in parallel to the sensor terminals, while stray inductance (should it exist) will appear in series.

*Input impedance.* Any device to which we connect a sensor, such as an amplifier, a meter or an oscilloscope, will have an *input impedance* which is the apparent ratio of the voltage applied to the device to the current it draws. Generally, we want this to be high and resistive for voltage measurements, and low for current measurements.

The same arguments as above apply for stray capacitance and inductance. A typical oscilloscope input is equivalent to a resistance of 1 MΩ in parallel with a capacitance of about 20 pF. Due to the fact that the capacitance is in parallel, it has no effect at low frequencies, but at MHz frequencies it is clear that the total input impedance $\mathbf{Z_T} = (1/R + 1/\mathbf{Z_C})^{-1}$ will be significantly reduced. Most scopes come with special probes designed to boost the impedance and hence prevent the probe from loading the circuit being measured.

### *3.1. Impedance matching*

When the source and load impedance are equal then we find that the maximum power is transferred from source to load. We will also find that this is the condition for minimising any reflections of signals. Note that, in the case of the voltage source, the source and load form a voltage-divider in such a way that the voltage measured in the load is only half the source voltage – see Figure 3.1. This is due to the significant current which the load (same low impedance as the source) is pulling from the source. This is an example of *loading* a source; we might say that by loading the sensor with a low-impedance we have "pulled down" the voltage. This is OK, as long as our intention is to transfer maximum power.

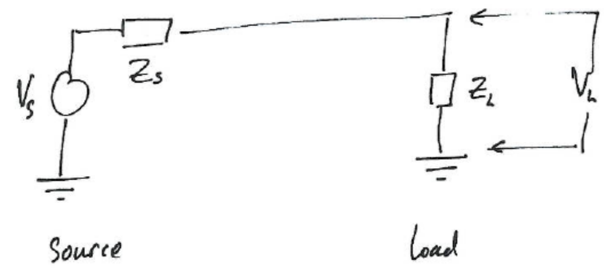


Figure 3.1: Impedance matching: the power transfer to the load is maximised when $\mathbf{Z_S} = \mathbf{Z_L^*}$ (or $R_S = R_L$ if both are purely resistive). Note that when the impedance is matched the source has been loaded, i.e. we now measure only a fraction of the voltage: $V_L = V_S/2$ – though $P_L = P_S$ in this instance.

The *maximum power transfer theorem* states that for source and load resistances $R_S$ and $R_L$ the maximum power is transferred for $R_S = R_L$. In the general case of complex impedances, we find that we need $\mathbf{Z_S} = \mathbf{Z_L^*}$. This results in equal power being dissipated in both source and load, since the voltage-divider splits the source voltage equally across source and load impedances. This is not very efficient, but we do achieve maximum power, which we sometimes wish to do. The theorem is easily proved; this will be left as a problem sheet question.

If source and load impedance are fixed and we want to achieve maximum power we can use an *impedance transformer*, as shown in Figure 3.2. This works for AC signals only, as it requires use of a transformer to magnetically couple the power from the primary to the secondary circuits. A transformer consists of two coils of wire (solenoids) wound together often on a core of high magnetic permeability. The inductance of each solenoid is proportional to the square of the number of turns. We have $L \propto n^2$ and $Z \propto L$ for both the primary and secondary sides of the transformer. Since the transformer is entirely inductive (reactive) it couples power from primary to secondary sides *with no loss* (in practice there is some small resistance in the wire which results in ohmic loss, of course; however, transformers can be very efficient). By choosing the number of turns on the primary side to give the same impedance as the source we couple max power from source into the transformer. By choosing the number of turns on the secondary winding to give the same impedance as the load we couple all the power out of the transformer into the load. As if by magic, the transformer has 'transformed' the impedance of the load into something which matches the source. In practice, this works quite well for frequencies in the kHz to MHz range.

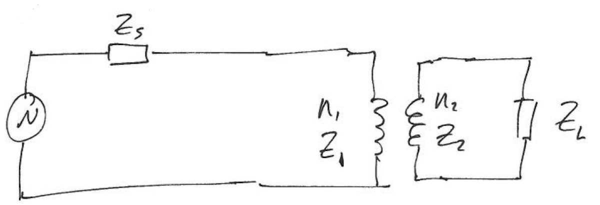


Figure 3.2: Impedance transformer: stepping up or stepping down impedances.

*Efficiency.* We can define *efficiency* as the ratio of load power to total power: $\epsilon = P_L/P_T \times 100\%$. Maximum power transfer results in only 50% efficiency! This seems counter-intuitive, but consider a battery with $R_S = 1\ \Omega$: if we connect a 1 $\Omega$ load we will get the maximum power we can out of the battery. Both battery and load will get very hot, and the battery will be exhausted very quickly. Half of the chemical energy has been wasted heating the battery. We can make better use of the stored energy by releasing it more slowly into a higher-resistance load. It can easily be shown that:

$$\epsilon = \frac{R_L}{R_L + R_S}\,.$$

Hence, for maximum efficiency we want either a low $R_S$ or a high $R_L$. Either will deliver improved efficiency, at the expense of reduced overall power. In other words, it is important to understand that maximum power transfer and maximum efficiency are different things. We can increase the efficiency by increasing the load resistance; however, this decreases the overall power delivered by the source. If we want instead to maximise the absolute amount of power transferred to the load, we should match the two impedances.

### *3.2. Impedance bridging*

In many situations we do not wish to transfer maximum power from source to load. For example, for many voltage-source sensors we just wish to measure the voltage which the sensor produces. Indeed, drawing any current from the sensor is a problem, since the loading effect will drop voltage across the sensor's source impedance. In this case, we will use *impedance bridging* where a low-impedance source is connected to a high-impedance load. For example, measuring the voltage of a battery we use a digital multimeter (DMM) with an input of 1 M$\Omega$ or so. The efficiency here is very high and, in practice, we transfer negligible power from the battery into the meter. The meter is a voltage-sensing device; we do not wish it to draw any more current than is strictly necessary for accurate operation. *Impedance bridging* is the norm for instrumen-

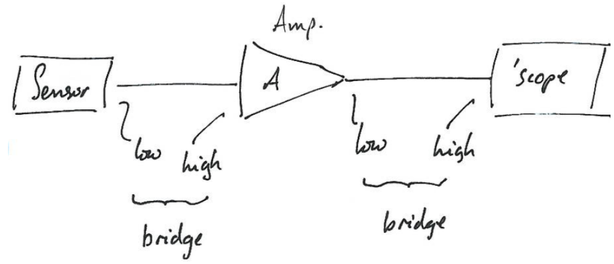


Figure 3.3: Impedance bridging to preserve the amplitude of a voltage signal. In this regime the source impedance is low and the load impedance is high, so that the efficiency approaches 100%.

tation applications where we simply wish to move a voltage signal around without transferring significant amounts of power – a typical example is illustrated in Figure 3.3. The exception to this is where we need to *impedance match* specifically to avoid signals being reflected (e.g. if the signal out of the sensor is very fast, such that reflections from a mismatched amplifier would distort the signal shape or even appear as new, delayed signals).

## 4. Transmission lines

In normal circuit analysis (e.g. Appendix I on Circuit Theory) we consider that voltages and currents propagate instantaneously. For many practical purposes this assumption is valid, i.e. we can assume for any piece of wire that the voltage at one end will be the same as the voltage at the other end. However:

- The wire will have some resistance, so if it is very long we may have a small voltage drop due to simple ohmic loss. In general, silver-plated copper wire is common-place and ohmic loss is negligible (and usually can be ignored);
- The voltage and current signal will actually propagate down a cable at about two-thirds of the speed of light. Since modern electronics operates at very high frequencies, and often we want to transmit information over quite long distances, this turns out to be a much more important consideration.

For example, consider the wired ethernet link which connects through Blackett (dimension ~100 m). The data-rate is up to 1 Gb/s. It takes about 500 ns for information to travel from one end of the building to the other, in which time a "sender" will have sent about 500 bits (62.5 bytes) of data before any information is received at the far end. So, 500 bits of data are travelling simultaneously down the wire at any given time – in the form of coded pulses. For high-frequency signals, the speed of light is actually rather sluggish and something of a limiting factor. Consequently, for circuits where the signal must run over a considerable cable length, we must we must consider the signal to be a wave propagating along the wire. A rule-of-thumb says we must worry about this for wire lengths $L$ such that

$$L \approx \frac{\lambda}{10}\,.$$

There is a further complication: when $L$ is comparable to $\lambda$, then what we have built is an aerial: the wire will very effectively couple electromagnetic energy into the environment. Our signal source and wire are a transmitter! This also works the other way around: any electromagnetic radiation in the environment will couple into the wire, and we will pick this up as noise or interference.

For all the above reasons and more we often choose to send high frequency signals down coaxial cables rather than single wires.

### *4.1. Coaxial cable*

This is a type of wired connection where the signal is carried on a central conductor of radius $r$ surrounded by a sleeve of dielectric material with relative permittivity $\epsilon$. This is wrapped with an outer conductor (usually a braid) of radius $R$; this outer conductor is grounded. Since the inside-to-outside conductors form a capacitor, it is easy to see that the cable will have a capacitance which is dependent on the length of the cable. This is usually specified as a capacitance per unit length $C'$ where

$$C' = \frac{2\pi\epsilon_0\epsilon}{\ln(R/r)}\,.$$

Similarly, there is an inductance per unit length

$$L' = \frac{\ln(R/r)\mu_0\mu}{2\pi}\,,$$

where $\mu$ is the relative permeability of the dielectric, which is always 1 for non-magnetic materials.

As we might expect, $C'$ and $L'$ depend entirely on the geometry and materials of the cable. The cable acts like a waveguide: signals will propagate via the transverse electromagnetic mode, with a radial E-field and a circumferential B-field. This means that the E-field is entirely trapped within the cable, and will not radiate away like it might do from a plain wire. Also, the outer grounded shield conductor 'shields' the signal from any pick-up of external interference.

The actual capacitance and inductance of a cable of length $l$ are $C'l$ and $L'l$, which both tend to $\infty$ as $l \to \infty$. However, a propagating wave does not see the whole cable: this is analogous to a mechanical wave propagating along an infinite chain with a certain mass per unit length.

### *4.2. Characteristic impedance*

Because it is modelled as a capacitance and an inductance, the cable is a linear system, in that the applied complex voltage results in a proportional complex current, the ratio of the two being the *characteristic impedance* of the cable:

$$Z_0 = \frac{\text{applied sinusoidal voltage}}{\text{resultant sinusoidal current}}\,.$$

$Z_0$ is characteristic of the medium through which the signal propagates. This is analogous to the acoustic and electromagnetic impedances of materials which can also be considered as the ratio of some applied-to-resultant sinusoid. This understanding of characteristic impedance is only relevant for cables significantly greater in spatial scale than the wavelength of the signal propagating. You will have covered the propagation of waves on a transmission line in Year 2 E&M, and all textbooks cover this well. The voltage and current on an infinite coaxial cable can both be represented by the wave equation with the solution that any arbitrary-shaped signal will propagate down the line at speed

$$v = \frac{1}{\sqrt{L'C'}}\,, \tag{4.1}$$

and the characteristic impedance is given by

$$Z_0 = \sqrt{\frac{L'}{C'}}\,. \tag{4.2}$$

Consider an infinite length of coaxial cable. If we introduce a voltage wave into the end of the cable it will propagate down the cable at speed $v$, never to return. Simultaneously, the voltage wave will draw a current wave, and the ratio of the two, at any point, is the characteristic impedance $Z_0$.

You can easily conclude that, assuming unity permeability, $v = c/\sqrt{\epsilon_r}$, which is around $2/3\,c$ for a typical cable dielectric. So, signals travel at approximately 1 foot/ns in free space, and at around 2/3 of that speed in a typical coax cable. For a small circuit this may not be a problem, but if your fast sensor is, say, 10 m away from its front-end amplifier, this can be a significant time.

### *4.3. Input impedance of a matched cable*

First imagine a coaxial cable of infinite length. Seen from the outside, the infinite coax appears to 'swallow' the voltage signal (it disappears into the cable, never to return), so it acts the same as a resistor of value $Z_0$ in that it appears to dissipate all of the energy in the signal. Now imagine that the infinite cable is cut down to a finite length and terminated at the far-end with a resistor of value $Z_0$. When the signal reaches the far end of the cable, it encounters the resistor and is dissipated. No signal returns back up the cable. Therefore, a coax *terminated* with a resistor of value $Z_0$ (said to be a *matched cable*) has an input impedance which appears to be entirely real and have the value $Z_0$. Further, we cannot tell the difference between an infinite coax and a short one terminated with $Z_0$: from the sending end, they appear identical.

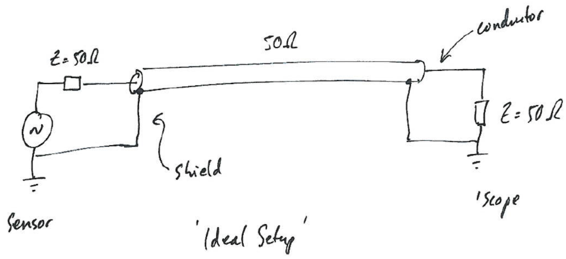


Figure 4.1: A matched transmission line delivers the power from the source to the load impedance with no reflections.

Note that the (ideal) coax cable is purely capacitive and inductive, so it does not dissipate any of the signal power. Also, the interpretation of $Z_0$ as being a real constant means that it is not frequency-dependent. So, the correctly terminated cable is "transparent", in that the signal passes down it with no reflection or absorption. The only effect of the cable is the time delay it introduces as the signal propagates down its length.

### *4.4. Mismatched cables*

If we join two cables with impedances $Z_1$ and $Z_2$ then, as expected, we will get reflections at the boundary according to

$$k_r = \frac{Z_2 - Z_1}{Z_2 + Z_1}\,, \tag{4.3}$$

where $k_r$ is the amplitude reflection coefficient (the ratio of incident wave amplitude to reflected wave amplitude). This corresponds to the Fresnel reflection of a light wave at normal incidence in optics: the square of this quantity gives the reflected intensity, and $1 - k_r^2$ is the transmitted intensity (and there is also an acoustic analogue).

The above equation can be derived by consideration of the boundary conditions at the junction. If the incident voltage amplitude is $v_i$ and incident current amplitude is $i_i$, then we have also a transmitted wave $(v_t, i_t)$ and a reflected wave $(v_r, i_r)$. At the boundary, we have

$$v_i + v_r = v_t\,,$$

since the boundary is conductive the potentials are the same either side of the junction. Also, by conservation of current at the junction:

$$i_i - i_r = i_t\,.$$

The reflected current wave is travelling in the opposite direction to the incident wave. We also have everywhere:

$$v_i = Z_0\, i_i\,. \tag{4.4}$$

Since $Z_0$ is real, voltage and current are in phase; the solution yields equation (4.3) for the reflected amplitude (and $k_r^2$ for reflected power). Note that we can only apply this rather simplified analysis to obtain equation (4.3) since the $v$ and $i$ waves are in phase, i.e. because $Z_0$ is real.

### *4.5. Transforming impedance*

Since the coax cable is truly a macroscopic medium through which an electromagnetic signal may propagate we can use exactly the same tricks which we saw applied in optics, where coated elements and interference coatings are often used.

*Intermediate impedance.* If we have a source with impedance $Z_1$ which we cannot change – such as an aerial (see example below) – which we need to connect to a non-matching load impedance $Z_2$, then we can insert a length of cable with $Z_0 = \sqrt{Z_1 Z_2}$ which will have the effect of minimising the overall signal reflection and therefore maximising transmission. An example from optics: magnesium fluoride ($MgF_2$) coatings have about the right refractive index to better match air (or vacuum) to glass.

*Quarter-wave section.* If we make the intermediate cable section exactly $\lambda/4$ in length, then we will obtain interference cancellation and hence we can approach a maximum of 100% transmission, albeit at a single frequency only. The same is true in optics: if the above $MgF_2$ coating has thickness $\lambda/4$, we can make the two reflections interfere destructively with each other and couple 100% of the intensity: we have thus made an anti-reflection coating at a particular wavelength $\lambda$.

### 4.6. Application example: the TV aerial cable

This is a simple example which serves to illustrate many advantages of using coax cables. The traditional aerial mounted on the roof is connected to the receiver using coax cable of $75\Omega$ characteristic impedance. This impedance matches that of the aerial quite well. The aerial – at its simplest just a length of wire – is tuned to pick-up frequencies in the desired range. We want to transfer maximum signal power from the aerial into the receiver, with minimum reflections which would distort and diminish the received signal. The aerial is tuned so as to reject pick-up of unwanted frequencies, e.g. shorter wavelengths such as mobile phones or longer wavelengths such as FM radio. The screened/grounded coax cable prevents these signals from being picked-up en-route to the receiver. Since the TV signal is entirely contained within the cable (like a waveguide) it does not couple capacitively to any nearby metal objects, such as the drain-pipe down which the cable may have been routed.

### 4.7. Real-world effects

The coax cables described above were assumed to be ideal or *lossless*. In practice, any substantial length of cable will have some ohmic resistance which will contribute to an overall reduction of the signal amplitude or power. This is called *ohmic loss* and is independent of frequency. For very high-frequency signals the dielectric material is not fully efficient, and hence the E-field will dissipate some power into the dielectric. This is called *dielectric loss* and is frequency-dependent. The net effect is usually quoted as an overall power attenuation factor per unit length. For example the typical type of RG-58 cable commonly found in labs has a characteristic impedance of $50\Omega$ and an attenuation factor varying between 0.11 dB/m at 50 MHz and 1.4 dB/m at 2 GHz.

### 4.8. Key points

These are the main points from this section:

- The characteristic impedance $Z_0$ is only a function of material and geometric properties of the cable;
- $Z_0$ is real and independent of frequency, yet the ideal line is lossless;
- The voltage and current on the line are everywhere in-phase, and their ratio is $Z_0$;
- A line terminated with a resistance $R = Z_0$ appears to be totally transparent: its input impedance is $R$;
- A mismatched line will result in reflections at the boundary with an ensuing reduction in signal power. The reflected power will be proportional to $k_r^2$;
- A line left open (or short-circuited) at the far end will result in 100% reflection of the signal; the returning signal will be inverted for the short-circuit case.

## 5. Digital signals

Most signals we encounter in physics are continuous functions of time, such as light intensity, temperature, pressure, etc. If we want a computer to interact with these variables we have to first sample and then digitise them. The process of *sampling* quantises time, while *digitisation* quantises the parameter being measured. Ultimately, this means we lose some information about the original signal, and we have to be careful in how we interpret digitised signals. This section will outline some of the basics; we postpone a discussion of the electronics implementation of some key circuits for analogue-to-digital conversion until Section 12.

Note that sampling (and digitisation) are not entirely phenomena of the computer age. Taking a temperature reading (every hour, every day, ...) is a process of sampling. Converting the height of mercury in the thermometer (a continuous, analogue parameter) into a number written down in a notebook is digitisation.

This section is based on Chapter 3 of the book "The Scientist and Engineer's Guide to Digital Signal Processing" by Steven Smith. This book is very readable and provides a good descriptive text though not always with as much mathematical backing as we might like. It is available online.[2]

**Learn it in lab**
The first lab session is devoted to the topics discussed in this section – you will see how signals can be properly sampled and digitised, and the effects of aliasing in detail. Conversely, you should read this section again before that particular lab.

### *5.1. Sampling*

As suggested above, sampling-and-digitisation is a two-stage process, as illustrated in the central diagram of Figure 5.1. Taking an analogue signal (typically a voltage) as an input, we must first sample it, which is mathematically equivalent to taking an instantaneous value of the function at some point in time. Usually, we want a series of samples with uniform time spacing. If the time between samples is $T_s$, then we can define a sampled version of the function as

$$f_s(t) = \sum_{n=-\infty}^{\infty} f(nT_s)\,\delta(t - nT_s)\,. \tag{5.1}$$

This says we take the function $f(t)$ and we multiply it with an infinite sequence of delta functions (mathematically this is the 'comb' function). What we get is an infinite sequence of delta functions, each one weighted by the instantaneous value of the function. The main point to take from this is that a *sampled signal* is mathematically very different from a continuous signal in that it is non-zero only for the values of time $t = nT_s$.

*Sample and hold.* In practice this mathematical representation is very far from reality, since:

- We know that it is actually very difficult to get an instantaneous value of a function since any real-world hardware will take a short but non-zero time to take a sample;
- As per Figure 5.1, we want to pass the sampled value into the next block which performs the digitisation. This can take some considerable length of time to do, so we want to value of the sample to persist.

The solution is the *sample-and-hold* circuit, whose job is to take a snapshot sample of the waveform and then hold that value on its output. Note that the output is still an analogue voltage. The circuit takes a sample when commanded to do so by an external *clock signal*. The clock is usually a square wave with period $T_s$. The sample typically occurs on each *rising edge* of the clock signal, so if we have a nicely stable clock signal we will have very regular sampling, which is very important for the quality of the sampled signal. The variability in the sampling period is called "jitter", and is usually very small. The circuit for the sample and hold is relatively straightforward, but we will look at this later.

As shown in Figure 5.1, changes at the input that occur between sample times are ignored, that is to say, sampling converts time (the independent variable) from continuous to discrete.

### *5.2. Digitisation*

In Figure 5.1 we can assume that the input signal can vary from 0 to 4.095 V which the *Analogue to Digital Converter (ADC)* will translate into the digital numbers 0 to 4095. The ADC produces an integer value for each of the samples. Only integer values are allowed, which means that the ADC has a *digital resolution* of 1 mV (the smallest change on the input which will guarantee a change in the output number). Therefore, the ADC performs a *quantisation*: it converts the voltage (the dependent variable) from continuous to discrete. The bottom panel of the figure shows the error that this process introduces called the *quantisation error*, defined as the difference between the sampled signal and its digital representation. It it presented in terms of digital numbers so the probability density function (PDF) shows that it is uniformly distributed between $\pm 0.5$. The fundamental digital number is frequently referred to as the *Least Significant Bit (LSB)*, for reasons we will discuss soon. The ADC is a rather more complicated circuit than the sample-and-hold, and again we will examine this later.

*Binary digits.* A 'bit' is a binary digit which consequently has only two values: '1' or '0'. In our ADC example we can have any digital number between 0 and 4095, that is to say 4096 distinct values, which are represented by a 12-bit binary number (since $2^{12} = 4096$). Table 5.1 gives some

[2] www.dspguide.com

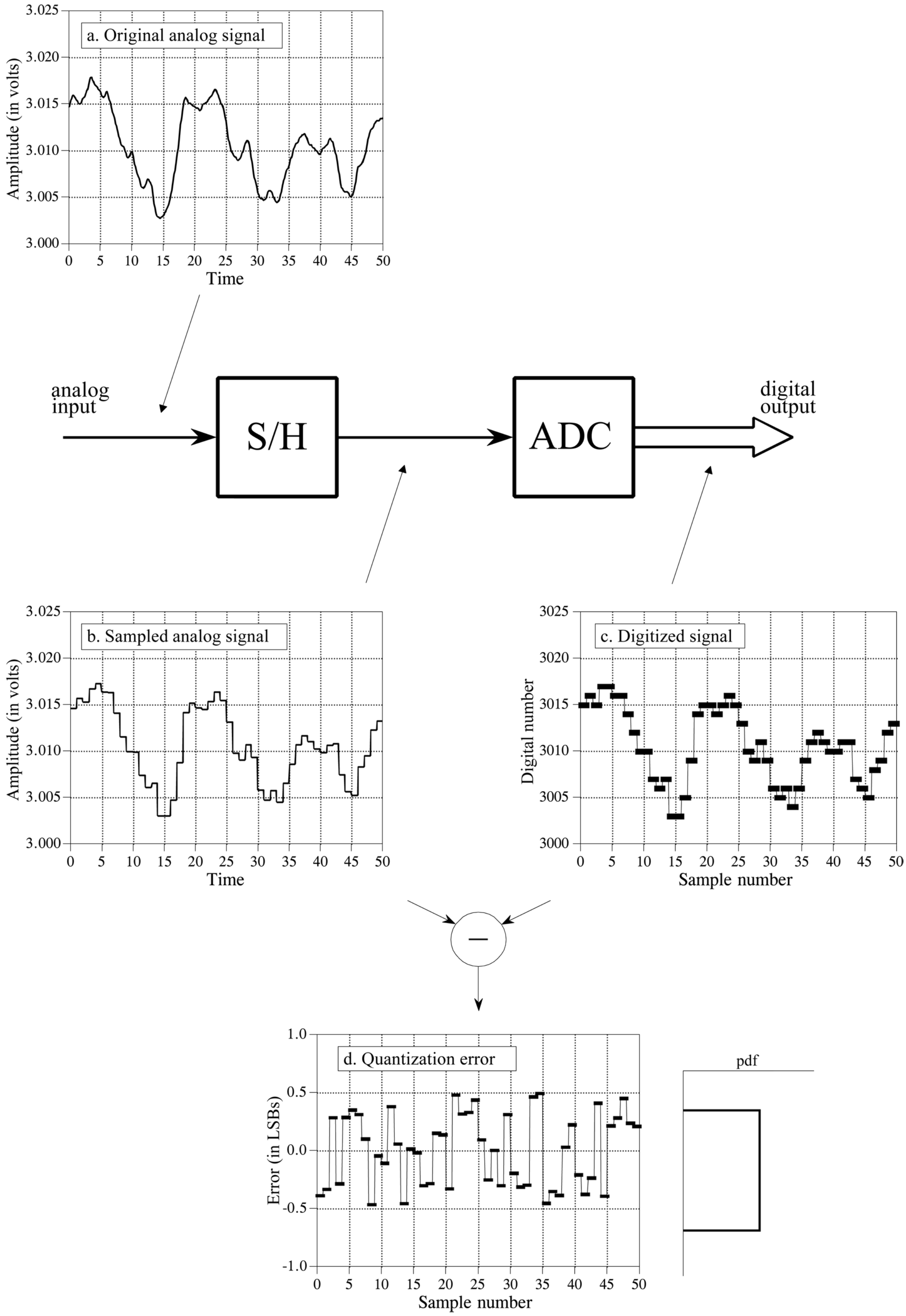


Figure 5.1: Waveforms illustrating the two-stage process of sampling and digitisation (from Smith, www.dspguide.com).

Table 5.1: Some 12-bit binary numbers and their decimal equivalents.

| binary | decimal |
|---|---|
| 000000000000 | 0000 |
| 000000000001 | 0001 |
| 000000001110 | 0014 |
| 111111111111 | 4095 |

examples. Digital electronics uses binary representation of numbers since this fits nicely with the fundamentally two-state 'ON=+5 V' and 'OFF=0 V' nature of digital logic.[3] The 'number of bits' is a measure of *digital resolution*. If we were to use a 14-bit converter over the same input range then the resolution would increase four-fold (each digital number now represents 0.25 mV). In engineering-speak, the LSB (right-most binary digit) is 0.25 mV.

*Quantisation error.* Quantisation results in nothing more than the addition of some random noise to our signal. As mentioned above and in Figure 5.1, this extra noise is usually uniformly distributed between ±0.5 LSB. This means we can define the noise statistically according to its mean $\mu$ and standard deviation $\sigma$:

$$\begin{aligned} \mu &= 0 \\ \sigma &= \frac{1}{\sqrt{12}}\,\mathrm{LSB}\,. \end{aligned}$$

Now, for distributions with zero mean, the RMS is the same as the standard deviation (check the definitions if you are unsure). If we want to reduce the RMS noise in our digitised signal, we need to "improve the resolution of the measurement by increasing the number of bits".

**Example: Quantisation noise and full-scale**
Passing an analogue signal through an 8-bit ADC adds an RMS noise of $1/\sqrt{12}$ LSB. Since the full-range (maximum) input is $2^8 = 256$, this is about 0.1% of the full-range value. If we switch to a 12-bit ADC this is reduced to 0.007% (much more acceptable, generally).

This understanding of quantisation noise is extremely powerful, because the RMS noise generated during digitisation simply adds in quadrature with any noise already existing in the analogue signal (and as we shall see later, there is always some noise in any signal).

[3] See Horowitz & Hill chapter 8.

**Example: Quantisation noise and signal noise**
Returning to our original example of the 12-bit converter with a resolution of 1 mV. Say the RMS noise measured in the analogue signal is 0.5 mV then this translates to 1/2 LSB in digital numbers and the total noise is $\sqrt{1/4 + 1/12} \approx 0.6$ LSB. Note how the digitisation increases the overall noise by a small amount. We might be tempted to upgrade to a 14-bit converter, but actually there is little point, since the total noise will always be dominated by that in the original signal.

As a final note on the topic, recall that the ADC returns only integer values but we are discussing here noise being a fraction of 1 LSB. We will only see this noise in the statistics of a large number of samples, each of which is an integer, but taken together they have non-integer statistics.

### *5.3. Aliasing and the sampling theorem*

We throw away a lot of information about a signal when we digitise it because we quantise both time and the parameter being measured. This is probably the single most important point to understand about digitised signals: information lost in the digitisation process is lost forever – there is no way we can recover the information about what the signal was doing between samples, nor can we say anything about the small amplitude details which are below the resolution of our ADC. In the case of the latter, we have seen how the digitisation of the signal can be characterised as extra noise added to the signal, which we can understand statistically. Now we will turn our attention to the effect of the former, that is to say the sampling of the signal and the quantisation of time. First we will look at the *Sampling Theorem*, which is actually a rule about how to ensure *proper sampling*, and then we will look at *aliasing*, which is the effect we get if we break the sampling theorem rule, resulting in *improper sampling*.

*The Sampling Theorem.* Put simply, a digital signal can only properly represent frequencies up to *one half of the sample rate (1/$T_s$, where $T_s$ is the sample period)*. This is somewhat intuitive. What is the minimum number of points we need to (very crudely) represent a sinusoid? It is worth trying to sketch this on a piece of paper, with maybe 10, 5, 3 or 2 points per cycle of the sinusoid. Join the points with a straight line. It should be clear that 2 is the absolute limit; it looks more like a triangle wave than a sinusoid, but the key point is that it represents the correct frequency. We will look at this in more detail in the next section. The formal statement of the sampling theorem (also known as the Shannon sampling theorem, or the Nyquist theorem) is that:

*A continuous signal can only be properly sampled if it does not contain frequency components above one half of the sampling rate.*

For example, a sample rate of 2000 samples/s (S/s) requires that the original signal only contain frequencies below 1000 Hz. Half the sample rate (here 1000 Hz) is called the *Nyquist frequency.* If signals above the Nyquist frequency are present they will be aliased, which means they will appear as new signals at frequencies below the Nyquist frequency. This is bad; it is often referred to as *improper sampling.*

*Aliasing.* This is best illustrated graphically, in the first instance. Figure 5.2 shows some some sinusoids before and after sampling. The continuous line is the input and the square markers are the sampled values. The question is: for each waveform input can we unambiguously recreate this from the samples? For the DC input (panel (a), a cosine of zero frequency!) we can certainly say "yes". For panel (b) we might have a 90 Hz signal sampled at 1 kS/s, so there are more than 11 samples per cycle of the signal. This is clearly proper sampling, according to Nyquist. For panel (c) we have only 3 samples per cycle. Imagine you take away the input (solid line) and join adjacent dots. Intuition tells us this is a roughly-drawn sinusoid at the correct frequency. It turns out that the human brain is very good at this sort of pattern recognition. More importantly, if we were to put these samples into a Fast-Fourier Transform program, it would also correctly report the frequency of the signal. This is still *proper sampling.* In (d) we push this further to just over 1 sample per cycle. Disaster: both the human and the FFT report a sinusoid at the wrong frequency. This is *improper sampling.* In fact, if the sample rate is 1 kS/s and the input frequency is 950 Hz, then we get a sinusoid of frequency 50 Hz in the digital data. In general, for an input frequency $f$ and a sample rate $f_s$, then if $f > f_s/2$ we get an alias frequency $f_a$ such that

$$f_a = f_s - f \, .$$

Now, the problem arises that our input signal may contain frequencies going to values many times $f_s$, so we need to generalise this to

$$f_a = |nf_s - f| \, , \tag{5.2}$$

where $n$ is an integer chosen to give a value of $nf_s$ as close as possible to $f$. This may seem a little confusing but we will see why this is in the next section. As an example of this, assume $f_s = 100$ Hz and the input signal contains all of the frequencies 25, 70, 160 and 510 Hz added together. The spectrum of the analogue signal would show all these frequencies. The spectrum of the digital signal shows the frequencies 25, 30, 40 and 10 Hz. The first is correct, but the last three are *aliases.*

But it gets worse. If we see a signal at 10 Hz in the digital data we have no means of knowing if the original analogue signal was 10 Hz, 90 Hz, 110 Hz, 190 Hz, etc. It could be that all of these signals were present, so all of the aliases would add on top of the real 10 Hz signal, thus destroying any knowledge we might need about the amplitude of the original 10 Hz signal. This illustrates a key point about aliasing: not only does it generate new false frequencies, it can also destroy information about the correct, lower frequencies. Because it is so important, we will study this in more detail in the first lab session.

*5.4. The frequency characteristics of sampled signals*

One important point to begin with is that a sampled signal is fundamentally unlike any other kind of continuous signal you will have come across before. According to equation (5.1), the original signal has been multiplied by a series of delta functions to create what we might call an 'impulse train'. It is non-zero for values of $nT_s$, but zero in-between. This gives it a very complicated spectrum, which we can see by taking the Fourier Transform of (5.1) to get[4]

$$F_s(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} F(\omega + n\omega_s) \, , \tag{5.3}$$

$$\omega_s = \frac{2\pi}{T_s} = 2\pi f_s \, .$$

This shows that the spectrum of the sampled signal is the same as the original signal, but repeated infinitely along the frequency axis. Figure 5.3 gives a graphical illustration of this. Panels (a) and (b) show the original signal and its spectrum. We can see that the signal is limited to a band of frequencies below the Nyquist frequency, so we should be able to sample it properly. In fact, we are sampling comfortably above this at about 3 times the highest frequency in the analogue signal. Note that this signal and its spectrum is highly stylised; real signals rarely have such neatly compact spectra, as we shall see later; however, this illustrates the principle here.

Panel (c) shows the sampled version of the signal, in the form of an impulse train, and in the spectrum (d) we can see the new repeating frequencies generated by the sampling process. The reason for this is not straightforward but once understood it does provide a rather satisfying explanation for aliasing. In equation (5.1) we can see that the original signal was multiplied by an infinite sequence of delta-functions (the so-called comb function). Now, the Fourier transform of the comb function happens to be another comb function.[5] Further, multiplication in the time-domain is equivalent to convolution in the frequency domain. Therefore, in the frequency-domain we expect that the spectrum of the sampled signal is the spectrum of the original signal convolved with a comb function.

[4]We will cover the integral Fourier transform in more detail in the next section. In general, the mathematics for sampled signals and the equivalent transforms into the frequency domain are rather involved and beyond the scope of this course. The result is quoted here to give an understanding of the behaviour of the sampled signal under the Fourier transform, but you would not be expected to know or derive this.

[5]See the table of Fourier transforms in Poularikas.

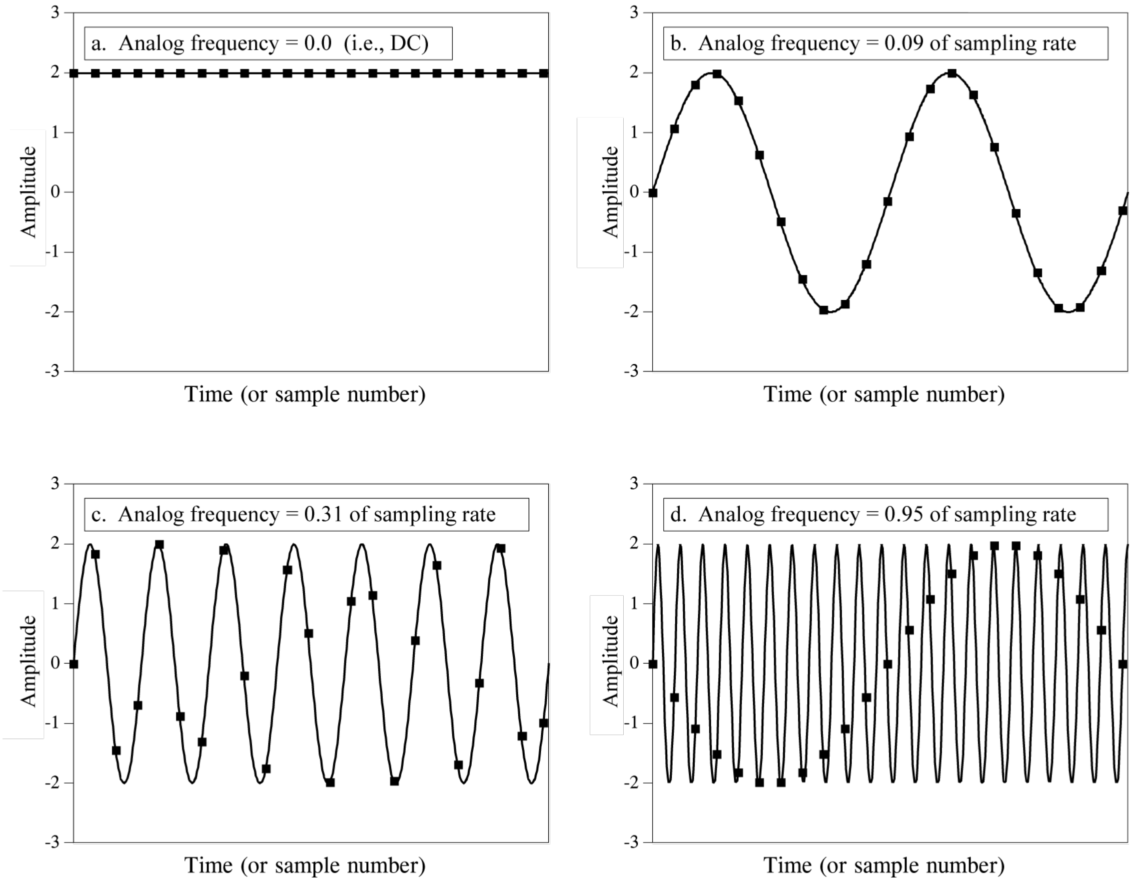


Figure 5.2: Graphical representation of proper and improper sampling (from Smith, www.dspguide.com).

This is why the spectrum repeats infinitely. Note that in the figure only the positive frequency range is shown. We know that the Fourier transform generates negative frequencies as well; this is what accounts for the part of the spectrum labeled "lower-sideband". The spectra "copies" repeat each multiple of the sample frequency. If we reduce the sample rate by a factor of two – panel (e) – then the spectra get closer together – panel (f) – and cross-over into each other. Recall that in the sampled signal we only properly represent signals up to the Nyquist frequency ($f_s/2$) then we see here how frequencies from the first repeated spectrum intrude into this range – this is aliasing.

As a final word on the topic, take another look at the spectrum of the properly sampled signal in panel (d). We can see that the sampled spectrum contains in the range 0 to $f_s/2$ *an exact copy of the original signal's spectrum.* This has two profound consequences:

1. For proper sampling, we have not lost any frequency information as a result of the sampling process.
2. If we take the the sampled signal and remove all the frequencies above $f_s/2$ then we will recover the original signal.

The latter can be done by the use of a low-pass filter. This technique is the basis of *digital-to-analogue conversion*, as we will see later.

### *5.5. Anti-alias filters*

We are not finished with aliasing yet. As seen in the previous section, if our signal is contained within a narrow band of frequencies below the Nyquist frequency, then all is OK. However, in practice this is rarely the case for two reasons:

1. Many real signals (either pulses or repetitive waveforms) can approach mathematically infinite waveforms (for example, the square wave). While no real signal is mathematically perfect, any signal with rapid changes ("edges") has a very wide frequency spectrum;
2. All signals contain noise, and noise is usually broadband, i.e. existing across the frequency spectrum.

In order to prevent these high frequency components aliasing into our digitised signal, it is usually preferable to remove them from the signal *before* sampling. Note that this is a compromise solution: if we filter out high frequency parts of our signal we certainly degrade it; however, the

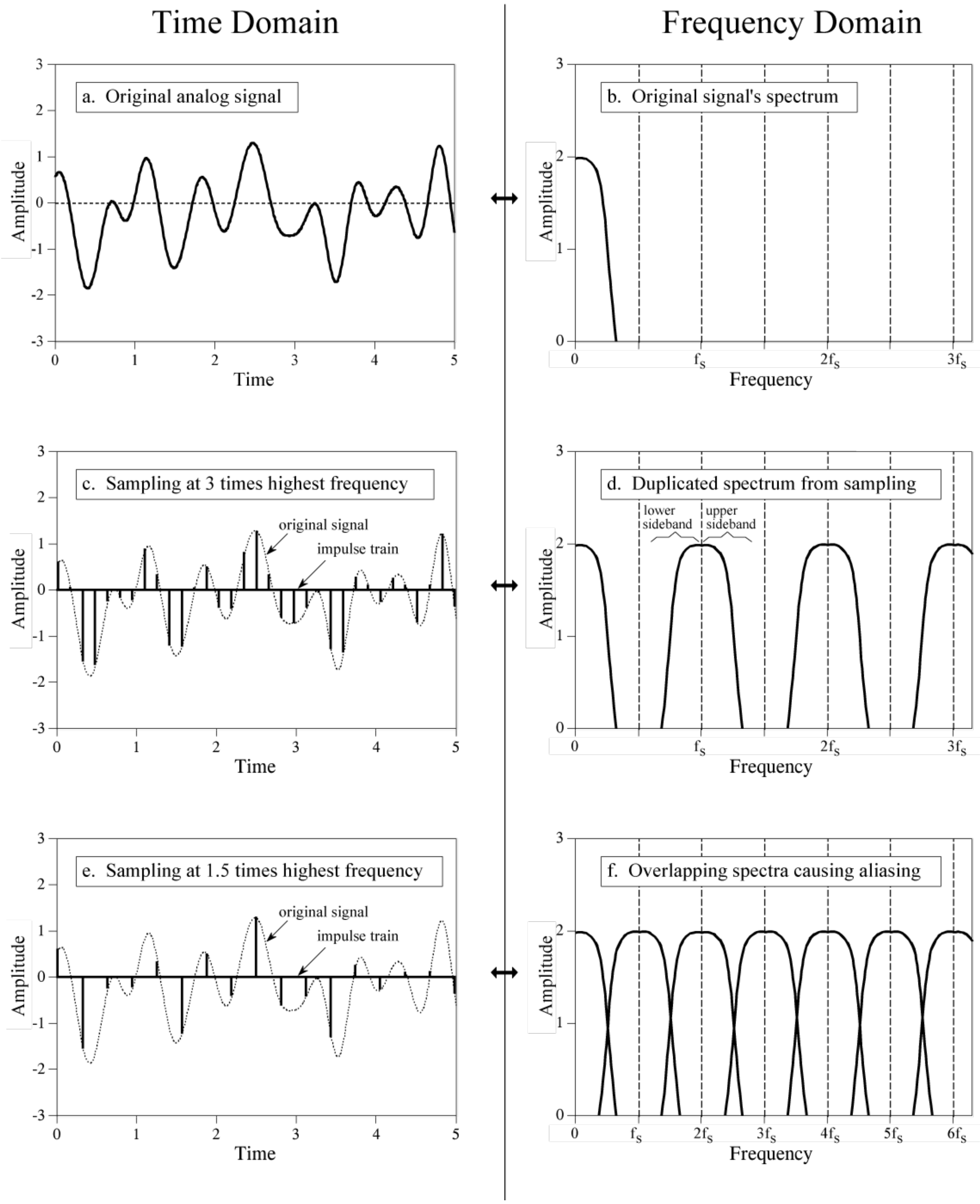


Figure 5.3: Sampled signals in the frequency domain (from Smith, www.dspguide.com).

problem of aliasing is so bad that this is usually a price worth paying. Figure 5.4 shows a Digital Signal Processing (DSP) system as it should be setup. Imagine this is an audio system; then the analogue input on the left is a voltage signal coming from a microphone. The first stage is a filter designed to remove frequency components above the Nyquist frequency. This is called the *anti-alias filter*. The signal is then sampled and digitised by the ADC and can then be stored and processed by the computer (central box). Everything to the left of the diagram is

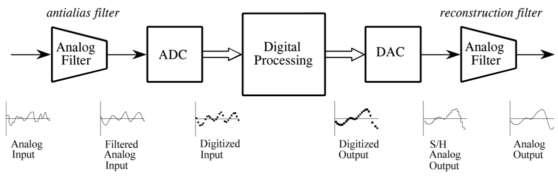


Figure 5.4: Correctly set up Digital Signal Processing system (from Smith, www.dspguide.com).

the recording system. The right side is for "playback". As mentioned above, we need a *Digital to Analogue Converter (DAC)* and another filter to properly reconstruct the original analogue signal, which then drives the speakers. It is worth noting that all of this, and of course much more, is contained within the modern mobile phone...

Entire books have been written on the subject of anti-alias filters, and electrical engineers receive entire lecture courses on filter design. For our purposes, including the lab sessions, we will use the simple RC low-pass filter discussed in Section 7. The ideal filter would have a very sharp cut-off, that is to say, it would allow through all frequencies below the Nyquist frequency and completely block anything above. Unfortunately, this is impossible. In the case of the RC low-pass filter we usually and choose the −3 dB cut-off frequency of the filter to be the same as the Nyquist frequency of the sampling. This is, however, a compromise, as can be seen in Figure 5.5. This filter would be suitable for sampling with Nyquist of 1 rad/s. Some frequencies below the Nyquist value are attenuated (undesirable) and also some frequencies above Nyquist will still exist (also undesirable). Filters with a sharper "cut-off" are possible; we will study this briefly when we consider active filters built with op-amps, later in the course.[6]

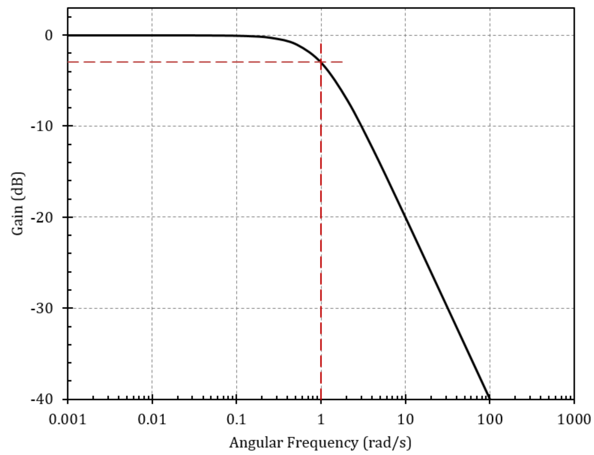


Figure 5.5: Response of an RC low-pass filter with a (−3 dB) cut-off frequency of 1 rad/s.

[6]See Smith chapter 3 at www.dspguide.com and also Horowitz & Hill chapter 4.

## 6. Fourier theory of signals

Generally, a signal is any time- or spatially-varying quantity. We are used to time-varying quantities such as voltage, current, light intensity and sound pressure. A spatially-varying signal might be, for example, an image. Typically, a signal represents something physical.

*Continuous and discrete signals.* Both the sound pressure and its electrical representation are signals in their own right, and both these types of signals are continuous functions of time. Generally, any physical quantity we may wish to deal with is represented by a continuous parameter such as mass, length, voltage, current, etc. However, the very act of measurement creates a discrete signal. Each time we make a measurement of a quantity we are generating a discrete signal. For example, the height of the sea measured every hour, or the temperature in a reaction chamber measured every second. These are examples of sampled signals. In physics and engineering we encounter mainly signals which have been both sampled and digitised. In this section we will be looking at *continuous signals*; however, it is important to always bear in mind that a discrete equivalent exists.

**Learn it in lab**
In the third lab session you will look at the representation of signals in the frequency domain, and characterise the components of a system in frequency space by measuring and analysing their *frequency response functions.* The Fourier treatment of signals introduced here (and extended in later sections to the more generic Laplace transform treatment) is at the heart of this topic.

### *6.1. Periodic continuous signals*

Figure 6.1 shows six common waveforms. The simplest is the last (f), which is the sinusoid. This might represent, for example, the position of a mass oscillating with simple harmonic motion. This figure shows both the time and frequency domain representations. For the latter, the relative amplitudes of the first six Fourier coefficients are given. By convention, $f$ is the fundamental frequency, and $2f$, $3f$ are the $2^{\text{nd}}$ and $3^{\text{rd}}$ harmonics, and so on. Our sinusoid of course has only the fundamental, but the rectified version (e) has both a "DC" (constant) component at 0 Hz but also lots of higher harmonics which are due to the sharp corners in the waveform. Saw-tooth (d), triangular (c) and square (b) waveforms are frequently seen in electronics. The triangular and square waves have the characteristic that all the even harmonics are zero. However, all three have the characteristic that their Fourier series is infinite, i.e. an infinite number of harmonics is required to accurately represent these waveform.

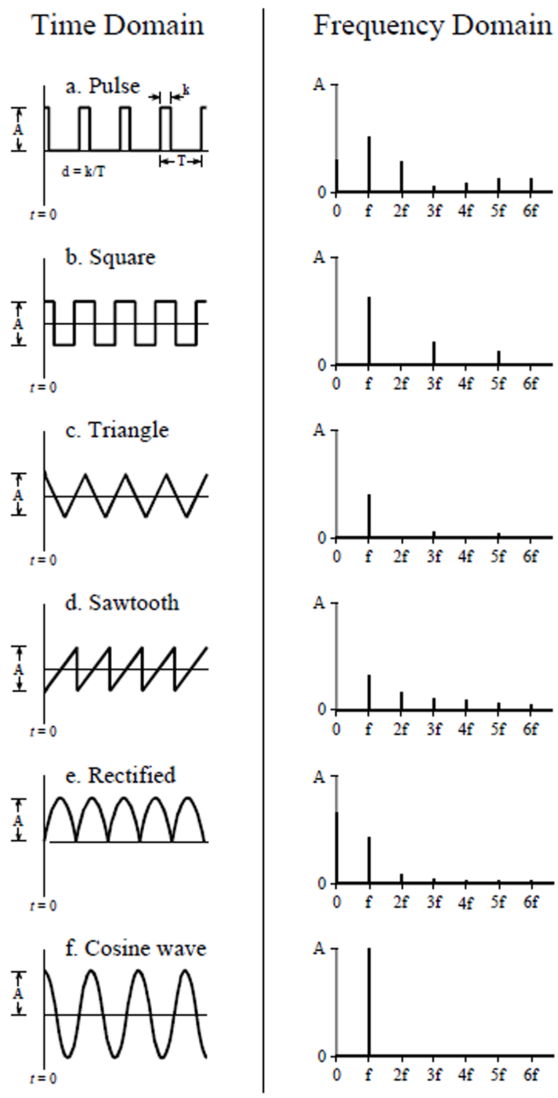


Figure 6.1: Common waveforms in physics or engineering.

#### *6.1.1. Fourier series of periodic waveforms*

Wherever there exists periodicity, we should seek a Fourier representation, and a Fourier understanding of signals is important in instrumentation as it allows us to predict a *linear system's* output in response to a periodic input. In practice, all physically realistic *periodic* signals obey the Dirichlet conditions[7] and are therefore transformable into a Fourier series. Hence, any signal $f(t)$ with period $T$ can be expressed as

$$\begin{aligned} f(t) &= \sum_{n=-\infty}^{+\infty} \alpha_{\mathbf{n}}\, e^{jn\omega_0 t} \qquad (6.1) \\ &= \sum_{n=-\infty}^{+\infty} |\alpha_{\mathbf{n}}|\, e^{j(n\omega_0 t+\phi_n)}\,, \end{aligned}$$

[7] See Poularikas section 3.2.

where $\omega_0 = 2\pi/T$ (note: this is a constant) and the $\alpha_\mathbf{n}$ are complex constants given by

$$\begin{aligned}
\alpha_\mathbf{n} &= \frac{1}{T}\int_{t_0}^{t_0+T} f(t)\, e^{-jn\omega_0 t}\,\mathrm{d}t && (6.2)\\
&= |\alpha_\mathbf{n}|\, e^{j\phi_n} \\
&= |\alpha_\mathbf{n}|\cos\phi_n + j|\alpha_\mathbf{n}|\sin\phi_n\,. && (6.3)
\end{aligned}$$

Since the $\alpha_\mathbf{n}$ are determined from $f(t)$, and vice versa, they are known as a *Fourier transform pair.* At any time $t_0$ the Fourier expansion of the function converges to $f(t_0)$ as long as the function is continuous at $t_0$. If the function is discontinuous, then the Fourier expansion converges to a point mid-way between the discontinuity. If $f(t)$ is real, then

$$\begin{aligned}
\alpha_{-\mathbf{n}} &= \alpha_\mathbf{n}^* \\
f(t) &= \alpha_0 + \sum_{n=1}^{+\infty}\left[(\alpha_\mathbf{n}+\alpha_\mathbf{n}^*)\cos n\omega_0 t\right. \\
&\qquad \left. + j(\alpha_\mathbf{n}-\alpha_\mathbf{n}^*)\sin n\omega_0 t\,\right], && (6.4)
\end{aligned}$$

which can be written in trigonometric form as[8]

$$\begin{aligned}
f(t) &= \frac{A_0}{2} + \sum_{n=1}^{+\infty}(A_n\cos n\omega_0 t + B_n \sin n\omega_0 t) \\
A_0 &= \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\,\mathrm{d}t \\
A_n &= \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\cos n\omega_0 t\,\mathrm{d}t \\
B_n &= \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\sin n\omega_0 t\,\mathrm{d}t\,, && (6.5)
\end{aligned}$$

or

$$\begin{aligned}
f(t) &= \frac{A_0}{2} + \sum_{n=1}^{+\infty} C_n\cos(n\omega_0 t + \phi_n) \\
C_n &= \sqrt{A_n^2 + B_n^2} \\
\phi_n &= \tan^{-1}\left(\frac{B_n}{A_n}\right)\,. && (6.6)
\end{aligned}$$

*The square wave.* The Fourier expansion of a square wave of unity amplitude and unity period is

$$f(t) = \frac{4}{\pi}\left(\sin t + \frac{\sin 3t}{3} + \frac{\sin 5t}{5} + \frac{\sin 7t}{7} + ...\right)\,, \quad (6.7)$$

i.e. all the $A$'s are zero, as are all the even $B_n$. Figure 6.2 illustrates the first nine coefficients and Figure 6.3 shows the square wave reproduced from the first 2, 3 and 9 *non-zero* coefficients. As $n$ increases so does the fidelity, though we are always left with an overshoot of about 10% at the edges (Gibbs' phenomenon). Fidelity is worst at the edges, and this improves rapidly with $n$. From this we know that sharp edges are represented by high-frequencies in the expansion. We see this if we run a square wave through a low-pass filter: the edges are "rounded off" (see also the transient response of the RC filter in the next section).

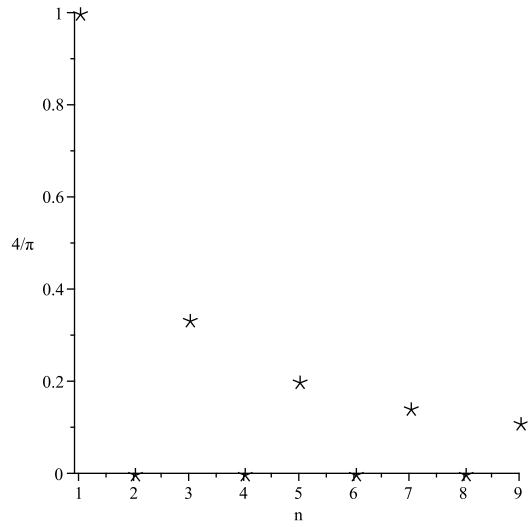


Figure 6.2: Square wave amplitude coefficients.

Related is the fact that the higher harmonics in the expansion contribute only to the detail of the waveform. We can see in Figure 6.3 where the centre plot shows a passable square wave from just the fundamental plus the 3$^{\text{rd}}$ and 5$^{\text{th}}$ harmonics (the first 3 non-zero coefficients). Figure 6.4 plots the power spectrum (amplitude coefficients squared, normalised to the power of the fundamental). The higher harmonics contribute a tiny fraction of the overall signal power.

*The triangular wave.* The Fourier expansion of a triangular wave of unity amplitude and unity period is

$$f(t) = \frac{8}{\pi^2}\left(\sin t - \frac{\sin 3t}{9} + \frac{\sin 5t}{25} - \frac{\sin 7t}{49} + ...\right)\,. \quad (6.8)$$

Figure 6.5 shows the first 9 coefficients and subsequent expansion. Note that:

1 There are negative coefficients;
2 The coefficients are similar to those of the square wave but scaled by $1/n$. This makes sense when we consider the action of the low-pass filter on the square wave: the low-pass filter has a response $\propto 1/f$ for frequencies higher than the cut-off frequency.

### 6.1.2. Symmetry, spectra and energy

An even function has symmetry about the $t = 0$ axis (i.e. it is the same under reflection about the vertical axis, e.g. $\cos(t)$), while an odd function has rotational symmetry

[8] Note that you would not be expected to reproduce these formulae for an exam!

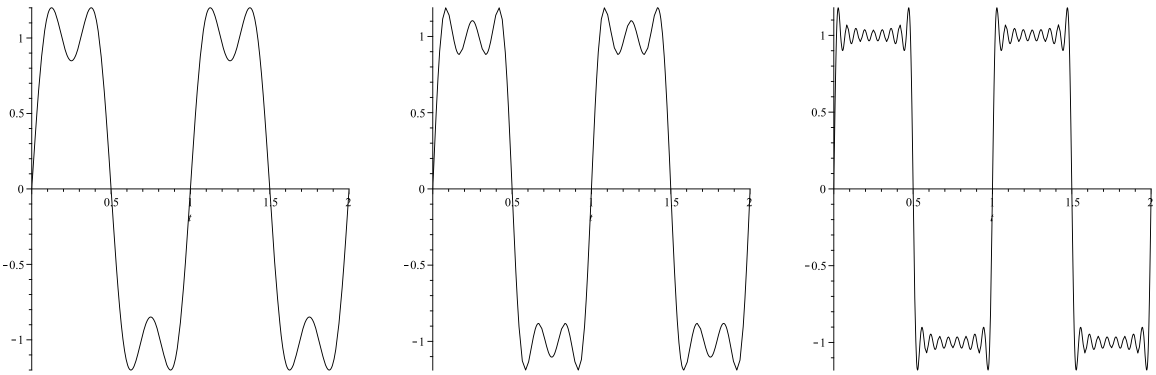


Figure 6.3: Square wave reproduced from the first 2, 3 and 9 non-zero Fourier coefficients.

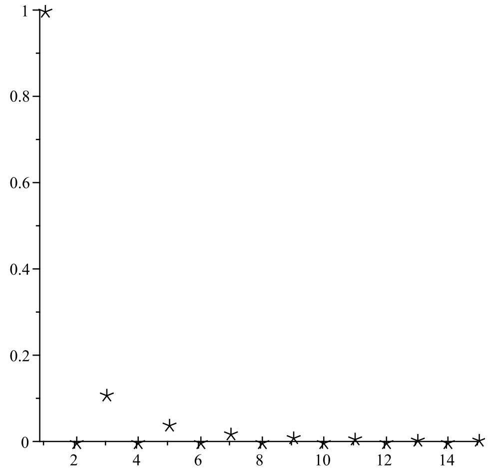


Figure 6.4: Power spectrum for the square wave.

about the origin (unchanged after rotation 180° around the origin, e.g. $\sin(t)$). Table 6.1 summarises these symmetries which we can exploit (noting that many functions have no overall symmetry).

Table 6.1: Symmetries and Fourier coefficients.

| symmetry | functional form | $A_0$ | $A_n$ | $B_n$ |
|---|---|---|---|---|
| even | $f(t) = f(-t)$ | exist | exist | 0 |
| odd | $f(t) = -f(-t)$ | 0 | 0 | exist |

We should point out that symmetry can be a matter of phase. For example, in our expansion of the square and triangular waves above we see only sine terms (i.e. the $B_n$), suggesting that we started with an odd function. This is not what we represented in Figure 6.1, where these functions were even – but if we can shift periodic waveforms along in time (in the same way that we can turn a sine into a cosine) this can simplify things.

Typically, we might specify a spectrum of a signal by its coefficients $C_n$ (known as the *amplitude spectrum*) and their corresponding phases $\phi_n$ (known as the *phase spectrum*). The spectra studied in the previous sections are special cases with zero phase spectra. As pointed out above, choosing an appropriate origin for the representation of a signal can significantly simplify its expansion.[9]

*The energy relation.* Parseval's theorem applied to a signal $f(t)$ expanded as a Fourier series gives

$$\frac{1}{T}\int_{t_0}^{t_0+T} f(t)^2\,\mathrm{d}t = \frac{A_0^2}{4} + \sum_{n=1}^{+\infty} \frac{C_n^2}{2}\,. \tag{6.9}$$

If we consider $f(t)^2$ to be the power in the signal, then the LHS of equation (6.9) is the average power in the signal over the period of 1 cycle. The RHS is the power of the signal in the frequency domain, expressed as a sum of the power of the individual frequency components. Equation (6.9) is known as the *energy relation* since it tells us that energy measured in the time domain is the same as the energy measured in the frequency domain, a consequence of conservation of energy.

## *6.2. Realistic signals*

The Fourier series is excellent for describing the periodic, repetitive signals frequently encountered in physics and engineering, such as the square, triangular and sawtooth waveforms and indeed any arbitrary waveform – *so long as it has a repeat period* $T$. We see in the Fourier series of the square and triangular waveforms a fundamental signal at the repeat period $T$ which we can recognise in both time and frequency representations, and we can see that the fundamental carries most of the signal power. The Fourier series is very useful where we have a "steady-state" periodic signal input into a system. By *steady state* we mean to imply that the signal has existed long enough that it can be reasonably described as a Fourier series. Mathematically, this would mean that the signal should have existed for all time, which is of course not realistic; however, if the signal has existed long enough for any transient effects to have died away, then this is a reasonable

[9] See Poularikas section 3.3.

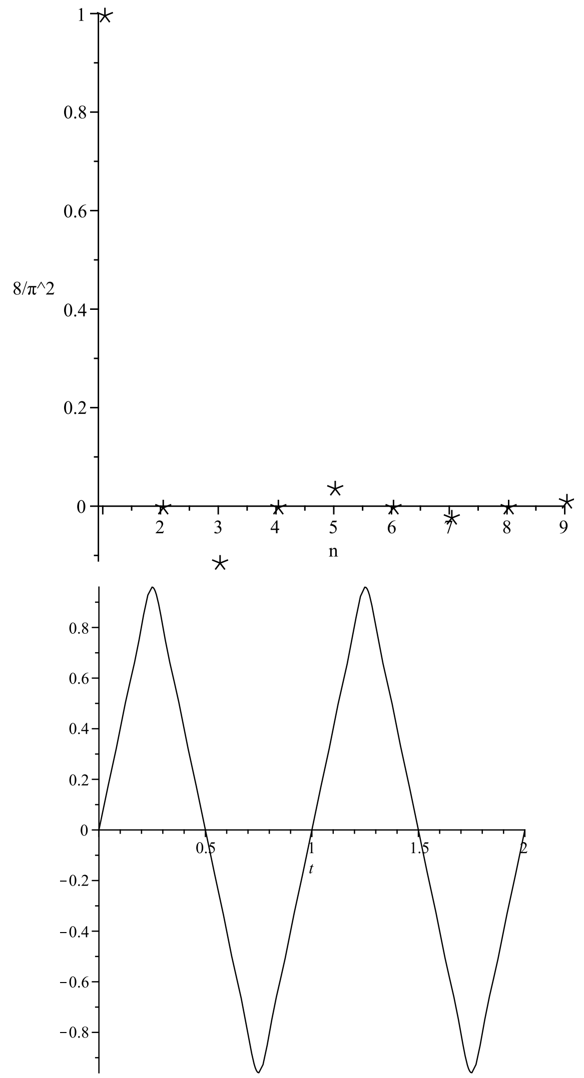


Figure 6.5: Triangular wave coefficients and waveform.

assumption. By *transient effects* we mean the initial response of the system to a sudden start-up of the signal, for example if we take a system and suddenly apply a square wave input then the initial response, say the first few cycles of the square wave, may be rather different from the behaviour after a thousand cycles. As a simple example imagine pushing someone on a playground swing: the periodic push is the input but it takes some number of cycles to build up to the full amplitude. We will come back to these ideas of transient effects later on and see how to deal with them; however, for now it is important to understand that the Fourier series is an excellent way of describing *periodic signals* that have existed for a time much longer than any *time constant* of the system being studied. Conversely, we must recognise that no signal persists infinitely in time, and therefore even a pure sinusoid truncated in time will exhibit some measure of spectral width.

### *6.3. Non-periodic signals*

An example of a non-periodic function is the *unit pulse* function (Figure 6.6) which has width $2a$ and height 1:

$$P_a(t) = \begin{cases} 0, & |t| > a \\ 1, & |t| < a \end{cases} . \tag{6.10}$$

It can be scaled, e.g. $hP_a(t)$, which may represent a voltage $h$ applied for time $2a$. To find the frequency "content" of a non-periodic function such as this rectangular pulse $P_a(t)$ we need the *Fourier integral*, which is in essence the *limiting case* of the Fourier series as the period $T \to \infty$.

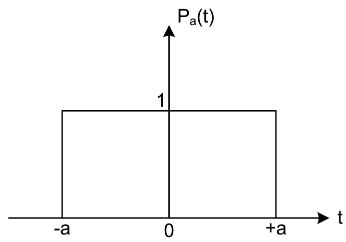


Figure 6.6: The unit pulse function.

#### *6.3.1. Continuous functions*

The Fourier transform of a *continuous* function $f(t)$ is written $\mathcal{F}\{f(t)\} = F(\omega)$ and the inverse is $f(t) = \mathcal{F}^{-1}\{F(\omega)\}$. So, we have the *forward transform*:

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{+\infty} f(t)\, e^{-j\omega t}\, \mathrm{d}t\,, \tag{6.11}$$

and the *inverse transform*

$$f(t) = \mathcal{F}^{-1}\{F(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega)\, e^{j\omega t}\, \mathrm{d}\omega\,. \tag{6.12}$$

$F(\omega)$ is the *spectrum function* of $f(t)$. Since $F(\omega)$ is usually complex, we can represent it as a plot of the real and imaginary parts versus $\omega$. It is usually more meaningful to plot the absolute value

$$|F(\omega)| = \sqrt{\mathcal{R}\{F(\omega)\}^2 + \mathcal{I}\{F(\omega)\}^2}$$

and the argument

$$\operatorname{Arg} F(\omega) = \tan^{-1} \frac{\mathcal{I}\{F(\omega)\}}{\mathcal{R}\{F(\omega)\}}\,,$$

in which case $|F(\omega)|$ is called the *amplitude spectrum* and $\operatorname{Arg} F(\omega)$ is the *phase spectrum*.

#### *6.3.2. Real functions*

In addition to continuous, physical signals are also *real* functions of time. Consider the function $f(t) = u(t)\, e^{-t}$ shown in Figure 6.7, which introduces another useful function: the *unit step* $u(t)$, which is zero for negative times and unity otherwise. The exponential pulse has the Fourier transform

$$F(\omega) = \frac{1}{1 + j\omega} \, .$$

The real, imaginary, magnitude and argument representations of the spectrum are given in Figure 6.8. There are some rules for real functions which can simplify their transforms.

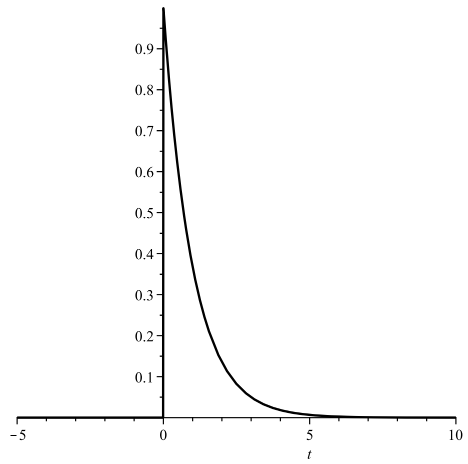


Figure 6.7: The exponential pulse, $f(t) = u(t)e^{-t}$.

*Any real function.* The reflected form of the spectrum (i.e. reflected about $\omega = 0$) is the complex conjugate of the spectrum:

$$F(-\omega) = F^*(\omega) \, .$$

*Real and even functions.* The imaginary part of the spectrum is zero:

$$\begin{aligned} \mathcal{R}\{F(\omega)\} &= 2\int_0^{+\infty} f(t)\cos\omega t \, \mathrm{d}t \\ \mathcal{I}\{F(\omega)\} &= 0 \, . \end{aligned}$$

*Real and odd functions.* Conversely,

$$\begin{aligned} \mathcal{R}\{F(\omega)\} &= 0 \, . \\ \mathcal{I}\{F(\omega)\} &= -j\int_{-\infty}^{+\infty} f(t)\sin\omega t \, \mathrm{d}t \end{aligned}$$

### 6.3.3. Properties

Fourier transforms have these important properties:

*Linearity.*

$$\mathcal{F}\{af_1(t) + bf_2(t)\} = aF_1(\omega) + bF_2(\omega) \, .$$

*Scaling.*

$$\mathcal{F}\{f(at)\} = \frac{1}{|a|} F\left(\frac{\omega}{a}\right) \, .$$

*Derivative.*

$$\mathcal{F}\left\{\frac{\mathrm{d}^n f(t)}{\mathrm{d}t^n}\right\} = (j\omega)^n F(\omega) \, .$$

## 6.4. Some common signals

The *unit pulse* function and the *delta* function are both important signals for instrumentation applications. We may be trying to measure the height and/or width of a pulse generated by an experiment. In order to understand how our instrument will react to the pulse, we need an understanding of the pulse's frequency content. The delta function, because of its special frequency content, is frequently used as a "test" input into a system in order to determine a system's *impulse response*. We will deal with this later, for now it is important to appreciate the frequency content of these signals.

*Pulse function.* The Fourier transform of the pulse function (time-domain) is the "sinc" function[10] (frequency domain), and the Fourier transform of the sinc function (time domain) is the pulse function (frequency domain):

$$\begin{aligned} \mathcal{F}\{P_a(t)\} &= \mathrm{sinc}_a(\omega) \\ \mathcal{F}\{\mathrm{sinc}_a(t)\} &= P_a(\omega) \, . \end{aligned}$$

This result means that the pulse function, which is finite in time, has an infinite range of frequencies associated with it. Conversely, the sinc function in the time domain, which exists for all time $-\infty < t < +\infty$, has only a finite range of frequencies. Since all instrumentation has frequency-dependent behaviour, the implications of this are important. It is well worth sketching these for a few different values of $a$ since this is an important topic.

*Delta function.* The Fourier transform of the delta function is a constant, and *vice-versa*:[11]

$$\begin{aligned} \mathcal{F}\{\delta(t)\} &= 1 \\ \mathcal{F}\{A\delta(t)\} &= A \\ \mathcal{F}\{A\} &= 2\pi A\delta(\omega) \, . \end{aligned}$$

Once again the implications are important. The delta function contains all frequencies in equal measure – see Figure 6.9. A physical example of a delta function input to a system is hitting a mass with a hammer. The hammer transfers a defined amount of energy to the mass in a very short period of time (ideally applying infinite force for an infinitely short period of time such that the total momentum transferred is 1). This means that the hammer blow excites all frequencies *simultaneously*. We can use this to test a system's frequency response. Mechanical engineers do this in practice: strike the object to be tested with a hammer and record the spectrum of frequencies which results. The peaks in the spectrum are where the object resonates.

[10] Recall $\mathrm{sinc}(x) = \sin(x)/x$, and we define $\mathrm{sinc_a}(x) = \sin(ax)/x$.

[11] A simple way to remember this is to consider the delta function as an infinitely narrow Gaussian: the Fourier transform of a narrow Gaussian in one domain is a wide Gaussian in the other domain, and an infinitely wide Gaussian is a constant.

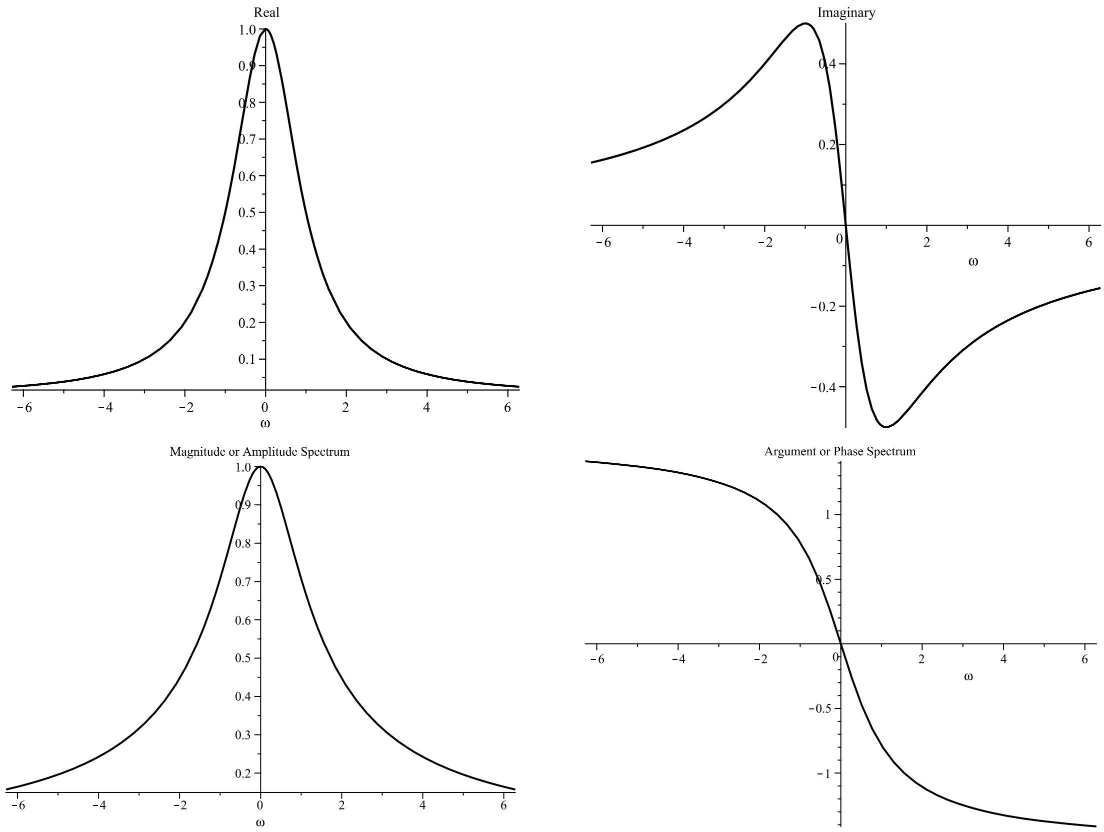


Figure 6.8: $F(\omega)$ representations in terms of $\mathcal{R}\{F(\omega)\}$ and $\mathcal{I}\{F(\omega)\}$ (top two panels) and magnitude and phase spectrum (bottom two panels). This refers to the transform of the exponentially-decaying pulse shown in Figure 6.7.

*Sinusoid.* The sinusoid contains a single frequency, and the frequency spectrum is represented by the delta function. Mathematically, it exists at both $\pm\omega_0$, though only positive frequencies exist in reality. It is useful to sketch the transform given here:[12]

$$\mathcal{F}\{A\cos(\omega_0 t)\} = \pi A[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]\,.$$

Naturally, all real signals are *finite* in duration, and so even a sinusoidal signal will have a finite (albeit it very small) spectral width. A real-world sine wave may be constructed by multiplying the ideal (infinite) one by a unit pulse, which kills the amplitude outside of some time $|t| > a$. What does the frequency spectrum look like for this real sinusoid? In this case we actually know the Fourier transforms of the individual signals (a delta function and a sinc function), and hence we can use the *Convolution theorem* that states that *the Fourier transform of the convolution of two functions is equal to the point-wise product of their Fourier transforms*: i.e. point-wise multiplication in the time domain equals convolution in the frequency domain and *vice-versa*. So, in this case we expect the delta functions at $\pm\omega_0$ in the frequency domain to be replaced by sinc functions centred at those frequencies. It is worth working through and sketching this example, as it illustrates an important point as well as a typical use of the convolution theorem.

[12] Figure 4.26 of Poularikas gives a very nice pictorial representation of the Fourier representation for many other common signals. It is well worth browsing this to get a feel for the behaviour of signals in the frequency domain.

### *6.5. Pulsed signals: limiting cases*

Figure 6.10 shows the Fourier transform of the pulse function $P_a(t)$ for various values of $a$. The spectrum is the sinc function, as discussed above. Observe that the lowest frequency at which the function crosses the time axis is at $\omega = \pi/2a$. This means that as $a$ increases (wider pulse) the frequency spectrum gets narrower. In the limit $a \to \infty$, as expected from the previous section, the frequency function becomes infinitely narrow and tends to a delta function.

We can state this as a general rule: *very short pulses (in the time domain) have very wide spectra, whilst very wide pulses have nicely compact spectra.* We can formalise this general rule using fundamental physical considerations deriving from the uncertainty principle. Ultimately, this leads to a dimensionless quantity called the *time-bandwidth product.* Note that the maximum height of the spectrum is at zero frequency and is given by

$$\begin{aligned} F(\omega) &= 2\frac{\sin a\omega}{\omega} \\ \lim_{\omega\to 0} F(\omega) &= 2a\,. \end{aligned}$$

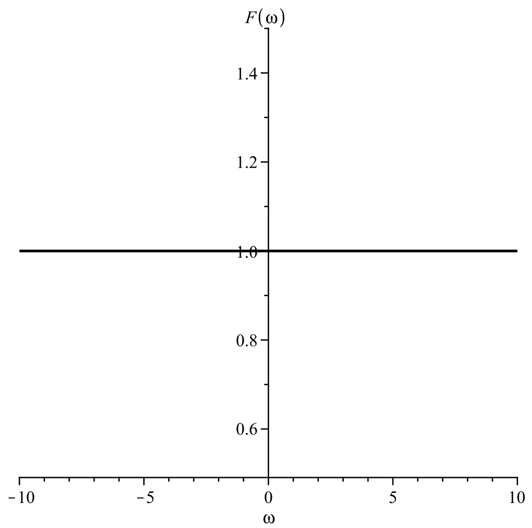


Figure 6.9: Fourier transform of the delta function, $\delta(t)$.

So, as the pulse gets narrower, its spectrum becomes both wider and less tall. This is also physically intuitive: as the pulse contracts in time, its energy decreases, therefore in the frequency domain the total energy must also be conserved by the spectrum becoming wider but lower in amplitude (see also Parseval's theorem).

Finally, it is worth considering what would happen if, as the pulse gets narrower, we let it get taller such that the total *area* of the pulse remains 1. Then we would find that as $a \to 0$ the pulse height $\to \infty$, so we have a delta function in the time domain.[13] In the frequency domain we will see that this is a constant (see Figure 6.9) and from the above arguments and Figure 6.10 it is clear that this is the limiting case of the sinc function as $a \to 0$.[14]

## 6.6. *Bandwidth*

There is an intimate relationship between the duration of a pulse in the time domain and its range of frequencies in the Fourier transform. The latter is generally known as the *bandwidth* of the pulse. The practical importance of this comes from the fact that all instruments have a finite bandwidth, hence we need to be sure that this bandwidth is compatible with the spectral content of the signal. It is tempting to describe the width of a pulse in the time domain as its non-zero time range; however, many signals we can describe mathematically – such as the Gaussian pulse – would technically have infinite duration, even if this is not physical. An easier way to get a "measure" of the pulse is to take the Full-Width at Half Maximum (FWHM) – see Figure 6.11.

[13] One can also think of a delta function as an infinitely narrow rectangular pulse.

[14] There is a nice worked example of this in Poularikas (Example 4.19 in section 4.3). It is well worth looking this up.

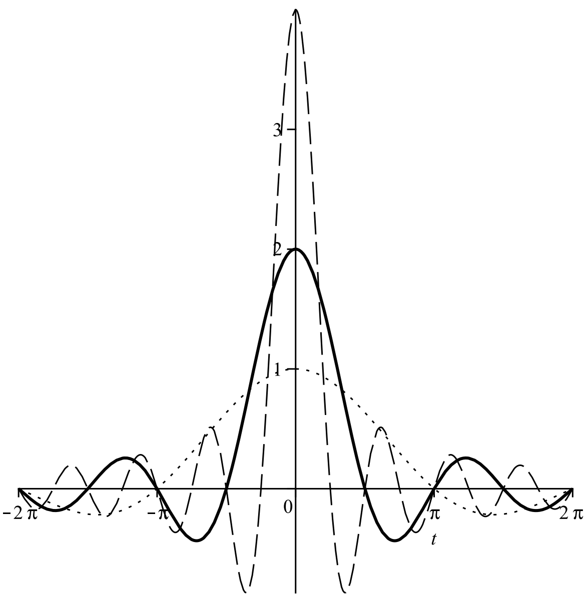


Figure 6.10: Fourier transform of the pulse function, $P_a(t)$ for $a = 1$ (solid line), $a = 2$ (dashed line) and $a = 1/2$ (dotted line) [Note: horizontal axis should be labeled $\omega$].

### 6.6.1. *Time-bandwidth product*

A central theme in instrumentation, communications, optics and other branches of physics is that a signal that has a finite duration in time must therefore have associated with it some spread of frequencies. For any pulse shape we care to choose, with a given duration (FWHM), we can calculate the frequency spectrum by taking the Fourier transform and thus obtain its *spectral width* (FWHM). For any given pulse shape, the *time-bandwidth product* $\Delta\nu\Delta t$ is a constant that we can calculate. This is analogous to what you find in atomic, nuclear and particle physics eigenstates: the energy width of a particular state is inversely proportional to its lifetime and, conversely, only a truly stable state has zero energy width.

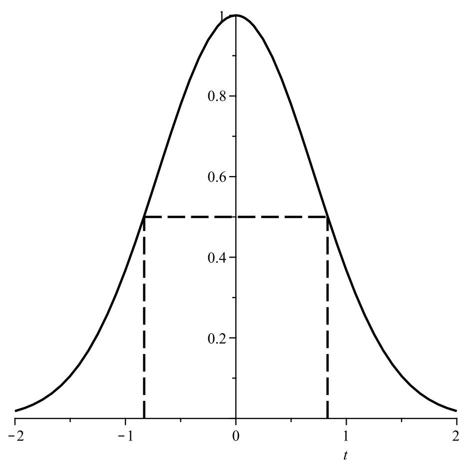


Figure 6.11: FWHM of a Gaussian pulse.

It is important to note that the definition of $\Delta t$ and $\Delta\nu$ (or $\Delta\omega = 2\pi\Delta\nu$) which we are using here is the FWHM definition, and that often this is applied to power rather than amplitude; this determines the constant we actually get – so it is important to specify this. If we take a Gaussian-shaped pulse then, mathematically, it has infinite duration in the time domain and also also an infinite range of frequencies. However, if we use the FWHM definition then we can calculate the time-bandwidth product to be finite. Note that if $\Delta t$ is the FWHM of the *power* in a pulse ($f(t)^2$), then the product must involve the width of the *power spectrum* ($|F(\omega)|^2$) – and in the Gaussian case this product is about 0.44, as in the example below.

The time-bandwidth product has interesting properties which make it a robust measure of the time and frequency spread of a signal: it is invariant under shifting the waveform in time and in frequency, and multiplication by a constant. It is essentially a property of the signal shape.

**Example: Ultrafast laser pulse**
The duration of a laser pulse is by convention taken to be the FWHM of the pulse's optical *intensity* (amplitude squared, $f(t)^2$), and its bandwidth is then the FWHM of the *power* spectrum (amplitude spectrum squared, $|F(\omega)|^2$). Using these definitions we can calculate the time-bandwidth product for any given pulse shape. For the Gaussian, this turns out to be about 0.44 (the proof of this is left as a problem sheet question). If we wish to produce a Gaussian-shaped pulse of duration 10 fs, we discover that we require a bandwidth of 44 THz!

*Application to very short pulses.* An alternative way of thinking about this is from the point-of-view of an instrument with limited bandwidth (as all instruments have, fundamentally). This means that the instrument can only reproduce signals over a finite range of frequencies, and ultimately this limits how short the pulse can be in the time domain. The time-bandwidth product is dependent on the shape of the pulse. Physically realisable pulses with values as low as 0.3 are possible, and some specialist lasers are able to generate pulses with durations close to this limit. This is not just of academic interest (see box below).

Real pulses are of course only approximations to the mathematical ideal. The "quality" of a very-short pulse is measured by how close we can get to the ideal time-bandwidth product.

*6.6.2. Fourier transform limit*

The *Fourier transform limit* gives us a useful tool that we can apply in the lab to test the validity of measurements such as the duration of an ultra-short laser pulse. We say that a pulse or signal is *transform-limited* if it contains (in the frequency domain) exactly the minimum range of frequencies required to support the pulse shape. Pulses with more frequencies than are required by the transform limit are physically possible, but those with less are not. One important result that follows from the transform limit is that that any signal that contains a sudden change (delta functions, step functions, square waves, and so forth) has associated with it a large spread of frequencies. Real instruments cannot deal with infinite frequency ranges: they always have some finite bandwidth. This means that while perfect pulses, steps, square-waves and so forth provide us with useful mathematical tools for analysing instruments, we never actually get to see them in real life.

**Application: Optical fibre communications**
To get the maximum data throughput on a optical fibre link it is necessary that the pulses (representing the data) are both short and very close together. Typically, a limiting factor is chromatic dispersion (frequency-dependent propagation speed). This results in the pulse shape getting distorted as it travels down the fibre, as different frequency components travel at different speeds. Ultimately, the problem is that the pulses merge together such that the receiver cannot decode the original train of pulses. For a given pulse duration, *transform-limited pulses* are those with the minimum possible spectral width. In optical communications, a transmitter emitting close to transform-limited pulses minimises the effect of chromatic dispersion, thus maximising the transmission distance.

Physics imposes a fundamental limit on how small the time-bandwidth product can be. We can show that this in both a purely classical formulation and (rather pleasingly) by using quantum mechanics too. If we apply a combination of a Fourier and an inverse Fourier transform to an arbitrary function of time $f(t)$, we find that there is a fixed relationship between the temporal width $\Delta t$ and the bandwidth $\Delta\nu$ such that the time-bandwidth product $\Delta t\Delta\nu$ must obey the inequality

$$\Delta\nu\Delta t \geq \frac{1}{4\pi}$$

(or, equivalently, $\Delta\omega\Delta t \geq \frac{1}{2}$). If we look to quantum mechanics, the energy-time uncertainty principle gives us an equivalent result:

$$\Delta E\Delta t \geq \frac{\hbar}{2}\,.$$

Dividing the latter result by $h$ we obtain the same transform limit which we can obtain classically (through the Fourier transform). The time-bandwidth product of a real pulse is always $> 1/4\pi$. By Fourier transforming real-world pulse shapes and calculating $\Delta\nu\Delta t$ we can obtain numerical values for these cases, e.g. the 0.441 product for the Gaussian pulse ($Ae^{-a^2t^2}$) or an even lower 0.315 for a $\mathrm{sech}^2(t)$ pulse.

## 7. RC circuits

Electronic filters can generally be classified into passive and active types. Active filters use op-amps and we will look at this later in the course. Passive filters are built using resistors (R), capacitors (C) and inductors (L).

**Learn it in lab**
You will build and characterise a simple RC low-pass filter in the first lab session and use a high-pass filter in the second session.

### *7.1. The RC circuit*

To see how to apply AC circuit analysis we will study the series RC circuit in some detail. This also provides a simple illustration of several important concepts for the course. Because the capacitor has an impedance which is frequency-dependent, the RC circuit is an example of the class of systems known as *filters*, which will be very important for our studies. It is also a *Linear Time-Invariant (LTI)* system (cf. Section 14).

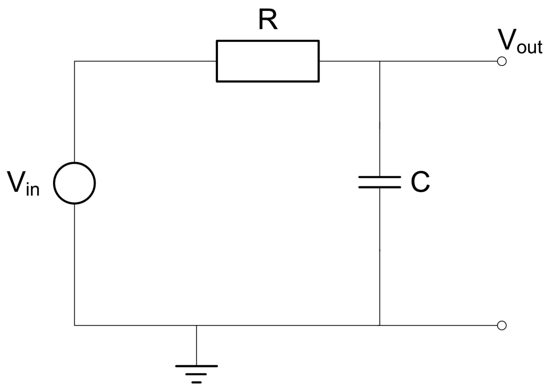


Figure 7.1: Simple RC circuit– The low-pass filter.

As shown in Figure 7.1, if we apply an input voltage across resistor and capacitor we can choose to take the output across either the resistor or the capacitor. Here we take the output across the capacitor. Consider the limiting cases of frequency: as $\omega \to \infty$, then the impedance of the capacitor tends to zero (effectively a short-circuit), so the entire input voltage appears across the resistor – by Kirchhoff's voltage law (KVL) – and $v_C \to 0$. Conversely, at DC (i.e. if we apply a constant voltage) then the impedance of the capacitor tends to $\infty$, the current flowing tends to zero, hence $v_R \to 0$, and the entire applied voltage appears as $v_C$. We will study two intermediate behaviours in more detail: first, the behaviour under steady-state AC conditions, and secondly the transient response.

#### *7.1.1. Steady-state behaviour*

In the following discussion, voltages and impedances are complex quantities. In the steady-state condition, we are working with periodic signals which have been present at the input for sufficiently long time that there is no remaining "transient behaviour" (which we will cover in the next section). The voltage $\mathbf{v_{in}}$ is an AC sinusoidal waveform and the voltage across the capacitor we label as $\mathbf{v_{out}}$. Complex Ohm's law gives

$$\begin{aligned} \mathbf{i} &= \frac{\mathbf{v_{in}}}{\mathbf{Z}} = \frac{\mathbf{v_{in}}}{R - j/\omega C} \\ \mathbf{v_{out}} &= -\mathbf{i}\frac{j}{\omega C}\,. \end{aligned}$$

We want to find the output in terms of the input, which we define as the *complex gain* (also denoted in bold-face to stress that this too is a complex quantity):

$$\mathbf{G} = \frac{\mathbf{v_{out}}}{\mathbf{v_{in}}} = \frac{-j}{\omega RC - j}\,.$$

Multiplying by the complex conjugate of the denominator and simplifying leads to

$$\mathbf{G} = \frac{1 - j\omega RC}{\omega^2 R^2 C^2 + 1}\,. \tag{7.1}$$

As anticipated, the gain is both frequency-dependent and complex (implying a phase change at the output). To visualise this we need to plot the magnitude and the phase of the gain:

$$|\mathbf{G}| = \sqrt{\mathbf{G} \times \mathbf{G}^*} = \frac{1}{\sqrt{\omega^2 R^2 C^2 + 1}}\,, \tag{7.2}$$

$$\begin{aligned} \phi &= \tan^{-1}\left(\frac{\mathcal{I}\{\mathbf{G}\}}{\mathcal{R}\{\mathbf{G}\}}\right) \\ &= \tan^{-1}(-\omega RC) \\ &= -\tan^{-1}(\omega RC)\,. \end{aligned} \tag{7.3}$$

#### *7.1.2. The Bode plot*

The gain and phase plots together constitute the *Bode plot* for the RC filter system. In Figure 7.2 the frequency axis has been normalised to units of $\omega_c = 1/RC = 1$ rad/s. For the Bode plot the gain magnitude is traditionally plotted log/log in dB, i.e. $20\log_{10}|\mathbf{G}|$. The phase is usually given in degrees.

#### *7.1.3. Phase interpretation*

The interpretation of the gain magnitude is clear: it is the ratio of the amplitudes of the output voltage (that across the capacitor) to that of the input voltage (applied to both the capacitor and the resistor in series). The interpretation of the phase is more difficult. We understand that there is a phase difference, but between what and what? And does it lead or lag? This can be interpreted in two main ways.

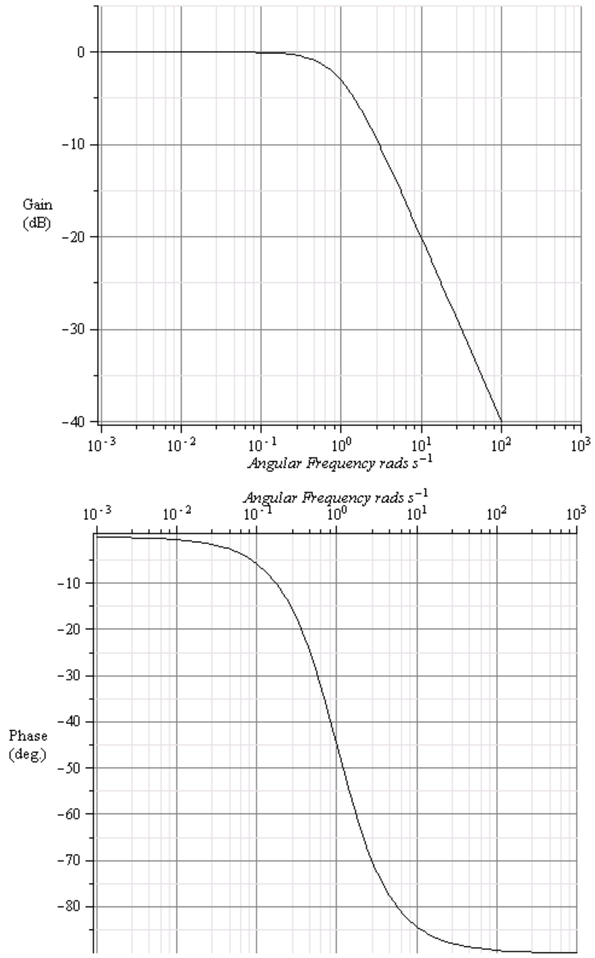


Figure 7.2: Bode plot of the low-pass (RC) filter.

*Argand diagram.* We can use the Argand diagram (Figure 7.3). At some instant in time we know that the voltage $v_R$ across the resistor (and the current through it) is along the real axis, i.e. it has zero phase angle. The voltage $v_C$ across the capacitor is always out of phase with the current, and we can see that it has a phase angle $-\pi/2$. The voltage on the input is along $\mathbf{Z}$, which is the equivalent (combined) impedance. The phase of the gain is defined as the angle by which the output voltage *leads* the input voltage, and here this is negative, i.e. $\phi = -\tan^{-1}(\omega RC)$.

*Phasor representation.* A simpler but perhaps less intuitive method is to consider the phasor representations of the quantities involved, and use

$$\mathbf{v_{out}} = \mathbf{G} \times \mathbf{v_{in}}\,.$$

We know that $\mathbf{v_{in}}$ lies along $\mathbf{Z}$ at $t = 0$, so we can write that the phase angle is (anti-clockwise from the real axis) $\theta = 3\pi/2 + \phi$ and, consequently,

$$\begin{aligned} \mathbf{v_{out}} &= |\mathbf{G}|\, e^{j\phi}\, |\mathbf{v_{in}}|\, e^{j\theta} \\ &= |\mathbf{G}|\, |\mathbf{v_{in}}|\, e^{j(\theta+\phi)}\,. \end{aligned}$$

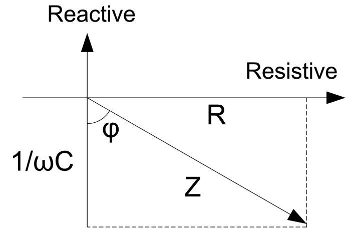


Figure 7.3: Argand diagram for the low-pass (RC) filter.

As we know from above, $\phi$ is negative, so it is clear that $\mathbf{v_{out}}$ lags $\mathbf{v_{in}}$ by $\phi$.

### *7.1.4. Low-pass filter*

Frequencies $\omega < \omega_c$ pass from input to output with little attenuation (or phase change); $\omega_c$ is the *cut-off frequency*, i.e. the frequency at which the gain has fallen by 3 dB (i.e. to $1/\sqrt{2} \approx 70\%$ of the maximum). If we imagine that the filter is driving a load resistance $R_{load}$ connected across its terminals, then the cut-off frequency is where the power delivered into $R_{load}$ has fallen by one-half compared to the flat or "pass-band" range of frequencies. For $\omega \gg \omega_c$ we can see that

$$|\mathrm{G}| \propto \frac{1}{\omega}\,. \tag{7.4}$$

On the Bode plot we see this as a slope of −20 dB per decade of frequency (a decade is a factor of 10). This is equivalent to −6 dB per octave (an octave is a factor of two in frequency). Both these figures are often quoted as the characteristic "roll-off" frequency behaviour for a first-order low-pass filter.

High frequencies are severely attenuated, hence the understanding of this circuit as a *low-pass filter*. As an example, were we to build a filter with $R = 1\,\Omega$, $C = 1\,\mathrm{F}$ and apply a 1 V signal at $\omega = 1$ rad/s, then the waveforms would be as per Figure 7.4.

Note that, by KVL, the voltage across the resistor has the opposite behaviour, so if we swap resistor and capacitor we have a *high-pass filter* (this is often called the "CR circuit", as we take the output over the resistor instead of the capacitor).

### *7.1.5. Transient response of the high-pass filter*

For situations where we do not have a "steady state", we need to consider the transient response of the circuit. This is common in instrumentation applications: something happens at time $t = 0$ and we need to observe how the system evolves for times $t > 0$. Consider the modified circuit in Figure 7.5. Note that in this figure we have swapped the positions of the resistor and capacitor. This is just because we want to observe the voltage across the resistor, but in terms of frequency-response this is a high-pass filter. If we set the applied voltage to be a constant $V$ and then close the switch at time $t = 0$ then $v(t) = u(t)V$.

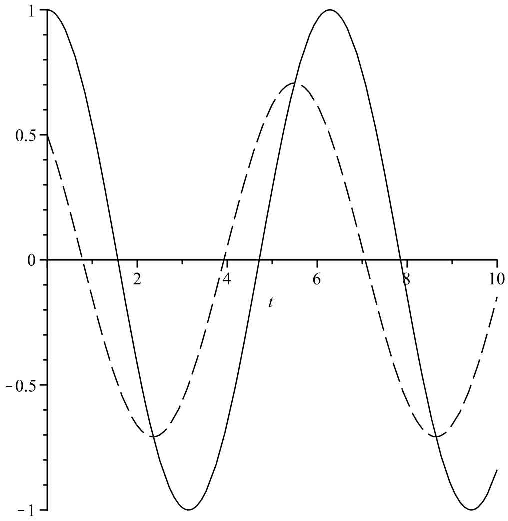


Figure 7.4: Low-pass filter waveforms (solid line is input, dashed line is output).

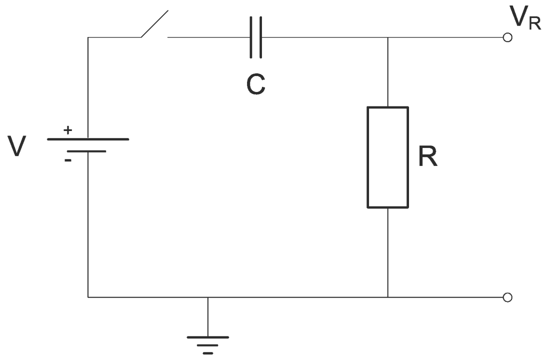


Figure 7.5: Switched CR circuit.

From now on we will just consider $t > 0$ so:

$$\begin{aligned} v(t) &= V \\ v_R(t) &= i(t)R \\ v_C(t) &= \frac{q(t)}{C}\,. \end{aligned}$$

Using KVL we find

$$v(t) = v_R(t) + v_C(t)\,,$$

and, taking the time derivative, we obtain

$$0 = RC\frac{\mathrm{d}i}{\mathrm{d}t} + i\,. \tag{7.5}$$

Note that we have set up a first-order ordinary differential equation. As a fundamental rule, there must be as many arbitrary constants in the solution to the equation as the order of the equation. This means we need additional information if we are to find a complete solution, and this information comes from a knowledge of the state of the system, usually taken at the initial time $t = 0$. You will have come across several methods for solving ODEs; the form of equation (7.5), where the function and its derivative must add to zero, suggests we attempt a trial solution of the form

$$i(t) = A\,e^{st}\,. \tag{7.6}$$

Substituting into (7.5),

$$RCsAe^{st} + Ae^{st} = 0\,. \tag{7.7}$$

yields

$$s = -\frac{1}{RC}\,.$$

To find the unknown $A$ we apply knowledge of the initial condition of the circuit. This is assumed to be *relaxed*, that is to say the capacitor is uncharged, therefore $v_C(0) = 0$, and we can take $v_R(0) = V$. A physical understanding of this is that, the instant the switch is closed, there is no charge on the capacitor so an instantaneous current $i(0) = V/R$ will flow. Substituting into equation (7.5) gives

$$i(0) = A = \frac{V}{R}\,.$$

So, the full solution is

$$i(t) = \frac{V}{R}\,e^{-t/RC}\,,$$

from which

$$v_R(t) = i(t)R = V\,e^{-t/RC}\,. \tag{7.8}$$

The exponential is characteristic of first-order systems. Figure 7.6 gives the response of a high-pass filter with $R = 10\,\mathrm{k}\Omega$, $C = 1\,\mu\mathrm{F}$ to a step input of 2 V. As $t \to \infty$ the capacitor achieves full-charge and hence $v_R \to 0$ as no current flows. Note that had we chosen a different initial condition then the solution would have been different.

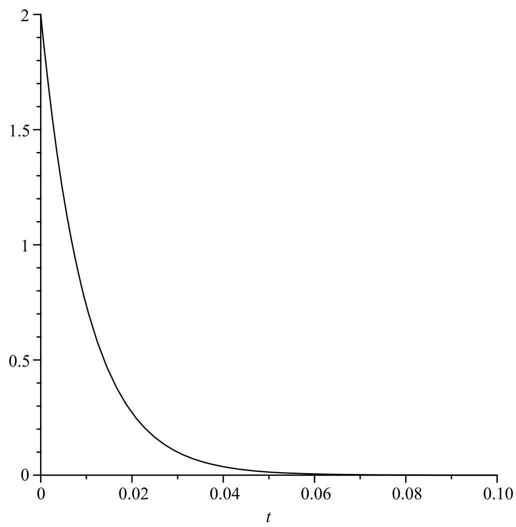


Figure 7.6: Time-domain response of high-pass filter to a step input.

The *time constant* of the system is RC. Note the relationship between the time constant and the cut-off frequency:

$$\omega_c = \frac{1}{\tau} = \frac{1}{RC}\,. \tag{7.9}$$

*Initial conditions.* There are three factors which determine how the voltage in our circuit evolves:

1. The differential equations which govern the circuit. These are entirely fixed and governed by the physics of the circuit;
2. The applied signal, in our case the voltage $v(t)$. This is frequently referred to as the *forcing function*; it is the externally applied stimulus which causes our system to respond;
3. The initial conditions of the system. Previously, we specified that the system was *initially relaxed*, that is to say the capacitor carried no initial charge. This is not always the case.

Imagine now that at some time after the capacitor is fully charged we open the switch. No current can flow round the circuit, therefore the capacitor remains charged: $v_C = V$. If we now close the switch again, nothing happens – no current flows and $v_R$ remains zero. Imagine that we now instantaneously change our forcing function to zero, i.e. $v(t) = 0$. Then, following the same methods as above, we will find that

$$v_R(t) = -Ve^{-t/RC}\,.$$

### 7.1.6. Integration and differentiation

If we now repeat this on/off sequence every $T$ seconds where, $T \gg RC$, then our forcing function is a square wave between zero and $V$ with period $T$. Plotted on a scale of several input cycles, the output looks like a sequence of spikes, positive and negative – see the upper sketch in Figure 7.7. The output is then *approximately* the time-derivative of the input signal. Mathematically, we can show this by

$$v_R(t) = iR = RC\frac{\mathrm{d}v_c}{\mathrm{d}t} \approx RC\frac{\mathrm{d}v_{in}}{\mathrm{d}t}\,.$$

For $\omega \gg \omega_c$ we find the impedance of the capacitor dominates over that of the resistor, therefore $v_{in} \approx v_C$.

Conversely, were we to go back to the low-pass configuration, but increase the values of $R$ and $C$ to make the time constant $T \ll RC$, we would find the circuit integrates the square wave to give a triangle output. We shall come back to this later on, but for now it is important to have a physical understanding of these RC circuits and how they behave in both the time and frequency domains. The principal characteristics are summarised in Table 7.1.

### 7.1.7. Physical analogues

The RC circuit is a good analogue for physical systems where the rate of change of a parameter is proportional to the magnitude of the parameter itself – see equation (7.5). For example, many thermal calculations apply the knowledge that heat-flow (and hence rate of change of temperature) is proportional to temperature difference. In mechanics, we might have a situation where a friction-force

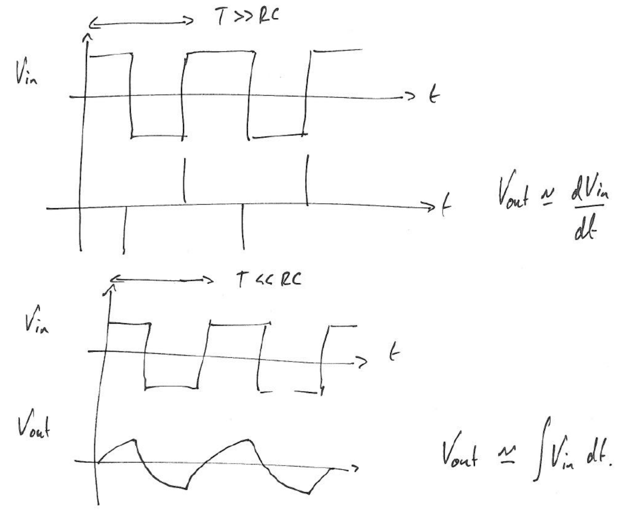


Figure 7.7: Differentiation of a square wave with a frequency above the cut-off of an RC high-pass filter (upper panel) and integration of the same signal for a frequency below the cut-off of an RC low-pass filter (lower panel).

(hence rate of change of velocity) is proportional to velocity. Such systems can be modelled by RC circuits, and indeed in the past before the use of numerical methods and computational models, it was common practice to model physical systems with such “analogue computers”.

### 7.1.8. Linearity of the RC filter

The reason we have spent this much effort studying the RC filter is that it is a *Linear Time-Invariant (LTI) system* (see Section 13). We will see how we can predict the behaviour of linear systems not just under sinusoidal inputs but with any arbitrary input. This will become a key theme in our studies, and in later parts of the course we will show how to use Fourier and Laplace transforms to gain easy solutions to some seemingly very complex problems in systems analysis, with the important proviso that all the component parts of the system are themselves linear.

To summarise what we know about the RC filter we start with the complex gain (here for the low-pass filter) given by:

$$\mathbf{G} = \frac{1 - j\omega RC}{\omega^2(RC)^2 + 1}\,. \tag{7.10}$$

- The filter gain is a complex quantity and also a function of frequency. The plot of $\mathbf{G}(\omega)$ is the Bode plot;
- For any frequency $\omega$ the gain gives us the output sinusoid $\mathbf{v_{out}}$ amplitude and phase compared to the input $\mathbf{v_{in}}$;
- Output and input sinusoid have the same frequency $\omega$ but an adjusted amplitude and phase;
- The behaviour of the RC filter is governed by first order ordinary differential equations (for example equation (7.5)), hence it is a first-order linear system;

Table 7.1: Summary of RC circuit characteristics.

| Behaviour | Low-Pass (RC) | High-Pass (CR) |
|---|---|---|
| Measure output across | C | R |
| Time constant | $\tau = RC$ | $\tau = RC$ |
| Cut-off frequency | $\omega_c = 1/RC$ | $\omega_c = 1/RC$ |
| $\omega \ll \omega_c$ | flat gain 0 dB | gain slope +20 dB/decade (differentiates) |
| $\omega \gg \omega_c$ | gain slope −20 dB/decade (integrates) | flat gain 0 dB |

- The time domain response to a step input is the familiar exponential form with a characteristic time constant.

Here is where our understanding of signals as complex quantities becomes useful. If we write the input as a sinusoid

$$\mathbf{v_{in}} = A\, e^{j\theta}\, e^{j\omega t}\,,$$

and the gain (evaluated at frequency $\omega$) as

$$\mathbf{G} = G\, e^{j\phi}\,,$$

then

$$\mathbf{v_{out}} = \mathbf{G}\, \mathbf{v_{in}} = G\, e^{j\phi}\, A\, e^{j\theta}\, e^{j\omega t}\,.$$

As was stated previously, the time dependent part of the expression is frequently omitted since it is common to input and output. We can do this because the *linear system always preserves the frequency of the input.*

$$\mathbf{v_{out}} = G\, A\, e^{j(\theta+\phi)}\,.$$

That is to say, the output is the input scaled by a factor G (the magnitude of the complex gain **G**) and rotated by an angle $\phi$ (the phase angle of the complex gain). Now comes the most important point which will be essential for our future studies. For a linear system, the output is a linear superposition of the input. If our input is composed a several sinusoids, we can apply the above calculation for each input component and sum the results to get the output. Since Fourier tells us *any* input can be decomposed into sinusoids, *this means that the complex gain tells us what the output will be for any arbitrary input waveform.* This is important, as it allows use to use Fourier (and related) techniques to solve problems.

### *7.2. The RLC circuit*

The *RLC circuit* shown in Figure 7.8 is the electrical analogue of the damped mechanical oscillator. In fact, we can use an analogue electronic system to simulate a complex mechanical system because of this equivalence. Whilst this is quite a powerful technique, it has been superseded by numerical computer simulations these days. In the RLC circuit we have the equivalence with the spring balance or damped harmonic oscillator.

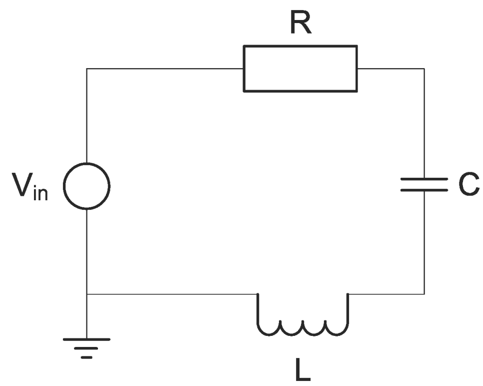


Figure 7.8: The RLC circuit.

As with the mechanical spring-balance we can determine the differential equations which govern the behaviour through use of some physical laws of conservation; in this case we use KVL:

$$v_R + v_C + v_L = v_{in}\,.$$

which we can re-write as

$$iR + L\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{1}{C}\int_{-\infty}^{t} i(t')\mathrm{d}t' = v_{in}\,.$$

If we take $v_{in}$ to be a step input at $t = 0$ then, for all times $t > 0$, we can differentiate and write

$$L\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + R\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{1}{C}i = 0\,.$$

Therefore, the series RLC circuit shows the same sort of transient response as the mechanical oscillator with undamped natural frequency $\omega_0$ and damping ratio $\xi$:[15]

$$\begin{aligned} \omega_0^2 &= \frac{a_0}{a_2} = \frac{1}{LC} \\ \xi &= \frac{a_1}{2\sqrt{a_0 a_2}} = \frac{R}{2}\sqrt{\frac{C}{L}}\,; \end{aligned} \tag{7.11}$$

We can understand the frequency-domain behaviour of the circuit best using the complex impedance approach given

[15]We define the $a$ constants in Section 14, unimportant here.

previously. The total impedance of the circuit is given by

$$\mathbf{Z} = R + j\omega L - \frac{j}{\omega C}\,.$$

Since $R$ is fixed, the circuit has minimum impedance when

$$j\omega L = \frac{j}{\omega C}\,,$$

which is the same result as equation (7.11). Further, since $\mathbf{Z}$ is a function of frequency, it is interesting to plot the response of the system as a function of frequency. Figure 7.9 shows the current flowing in the circuit as a function of frequency. For this figure, we take $L = R = C = 1$ and the applied voltage is 1 V in amplitude and swept in the range $\omega = 0.01...100$ rad/s. The *resonance* at $\omega_0 = 1$ rad/s is clearly visible.[16] At the resonance, the reactive part of the impedance is zero, so we have only resistive impedance, hence the current is 1 A.

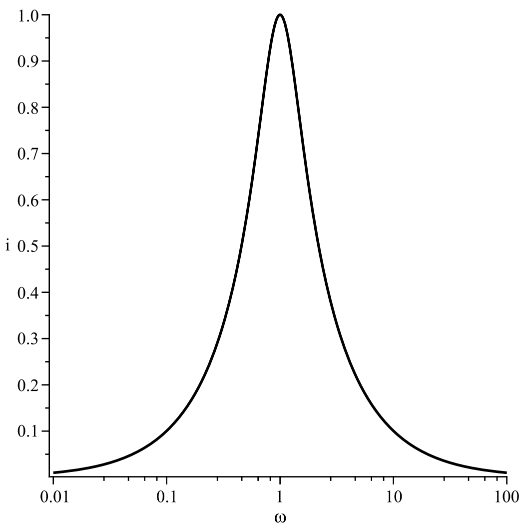


Figure 7.9: Current flowing in RLC circuit as a function of frequency.

At low frequencies the impedance $|\mathbf{Z}| \to \infty$ because of the capacitor, and at high frequencies $|\mathbf{Z}| \to \infty$ because of the inductor. Therefore, the overall shape of the curve is physically understandable. The mechanical oscillator would also show a similar behaviour in response to a swept-sinusoidal forcing function (force). Moreover, the RLC circuit will show similar time-domain response to a step input.

[16]Note that the *natural frequency* is the *un-driven* oscillation frequency, as we saw for the spring-balance where $y$ = constant. For the RLC circuit we found the resonant frequency under driven conditions (swept sinusoid input). In general, for second-order systems the natural frequency and resonant frequency are close but not identical. But the RLC circuit is one where they are identical.

## 8. Filters

A filter is used to modify the frequency content of a signal: usually, either to reject some unwanted component, such as interference at a particular frequency (e.g. mains at 50 Hz, or DAB radio near 200 MHz), or more generally to remove broad-band noise over a range of frequencies which are above/below the signal we are interested in. Passive filters are made from the R, L and C components. They offer no amplification – the output signal can never be larger than the input – hence the maximum *gain* of a passive filter is 0 dB.

### *8.1. Filter types*

*Low-pass/high-pass.* We have studied these in some depth, and we have seen how to implement simple versions of these with RC and CR circuits. The *bandwidth* of the low-pass filter is easy to define: from 0 Hz to the *cut-off frequency* ($-3$ dB point). The high-pass configuration (assuming the same cut-off) *blocks* this range of frequencies and *passes* frequencies above.

*Band-pass.* The band-pass filter has both a lower and a higher cut-off: it passes frequencies in-between. Therefore, it has a clearly defined bandwidth, and we want it to be flat in the *pass-band* region. In its simplest form the band-pass filter can be a RC-CR combination with appropriately chosen components.

*Band-stop.* Similarly, we can design a filter to reject a range of frequencies.

*Notch filter.* A variation on the previous two is the notch filter which is designed to pass (or reject) a particular frequency. In contrast to the band-pass/stop, the notch filter is designed to be as narrow as possible so that it only filters a particular frequency. These are used to select (or reject) a particular frequency component contained within a signal. Notch filters can be made using the RLC resonant circuit as seen in Section 7.

### *8.2. Filter characteristics*

The ideal filter rejects all frequencies outside the pass-band and passes all other frequencies perfectly. The ideal low-pass filter would be ‘square’, as shown in Figure 8.1. The pass-band is flat, the *knee* is perfectly sharp, and the slope of the *roll-off* is infinite.

We have seen that the passive RC filter, when used in either low- or high-pass configuration, has a gain roll-off of 20 dB/decade (see Figure 5.5, e.g. between 10 rad/s and 100 rad/s). This is equivalent to 6 dB/octave, and it is a useful exercise to check that this is so. This behaviour arises because – for frequencies well above the cut-off – the gain has a $1/f$ characteristic. In the first lab exercise the RC filter is used as an anti-aliasing filter, which exposes its limitations. Frequencies above the Nyquist are attenuated, however they are still present, and consequently they alias.

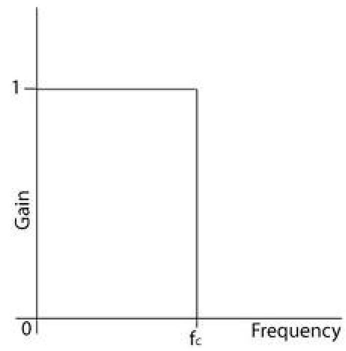


Figure 8.1: The ideal low-pass filter.

### *8.3. Higher order filters*

We can chain together two RC filters to make a second-order filter which should have a roll-off of 40 dB/decade. Three or more would increase the effect. This requires some care if we are to avoid *loading*. The filter is a frequency dependent voltage divider. Ideally, the signal source will be low-impedance (e.g. a signal generator) and the filter will pass the signal to a high impedance load (e.g. an oscilloscope). The filter’s impedance must not be too low otherwise it will load the source. The load (scope) must have a higher impedance otherwise it will load the filter.

Consider two chained low-pass filters. The RC of the second filter is actually in parallel with the C of the first. So, if this is not to load the first filter, we must make the impedance of the second filter somewhat higher than the first. A factor of ten is a good rule of thumb. Note that we can achieve this even for two filters with the same cut-off since it is the product of R and C which is important, not the individual values. In practice it is hard to make good sharp and steep filters with purely passive components.

### *8.4. Introducing active components*

When loading is a problem, we can use a buffer, which is an active electronic device (typically an op-amp). We will look at the design later in the course. The buffer has a high input impedance, low output impedance, and a gain of 1. It can be used to isolate two circuit blocks from each other to prevent the second from loading the first. This means we could – in principle – chain together as many identical RC sections as we like without fear of loading. We would get a fast roll-off but the knee would still be quite blunt.

If we are going to use op-amps then there are better active filters where the op-amp is part of the filter design and with remarkably few components we can achieve filters which are much better than the passive types. Some examples are shown in Figure 8.2. The problem is that the analysis of these circuits is fiendishly complicated. For the purposes of this course it is sufficient to know that:

- Active filters use op-amps as an integral part of the filter design;

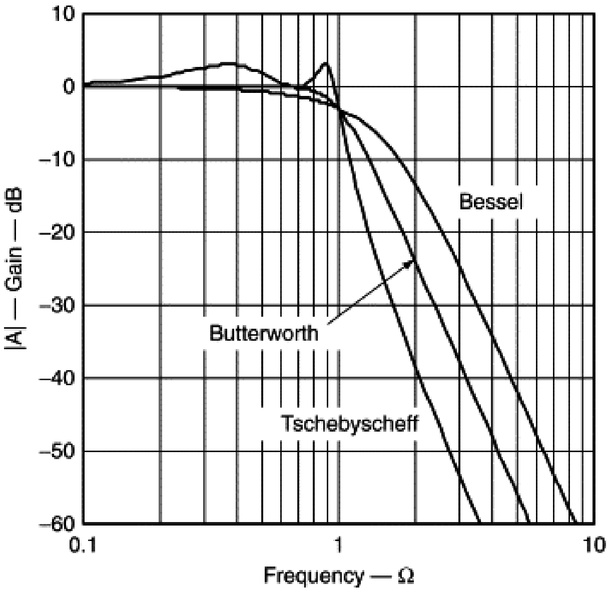


Figure 8.2: Some active low-pass filters (T. Kugelstadt, in *Op Amps for Everyone* (3$^{\text{rd}}$ edition), 2009). Typical features are the flat pass-band of the Butterworth filter and the sharper "knee" of the Chebyshev filter at the expense of some pass-band ripple.

- There are several standard designs to choose from; the user needs to look-up the correct component values to use in order to achieve the desired performance;
- Flatness of the pass-band is often compromised to achieve a 'sharp knee' or a 'fast roll-off;
- The op-amp provides buffering, so the active filter will not load the source;
- Signal amplification is possible.

We will come back to this topic again later in the course.

## 9. Negative feedback amplifiers

If you are enthusiastic about semiconductor physics, it is perfectly feasible to understand the operation of basic semiconductor devices such as diodes and transistors from fundamental physical principles. Then, using mainly transistors, it is fully possible to build all of the electronic circuits you are ever likely to need in a lab environment. The problem is that many transistor circuits, such as a simple amplifier, look easy to build (you can see many transistor amplifier circuits in Horowitz & Hill; they are not particularly complicated), but they tend to be quite tricky to get right in practice (hence the title of that book: "The Art of Electronics"...). It is for this reason that the *operational amplifier* was developed, to take the pain and inconsistency out of discrete amplifier circuits.

**Learn it in lab**
You will build a simple inverting amplifier in the second lab session.

### *9.1. The operational amplifier*

The op-amp actually predates the transistor: the first op-amps used vacuum valves instead. These were used in early analogue computers where the designers needed simple circuits to perform mathematical *operations* (hence the name) such as addition, subtraction, integration, differentiation, as well as simple amplification. The op-amp provides repeatability and consistency, that is to say, we can take one op-amp out of a circuit (maybe because we have broken it) and replace it with another and the circuit will behave exactly as before. This is often not the case with transistors. The key point of the op-amp is that we need not know what is going on inside the device as long as we have fully characterised its behaviour. We can then use the op-amp to build really quite complex circuits which have a good chance of working first time. There are also a huge number of different op-amps available on the market with many different kinds of specialist properties. It is for these reasons that physicists tend to prefer the op-amp when constructing circuits. As Figure 9.1 shows, the op-amp package integrates quite a lot of complexity into a tiny physical circuit – the details of which the end-user (or the student!) – need not be concerned with.

*Active versus passive.* All the electronics we have looked at so far has been *passive circuits* composed of resistors, capacitors, etc, which are not able to boost a signal amplitude or power. By contrast, op-amps are *active devices* which are themselves connected to a power supply, which enables them to actively boost the signal power – either by increasing the voltage, the current, or sometimes both.

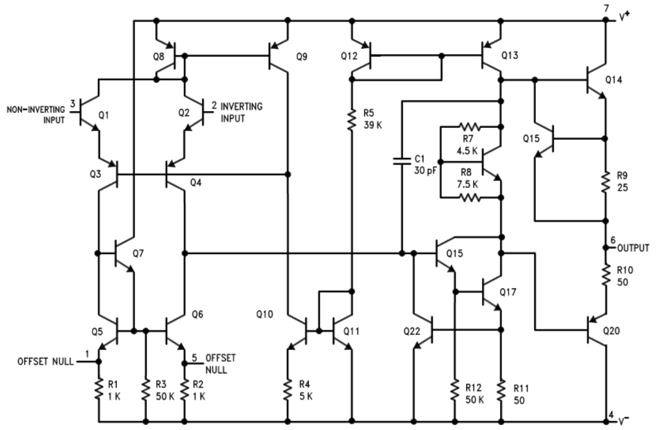


Figure 9.1: Functional circuit of the LM741 op-amp from Texas Instruments (from datasheet: https://www.ti.com/product/LM741).

### *9.2. Characteristics*

The basic op-amp is a *differential amplifier* with a gain $A$ known as the *open-loop gain*. It requires a minimum of 5 connections:

- *Non-inverting input*, often called $V_{in+}$, and usually labeled simply '+';
- *Inverting input*, often called $V_{in-}$, labelled '−';
- An output $V_{out}$;
- Two power-supply connections $\pm V_{ss}$, often ±15 V.

Figure 9.2 shows the simplified circuit (with somewhat different labels, in particular the open-loop gain which is more commonly denoted by $A$).

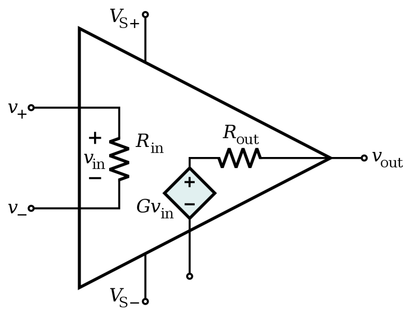


Figure 9.2: The simplified op-amp internal circuit and connections.

The device is governed by three key relations. Firstly,

$$V_{out} = A\left(V_{in+} - V_{in-}\right); \tag{9.1}$$

hence, the idea that this is a differential amplifier. Because it is an active device, the output cannot exceed the power supply "rails" so,

$$-V_{ss} \le V_{out} \le +V_{ss}\,. \tag{9.2}$$

If we do try to drive the op-amp above the power supply rails then we will see *clipping*, where the signal is sharply trimmed back to the $V_{ss}$ levels. Finally, while the op-amp is a fast device, the output cannot change instantaneously, and we find there is a maximum *slew rate*

$$S = \left[\frac{\mathrm{d}}{\mathrm{d}t} V_{out}\right]_{\max} , \tag{9.3}$$

typically of the order 1 volt/$\mu$s.

The gain $A$ is typically high (at least 100,000), the input impedance is real and high ($> 1\,\mathrm{M\Omega}$) and the output impedance is low (tens of $\Omega$). These facts, and the above relations, are all that is needed to understand the behaviour of any op-amp circuit.

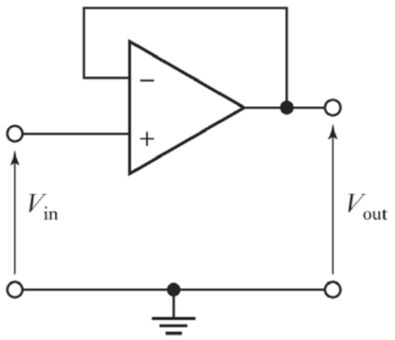


Figure 9.3: The voltage follower, or "buffer".

*9.3. The comparator*

The simplest op-amp circuit is the comparator. This is very effective for comparing two voltage levels and deciding which is greater. Due to the very high gain $A$ and slew-rate $S$, the op-amp will respond almost instantaneously to any potential difference greater than a microvolt between the two inputs by going straight to one of the two power supply voltages. This is fully consistent with equations (9.1), (9.2) and (9.3). Since the output of the op-amp has the capability to drive some power, we can do something useful with this such as controlling a switch or a relay. For example, if the two voltage inputs represent temperature in an experiment we can do:

IF
    temp_A > temp_B
THEN
    switch_off_heater

*9.4. The voltage-follower or buffer*

Simply by looping the output voltage $V_{out}$ back to the inverting input we make a circuit where, to a very good approximation,

$$V_{out} = V_{in} . \tag{9.4}$$

This is illustrated in Figure 9.3. Note that again this is fully consistent with equations (9.1), (9.2) and (9.3). Of course, when we look in detail we see that the output takes some time to track the input due to the finite slew-rate, and also there is some small ($\mu$V) difference between $V_{out}$ and $V_{in}$ due to equation (9.1); however, to a very good approximation, we will find that equation (9.4) is valid for most practical purposes.

The practical application of a circuit where $V_{out} = V_{in}$ comes when we consider that the input resistance of the circuit is very high, so it draws negligible current, while its output resistance is very low. The buffer is used as a kind of impedance transformer. This is useful where we have a sensor source which is 'weak' and easily loaded (typically high source resistance). Typically, we will *buffer* or *stiffen* the source using a voltage-follower. The op-amp is then able to drive any current needed for a low-impedance load or a transmission line without pulling any current out of the source. This avoids the problem of loading the sensor.

*Feedback.* When we connect the output (or some fraction of it) back to the input of a device then we have created a *feedback loop* – see Figure 9.4. In the case of the voltage-follower we have returned all of the output back to the input which is the same as creating a feedback amplifier with a gain of 1. In practice, all real-world op-amp circuits (except the comparator) use this *negative feedback*. So, we can see how the voltage-follower nicely illustrates the concept of the *ideal amplifier*.

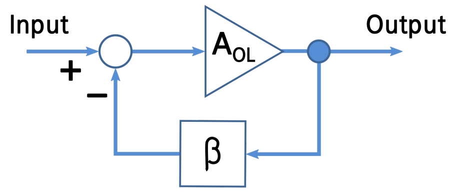


Figure 9.4: Negative feedback.

*9.5. The ideal amplifier*

Table 9.1 summarises the ideal properties of a perfect amplifier, compared to a readily available device such as the LM741, which will be used during labs. Most of the parameters for the ideal amp are self-explanatory; however, to understand why we want $A \to \infty$ we need to study feedback in more detail. Nevertheless, even in the simple case of the voltage follower, we can see that for $A \to \infty$ we will find $V_{out} \to V_{in}$.

Table 9.1: Ideal amplifier.

| Behaviour | Ideal amplifier | Real amplifier |
|---|---|---|
| $A$ | $\to \infty$ | $2 \times 10^5$ |
| $R_{in}$ | $\to \infty$ | $> 10^6\,\Omega$ |
| $R_{out}$ | 0 | $\sim 50\,\Omega$ |
| $S$ | $\to \infty$ | 0.5 V/$\mu$s |

### 9.6. Negative feedback

Feedback as applied to operational amplifiers is covered very well in both Horowitz & Hill (Chapter 3) and Diefenderfer & Holton. A illustrated in Figure 9.4 above, the general approach is this: taking an amplifier with very high open-loop gain $A$ and applying negative feedback of *feedback fraction* $\beta$ will result in an amplifier whose overall *closed-loop gain* is purely a function of the *feedback network*. Given that $A$ is very large, the closed-loop gain will be:

$$G = \frac{A}{1+\beta A} \approx \frac{1}{\beta}\,.$$

Since the feedback network is made from simple passives such as resistors, the closed-loop gain ($G = 1/\beta$) is going to be well determined, stable, and further the circuit will behave the same even if we blow our op-amp and have to put a different one in. You may wonder why we cannot just build transistor amplifiers with sensible values of gain in the first place. There are several answers to this, but the most convincing one is that transistors are tricky things which work best in a very high-gain setup, so it is just easiest to control the gain using feedback. The net result is that the high-gain feedback configuration has the following advantages compared to open-loop amplification:

- Increased linearity and wider bandwidth;
- Increased input impedance and reduced output impedance;
- Reduced gain (or gain brought under 'control');
- Reduced distortion.

Furthermore, since the op-amp is quite close to the 'ideal amplifier' *when connected in negative feedback*, two important *Golden Rules* can be applied:

#1: The output will do whatever is necessary to drive the potential difference between the two inputs to zero;

#2: The inputs draw no current.

These two rules make the analysis of op-amp circuits relatively straightforward, and circuits built using these assumptions and rules will conform to expectations most of the time.

### 9.7. Control theory

One aspect of feedback not covered in the course texts is *control theory*. A discussion of this may help to illuminate why feedback is such a clever and ubiquitous concept. Indeed, much of control theory is essentially dedicated to the study of feedback. A simple example also shows that feedback is not such a new concept. Consider again the steam engine. The speed of the engine depends on the amount of steam fed in from the boiler, and this is controlled by a valve called the throttle. If the engine is running too slowly we can increase the speed by opening-up the throttle. This is *open-loop control*. Now imagine we ask our engine to do some work such as threshing barley. As the farm-hand puts a pitch-fork of barley into the threshing machine, the engine is *loaded* and will slow-down. The engine operator can compensate for this by opening up the throttle to increase the power; however, once the barley is done then the load will diminish and now the engine will run fast – the operator sees this and closes down the throttle a little. Here we have applied feedback, but the feedback is done by having a human 'in the loop'. Much better to have an automatic system such as a *governor* which is a mechanical system which controls the speed of the engine – see Figure 9.5. If the engine slows down (load applied) then the governor opens-up the throttle, and vice-versa. These usually consist of a pair of rotating weights, which use the centripetal force to regulate the throttle. This is closed-loop control. The key point is that the governor allows the operator to set the desired outcome (i.e. speed) and let the system regulate itself. What is clear is that we can build a really powerful engine and then control this power using the feedback regulator. Under no load the engine will happily idle, and when a big load is requested the engine can respond.

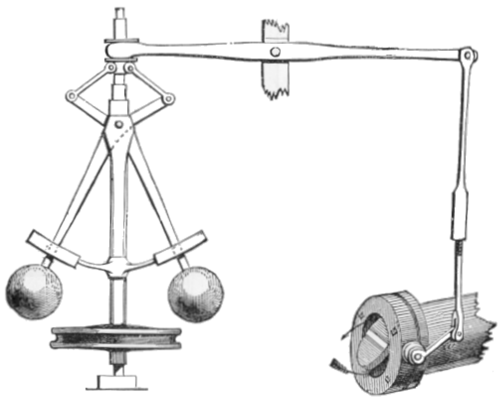

Figure 9.5: The centrifugal "fly-ball" governor: the balls swing out as speed increases, which closes the throttle, until a stable balance is achieved.

We can draw an analogy with the op-amp. Speed becomes output voltage. The governor becomes the feedback loop, whereby we control the net gain of the system. If the amplifier is connected to a low-impedance load then more output current is required and the feedback loop will respond accordingly in order to maintain the desired output voltage. This is another advantage of feedback: the feedback amplifier looks like a very nice voltage source since its output voltage is quite independent of load.

*Meaning of negative feedback.* Feedback is almost always negative in control theory. This means that the output acts so as to *reduce* the value seen at the input. If the steam engine runs too fast then the governor reduces the amount of steam. For op-amps we find that applying feedback to the negative or inverting input of the amp results in negative feedback according to equation (9.1). This also leads directly to an intuitive understanding that Golden Rule #1 is true: with negative feedback the output will always move to a level which makes the potential difference at the inputs very close to zero. You should ensure that you understand all the circuits in this section with this in mind.

### *9.8. Common op-amp circuits*

The most common op-amp circuits on the syllabus for this course are presented below. You are expected to understand and be able to analyse these (e.g. derive their gains). They are covered also in the usual textbooks.[17]

*Non-inverting amplifier.* The design in Figure 9.6 illustrates the gain $G = 1/\beta$ relationship, since the feedback fraction is due to the voltage divider formed from two resistors. Note that the voltage-follower or buffer is a non-inverting amplifier with $\beta = 1$. You should try to derive the gain of the non-inverting amplifier as a function of the input and feedback resistors $R_1$ and $R_2$.

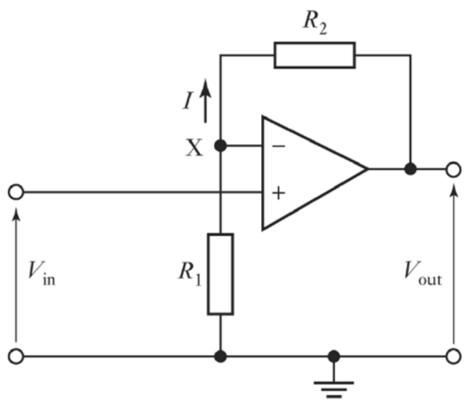


Figure 9.6: The non-inverting amplifier. Note that in this circuit we would avoid the crossing of lines by flipping the amplifier (so the inverting input is at the bottom) – a more common representation.

*Inverting amplifier.* A basic op-amp configuration is that shown in Figure 9.7, which is expanded on for many of the following designs. This also illustrates well the concept of the *virtual ground*. The virtual ground is not connected to ground (indeed it is isolated from ground by $\sim$M$\Omega$); however, it is always apparently zero volts, due to Rule #1, which makes circuit analysis easier. The output is inverted (phase shift 180 degrees with respect to input); however, for many AC signals (e.g. audio) this is not important.

[17]Many op-amp datasheets have example circuits such as these, e.g. https://www.ti.com/product/LM741. Although you are not expected to memorise any such datasheet, it is useful to look at some examples to see how parameters of real op-amps are specified.

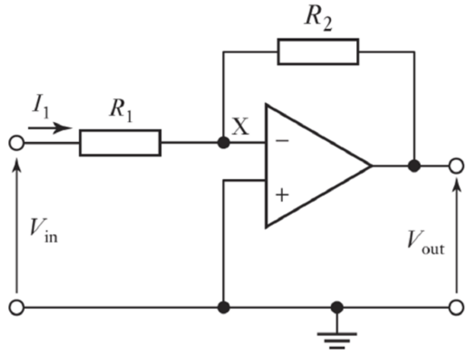


Figure 9.7: The inverting amplifier. Here, 'X' is a virtual ground.

*Summing circuit.* Kirchhoff's Current Law needs to be invoked here together with the virtual ground, to show that the input current is a function of both input voltages – see Figure 9.8. Whilst the output is inverted this can be fixed by a following inverting amplifier with gain $G = -1$. This circuit could be used, for example, to remove a zero-offset from a sensor's output.

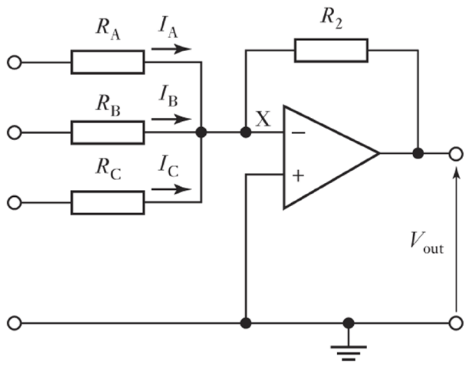


Figure 9.8: The summing amplifier.

*Integrator and low-pass filter.* Replacing the feedback resistance of the inverting amplifier with a capacitor – see Figure 9.9 (a) – makes the circuit reactive (frequency-dependent gain); in this case we integrate the signal. Compared with the passive RC integrator (low-pass filter) we discussed in Section 7, we have the advantage that here we can choose R and C to give us amplification too. Note, however, that the output cannot exceed the supply limits ($|V_{out}| < V_{ss}$). There are other applications for this circuit beyond filters; for example, you might integrate the output from an accelerometer to obtain the speed of an object and again to measure its displacement.

*Differentiator and high-pass filter.* With R and C swapped – Figure 9.9 (b) – we have the opposite behaviour: a differentiator and high-pass filter. The interpretation as a filter is intuitive: a low-frequency signal will be 'blocked' by the capacitor, and will not reach the amplifier. Once again the golden rules make it easy to derive the behaviour of this circuit: since ‘X’ is a virtual ground, the feedback current is $i_f = V_{out}/R$, and the input current is

$$i_i = C\frac{\mathrm{d}}{\mathrm{d}t}V_{in}\,.$$

Since no current enters the op-amp ($i_f = -i_i$) we have

$$V_{out} = -RC\frac{\mathrm{d}}{\mathrm{d}t}V_{in}\,.$$

We can instead use the capacitor impedance explicitly:

$$i_i = \frac{v_{in}}{|Z_C|} = v_{in}\,|j\omega C|$$

$$G = \frac{V_{out}}{V_{in}} = -\frac{R}{|Z_C|} = -\omega RC\,.$$

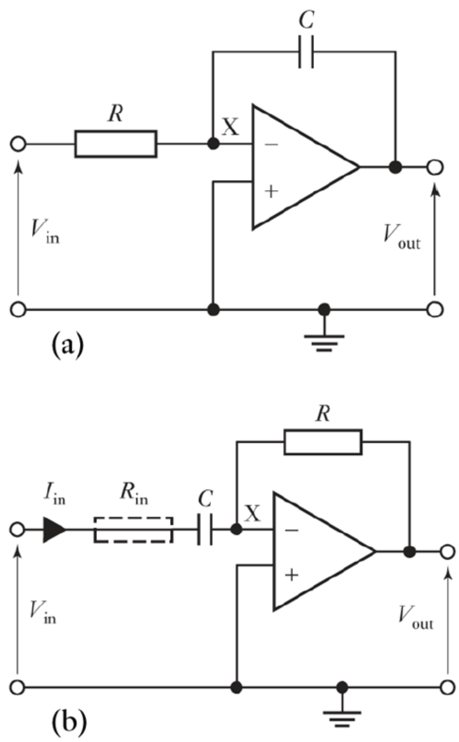


Figure 9.9: The integrating (a) and differentiating (b) amplifiers.

*Active filters.* Since op-amp filters present a relatively high impedance to the source and can give us amplification, we can chain together lots of such filters without loss of signal amplitude or loading the source. This is a complicated topic (see Horowitz & Hill Chapter 4) and for the purposes of this course it is sufficient to know about the three filter ‘figures of merit’: (1) *flatness* of the pass-band, (2) *sharpness* of the knee and (3) *steepness* of the roll-off. Even a simple op-amp filter is better than an RC filter, but to design these takes care. Typically, it is easiest to take one of the textbook designs (e.g. those in Figure 8.2) and set the parameters according to the required filter characteristics.

## 10. Differential signals

In all of the discussions so far to do with electrical signals we have considered our signal to be a voltage relative to ground potential (0 V). This means that we would measure the signals – e.g. with a digital multimeter (DMM) or oscilloscope – by connecting the black lead of the meter to ground and the red lead to the measurement point. These signals are known as *single-ended signals*, and can be carried from A to B on a single piece of wire, that is to say, if A and B are two different units (such as experiment and oscilloscope) then we only need one wire to transmit the signal. This assumes that the zero-volts ground at A and B is the same; this is not always the case, which is why it is good practice to connect the black lead of the scope probe to some ground reference near the point of measurement.

While the single-ended signal is of course always specified as being relative to ground potential, the *differential signal* has the subtle difference that it is relative to *some other potential* (which is itself usually not ground). Consider a sensor which has two terminals and produces an output voltage across these terminals, such as an accelerometer or a strain gauge. We are only interested in the difference between the two terminals, i.e. we do not care what the actual potential of either terminal is relative to ground. This is illustrated in Figure 10.1. We can define the differential signal as

$$V_D = V_+ - V_- \,.$$

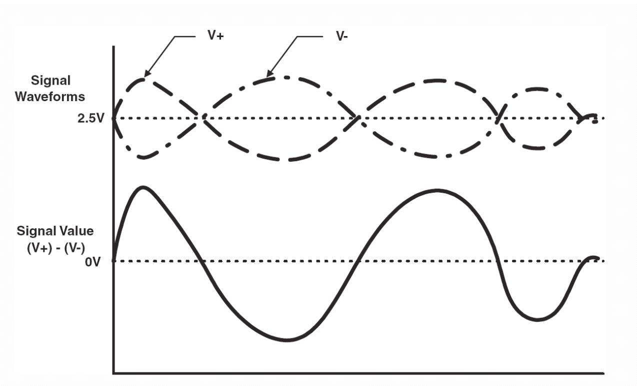


Figure 10.1: Differential signal.

*Common-mode signal.* The common-mode signal is defined as follows:

$$V_{CM} = \frac{V_+ + V_-}{2} \,.$$

This is the common level about which both terminals vary. In practice, $V_{CM}$ is the average value of the potentials at the two terminals. The main point to consider here is that $V_D$ represents the signal that we want to measure, while $V_{CM}$ represents some kind of background which we are not interested in.

Let us consider the readout of a set of strain gauges using the *Wheatstone bridge* shown in Figure 10.2 – a famous circuit used to measure very small resistance changes by balancing two legs of a bridge circuit. Strain gauges are resistive devices, typically a thin resistive pattern deposited on a malleable substrate. There are many strain-gauge readout configurations with anything between one and four of the resistors in this bridge replaced by such devices in order to measure compressive and tensile stresses in various directions. The point here is to make a very sensitive measurement across terminals A and B which reflects very small changes in resistance which will unbalance the bridge. If all resistors are equal then, under no strain the bridge is balanced and, if the bias voltage is, say, 10 V, then $V_A = V_B = 5$ V. Under some small strain the equivalent resistances of the two arms will vary, and $V_A$ will depart from $V_B$ by some small voltage (typically <1 mV) which we want to measure. So, the signal is differential ($V_D = V_A - V_B = 1$ mV) and must be measured over a large common-mode voltage ($V_{CM} = (V_A + V_B)/2 = 5$ V).

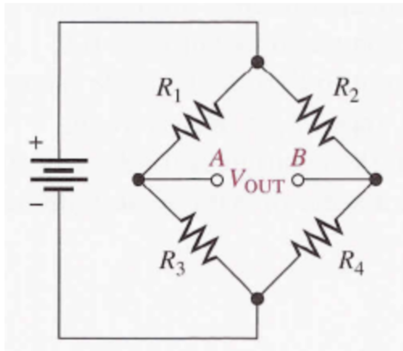


Figure 10.2: Reading a set of strain gauges ($R_{1..4}$ in this example) with a Wheatstone bridge.

*Noise rejection.* Another advantage of working with differential signals is illustrated in Figure 10.3. Here, the signal is represented by a differential pulse. Wires can act as aerials to pick up noise from radio waves or other signals elsewhere in the circuit – usually called *electromagnetic interference (EMI).* However, if the two wires which carry the differential signal are reasonably close to each other, then it is reasonable to assume that the amount of noise picked up will be the same on both. That is to say, the noise is a *common-mode signal.* We would like the amplifier to remove this noise. To measure a differential signal, we need a *difference amplifier.*

*Common-mode rejection.* Some main points to consider when choosing a difference amplifier are:

- Input isolation from ground (high);
- Differential voltage amplification (high);
- Common-mode voltage amplification (low).

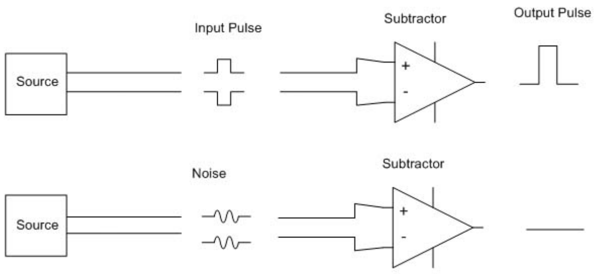


Figure 10.3: Noise rejection.

Since no amplifier is perfect, there will always be some common-mode voltage which gets through the amplifier and appears on the output. In practice, the output voltage is given by

$$V_{out} = A_D(V_+ - V_-) + \frac{1}{2}A_{CM}(V_+ + V_-)\,,$$

where $A_D$ is the *differential gain* and $A_{CM}$ is the *common-mode gain*. For any particular amplifier we can measure the *Common-Mode Rejection Ratio* (CMRR), which is usually given in dB and defined as:

$$\mathrm{CMRR} = 20\log_{10}\frac{A_D}{A_{CM}}\,.$$

The CMRR is a key figure-of-merit for a difference amplifier. The AD620 instrumentation amplifier from Analog Devices has a CMRR of 100 dB, which is very high performance. More specialist devices can achieve 120 dB or more.

*10.1. Difference amplifier*

The amplifier circuits we have looked at so far are suitable for single-ended signals. Typically, one of the inputs is connected to ground (the inverting amplifier, for example). It is not possible to use these to amplify differential signals since one side of the signal will be shorted to ground. Consider for example a complex experimental setup within which we want to measure the voltage across a single resistor (maybe because we want to be able to calculate the current through it). Connecting the inverting op-amp circuit across the resistor would connect one side to ground, probably stopping our circuit from working properly. The solution to this may be to attach some more resistors to the non-inverting input. The result is the *difference amplifier*, as shown in Figure 10.4.

*Circuit analysis.* Separating the top and bottom branches from the op-amp (Figure 10.5) and using Kirchhoff's Current Law we get

$$I_1 = \frac{V_1 - V_A}{R_1} = \frac{V_A - V_{out}}{R_f}\,,$$

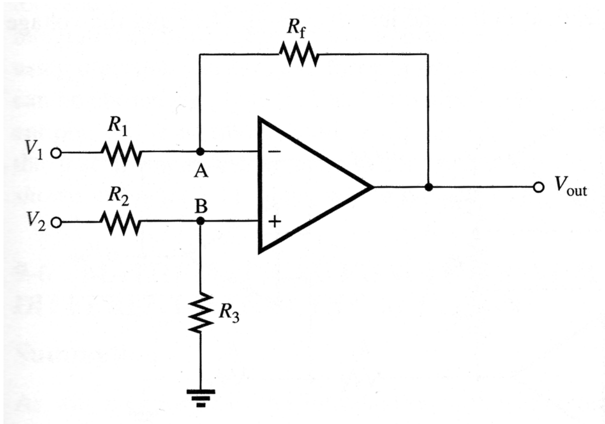


Figure 10.4: The difference amplifier (Diefenderfer Chapter 9); note the non-inverting input is not grounded.

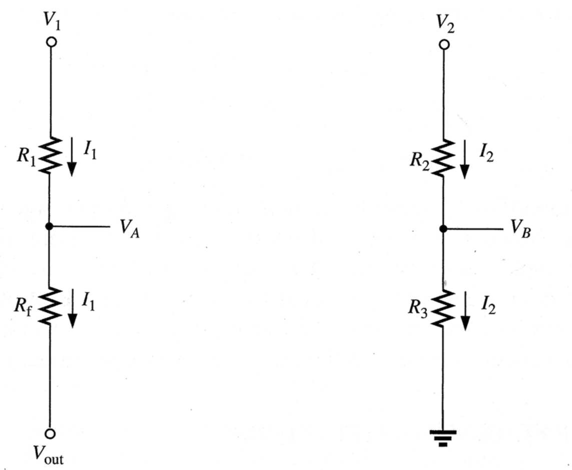


Figure 10.5: Circuit analysis for the difference amplifier (Diefenderfer Chapter 9).

$$V_A = \frac{V_1 R_f + V_{out} R_1}{R_1 + R_f}\,.$$

For the other branch we can do the same or, since the bottom of the circuit is connected to ground, we can just treat it as a voltage divider and write

$$V_B = \frac{V_2 R_3}{R_2 + R_3}\,.$$

Since Golden Rule #1 requires that $V_A = V_B$, we can solve for $V_{out}$ to yield

$$V_{out} = V_2\,\frac{R_3}{R_1}\,\frac{R_1 + R_f}{R_2 + R_3} - V_1\,\frac{R_f}{R_1}\,.$$

Now, if we choose to set $R_2 = R_1$ and $R_3 = R_f$ (i.e. both signals see the same input impedance) then we get:

$$V_{out} = (V_2 - V_1)\,\frac{R_f}{R_1}\,. \qquad (10.1)$$

The difference amp is commonly used as described, but for high-precision applications there are two problems:

1. If we fail to match $R_1 = R_2$ and $R_3 = R_f$ exactly, then we get a common-mode signal at $V_{out}$ as well as the differential signal that we want. Any common mode signal is an error signal, since it depends on how carefully you build the amplifier circuit;
2. It has a relatively low input impedance (at $V_2$ this is equal to $R_2 + R_3$). This can cause us to 'load' the signal we are trying to measure by drawing current from it. It is tempting to make the values of the resistors very large, say some M$\Omega$, but the inputs of the op-amp inevitably have some stray-capacitance, so we would have a time-constant effect which would slowdown voltage swings at the input. We need to keep $R_2 + R_3$ to be some k$\Omega$.

Recalling that op-amps are small and cheap, we can make use of the high-input impedance buffer or voltage follower circuit. The simplest thing to do would be to just attach a buffer to each of the inputs at $V_1$ and $V_2$. This completely solves the input impedance problem – the points in the circuit we are trying to measure are now isolated from our difference amplifier by the buffer impedances, which can be $\sim$G$\Omega$. However, the common-mode issue (problem (1) above) remains. The reason is that the difference amplifier is trying to do two jobs: both reject common-mode signals and amplify differential mode signals. We can improve this situation by moving some of the gain into the input stage, which relaxes a bit the requirement to match the resistors. This design is known as the *instrumentation amplifier*.

*10.2. Instrumentation amplifier*

The design of the classic *instrumentation amplifier* is shown in Figure 10.6. The circuit analysis for this is covered in Diefenderfer & Holton Section 9.5. For the purposes of this course though, it is not necessary to know the detail of the analysis. The output is given by

$$V_{out} = (V_2 - V_1)\,\frac{R_4}{R_3}\left(1 + 2\frac{R_1}{R_2}\right).$$

So, the amplifier produces a voltage output which is proportional to the difference between the two input voltages, and with a very high impedance at the inputs.

*Applications.* The instrumentation amplifier can be used wherever we want to measure the voltage across any circuit element without connecting either side to ground. For example, we might imagine that the input of a multimeter looks like the instrumentation amplifier because we can connect across any two points in a circuit without drawing any current from the circuit or shorting the connection point to ground.

The analogue inputs on the ELVIS units we will use in lab are instrumentation amplifiers. When we want to use them to make a single-ended measurement (e.g. making the Bode plot of the RC filter), we just connect the "$-$ve" input to ground.

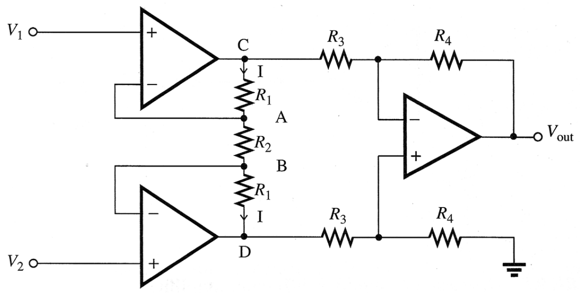


Figure 10.6: The instrumentation amplifier (Diefenderfer Chapter 9).

*Integrated circuit instrumentation amplifiers.* These days the semiconductor industry provides a 'ready-made' implementation on a single chip. One such device is the Analog Devices AD620 (Figure 10.7).[18] This has a CMRR (Common Mode Rejection Ratio) of 100 dB and a differential gain adjustable up to 10,000 set by an external gain resistor connected across pins 1 and 8 (this is $R_2$ in our circuit). All the other resistors are inside. The advantage of these devices is that the internal resistors have been matched to be as close as possible. This is done by laser-trimming the individual resistors for each part. Because they are manufactured in quantity, this process becomes cost effective and the AD620 retails for around £10.

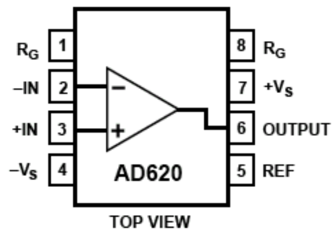


Figure 10.7: Pin connections for the Analog Devices AD620 instrumentation amplifier.

[18]See product datasheet and some standard circuit designs at `https://www.analog.com/en/products/ad620.html`.

## 11. Real-world op-amps

The ideal amplifier is a zero-order system with infinite gain, bandwidth and slew-rate. We saw in Section 10 that a "real" difference amp has some common-mode gain which is an imperfection due to non-matched resistors. For practical applications the real op-amp has the following characteristics:

*Finite gain and input impedance.* These non-ideal parameters result in the golden rules being only approximations, of course: in reality, the op-amp inputs will draw some non-zero current $i_{in}$ as a result of the finite input resistance $R_{in}$ of the device. It is easy to show (and you are encouraged to try) that the input impedance of the non-inverting input in a feedback design with closed-loop gain $1/\beta$ is $Z_i = V_{in}/i_{in} \approx R_{in}\beta A$. In older op-amp designs an extra resistance would often be added between the non-inverting input and ground, chosen such that the two inputs saw the same equivalent impedance – hence cancelling out any voltages between them to first order; modern op-amps have really very high input impedance so this is rarely needed. Since specialist op-amps with very high $A$ and $R_{in}$ are available, it is simply a matter of checking values of $\beta A$ and $i_{in}$. For the latter, this simply means we should not use very large feedback resistors; for the finite gain, see the problem sheet question on this topic.

*Input offset voltage.* Connecting the inverting and non-inverting inputs of an op-amp to ground should result in zero output; however, manufacturing imperfections lead to what we can think of as a small potential difference between the inputs. This can be $\sim$mV, and so feedback will automatically correct for this by adjusting the output, with the result that the closed-loop amp shows a zero-offset. Most op-amps have a couple of pins across which a "trim" resistance can be placed to compensate for this effect. The datasheet will give the details about how this is done and also the maximum limits and stability expected for the input-offset voltage.

*Finite bandwidth and slew rate.* These parameters are intimately linked, and very important to a full understanding of op-amp behaviour. The open-loop op-amp has a surprisingly limited bandwidth, which ultimately limits the speed with which it can respond to a fast-changing input.

### *11.1. Gain-bandwidth product*

The Bode plot for an op-amp such as the 741 (red line in Figure 11.1) shows that $A$ is only "very large" for the range of frequencies up to $\sim$10 Hz. Above 10 Hz the amp shows a gain fall-off $G \propto 1/f$ like a low-pass filter. By 1 MHz we have $A \sim 1$. Here we define the *bandwidth* $B$ as the range of frequencies from 0 to the $-3$ dB point. Under feedback we throw away the excess gain, so we find

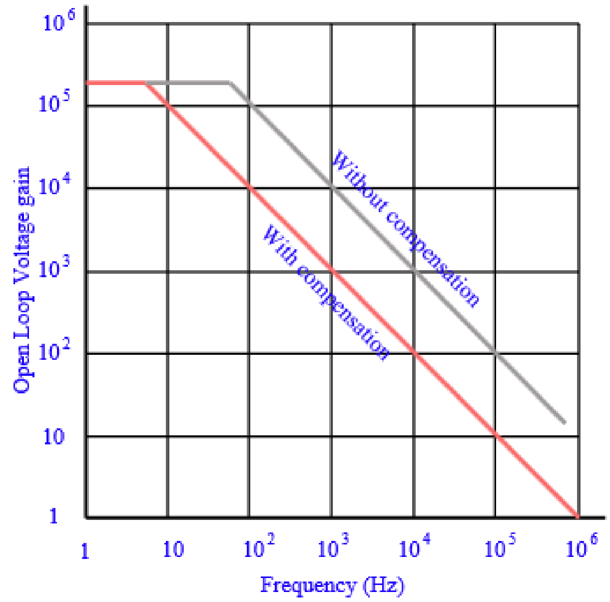


Figure 11.1: Open-loop gain response for the 741 op-amp (red); a similar device (748) without internal stability compensation is also shown (grey) (from www.electronics-notes.com).

that feedback widens the bandwidth of the amp and, above 10 Hz or so:

$$G \times B = \text{constant}\,.$$

This is illustrated by the red line in Figure 11.2. For example, if we have a 741 with $A = 1$ at 1 MHz, then $G{\cdot}B = 10^6$ and if we apply feedback with $\beta = 0.01$ then the closed-loop gain is 100, and hence $B = 10^4$ Hz. This is indeed the frequency where the closed- and open-loop gain curves intersect. The $G{\cdot}B$ product is a useful figure-of-merit for an op-amp.

### *11.2. Phase response*

Accordingly, the phase shift at the output is about $-45°$ at 10 Hz (i.e. output lags input) and continues to fall to about $-180°$ at about 1 MHz. Imagine a 10 Hz sine wave going into a unity-gain buffer. The output has a phase shift which looks like a time-delay. This is very important because we will use the op-amp with feedback.

### *11.3. Stability under feedback*

This requires a thorough understanding of how feedback really works, for which it is easiest to refer back to the block-diagram in Figure 9.4. Negative feedback works by subtracting some output from the input. This is easy to see when we think about DC signals going into the input. If we put 1 V on the input then the output reacts so as to feed back 1 V into the summing junction to counteract this, with the net result of 0 V at the input to the high-gain stage. However, consider a signal at 1 MHz where the phase change is $-180°$. When the input voltage is at its *positive* peak, the output is at its *negative* peak, so we actually *add* extra signal at the summing junction. This is

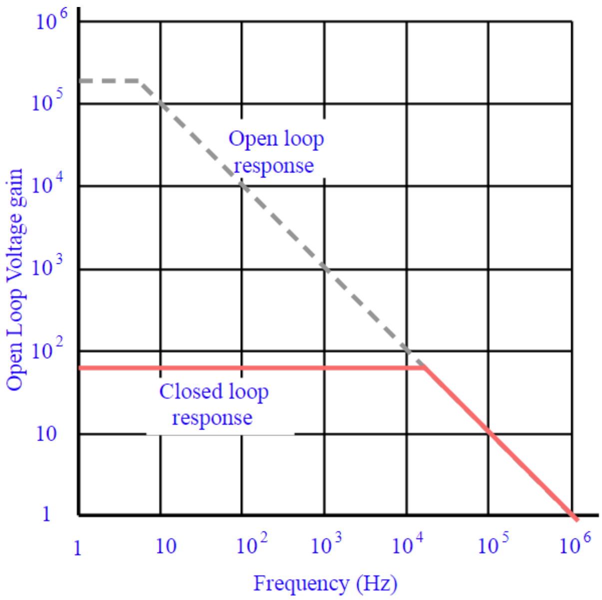


Figure 11.2: Gain-bandwidth product for the 741 op-amp (from www.electronics-notes.com).

*positive feedback* – which is disastrous. The enhanced signal at the amp input will just generate a bigger inverted output which just makes the problem worse, and soon the output will go to the supply rails. In practice, this usually results in the amp oscillating between $\pm V_{ss}$. Mathematically, for negative feedback we already saw in Section 9 that:

$$G = \frac{A}{1+\beta A}\,,$$

where we thought of $A$ as a constant; however, our understanding of real op-amps is that at high frequency $A$ appears to be negative (due to the phase shift).

*Stability criterion.* We must avoid at all costs the situation where the denominator goes to zero, i.e.

$$\beta A = -1\,,$$

otherwise $G \to \infty$. This is unstable, as well as being of no practical value (unless you want to build an oscillator!).

*Gain and phase margins.* The designer of the op-amp does not know what value of $\beta$ you will use. Hence, most amps have a *built-in gain roll-off* to ensure that $A < 1$ for $\phi = -180°$. This is the reason for the rather sudden gain reduction with frequency: to ensure stability with respect to high-frequency inputs. In practice, any such an *internally compensated op-amp* will be designed to ensure $A \ll 1$ way before $\phi = -180°$, just to be on the safe side – see gray curve in Figure 11.1. So, we can define the *gain margin* as the gain at $\phi = -180°$, while the *phase margin* is how far we get from $\phi = -180°$ when $A = 1$. These are best visualised graphically – see Bode plot in Figure 11.3. Typical desirable margins are 6 dB and 45° (although both >0 is in theory sufficient).

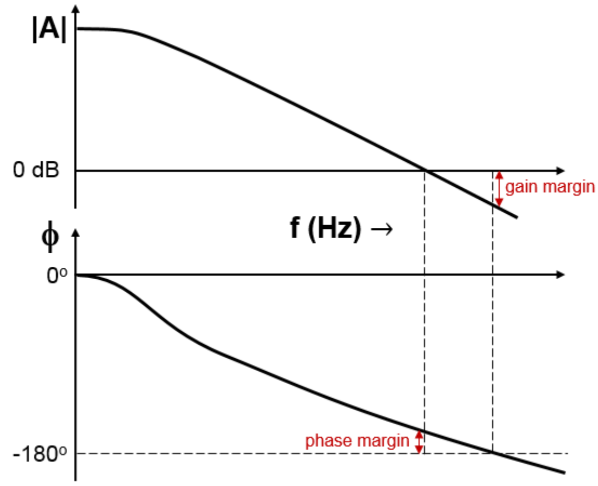


Figure 11.3: Gain and phase margins for stable operation of a negative feedback amplifier.

### 11.4. Final remarks

These considerations of stability under feedback apply to all situations where feedback is applied to a system or process, and we will encounter this again in a more mathematical and rigorous form later in the course. Since any feedback system has the potential to become unstable or oscillate, the question of stability becomes central to control theory. For feedback amplifiers we can assess the degree of stability we have through the stability criterion and the phase/gain margins. Note that many op-amps are sold without internal compensation; it is left to the user to ensure stability. For example the 748 is the *uncompensated* version of the 741 – shown by the grey line in Figure 11.1. With compensation we sacrifice gain for stability, so if we know what we are doing we can safely use the un-compensated device for greater performance at lower frequencies.

## 12. Analogue to digital conversion

We looked at sampling and digitisation of analogue signals in Section 5, but only now possess the required understanding of amplifiers to examine some concrete circuit implementations. These are to be understood rather than memorised. There are many implementations of these circuits, we look at one (the simplest) example of each.

### 12.1. Sample-and-hold circuit

We saw that converting a continuous signal into a digital form was a two-stage process: sampling and digitisation. Since the input signal can vary faster than our sampling speed, we need a circuit to hold the voltage while we digitise each sample – this is the job of the *sample & hold* circuit, such as the one shown in Figure 12.1.

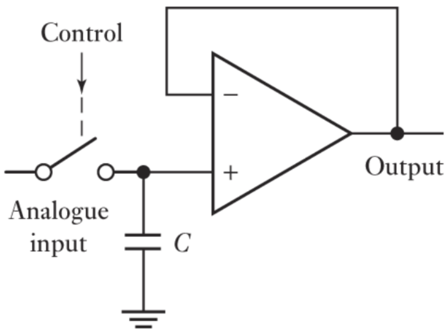


Figure 12.1: Concept for the sample & hold circuit.

The control signal comes from a 'sampling clock', here represented by opening the switch on the rising edge. The output will be held at the value seen by the input the instant the switch is opened, and it will remain there until we have measured it. The circuit is essentially a "switched" voltage buffer with a small capacitance $C$.

When the switch is closed then $V_+$ tracks the input whatever this is doing, and the output is equal to $V_+$. When the switch opens, the buffer's high input-resistance ensures that the output remains equal to $V_+$. We still need to figure out how to implement the switches. This we can be done with a pair of field effect transistors (FETs), which are readily available in integrated-circuit design (not covered in this course).

### 12.2. Analogue-to-digital converter

Now we are ready to convert the voltage to a number, using an *analogue-to-digital converter* (ADC or A/D). There are many IC implementations which are relatively inexpensive (a few pounds).

The simplest circuit is the *flash ADC* exemplified in Figure 12.2, where each comparator compares the input voltage with successively smaller reference voltages generated across the 'resistor ladder' (essentially a voltage divider with equal resistances). The conversion is near-instantaneous (hence the name), but clearly we would need

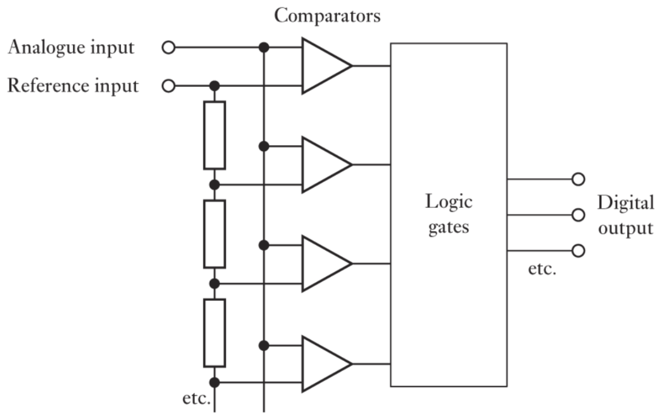


Figure 12.2: A 4-bit flash ADC.

pretty much one comparator for each digital value (e.g. $2^8-1 = 255$ comparators for an 8-bit device...). Another disadvantage is that all resistors need to be pretty equal or the device will be slightly non-linear. Having said that, this is still a popular design for low-resolution (e.g. 8-bit) ADCs, especially when speed is the key consideration. Other designs exist, such as the "successive approximation" ADC, the "sigma-delta" design, the "dual-slope" ADC, etc.

### 12.3. Digital-to-analogue converter

Frequently we need to convert digital signals back to the real world – and so we need a *digital-to-analogue converter* (DAC or D/A). For example, we may want to drive an output transducer such as a microphone. Some ADC technologies (e.g. successive approximation) also rely on a DAC to generate the intermediate stages of the ADC conversion.

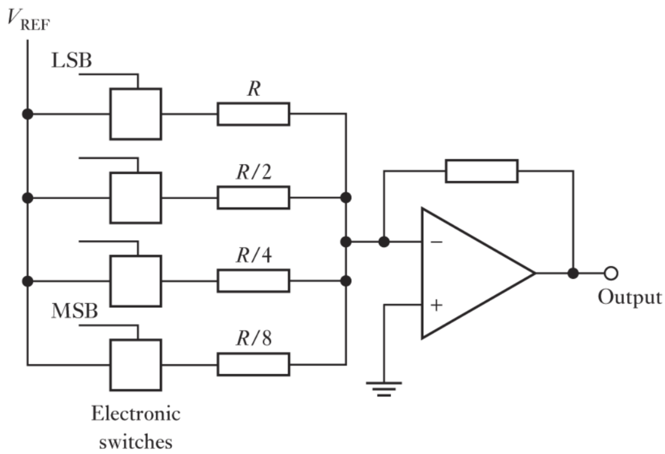


Figure 12.3: A 4-bit DAC.

A simple DAC circuit is that shown in Figure 12.3, at the core of which is a summing amplifier like the ones discussed in Section 9. Here we weight the 'bits' according to their significant-position in the binary scheme. Note the negative output: we could follow this with an inverting amplifier. This circuit has some advantages and disadvantages which are similar to those of the flash ADC design: it delivers near-instant conversion to an analogue voltage, but here, too, the resistors must be very accurate (exact power-of-2 ratio) or we will get non-linearity. As with ADC, many types exist and we usually buy an integrated circuit for the job.

As a final comment, these are unipolar designs – they operate with input/output voltages in the range $0 - V$. For bipolar input/output voltages in the range $\pm V$ these designs can easily be extended to allow bipolar signals – we just have to be careful how negative numbers are encoded in our binary scheme (we will not go into this in detail).

## 13. The phase sensitive detector

Sometimes we need to generate a *rectified signal*, i.e. a signal where the current only ever flows in one sense. Typically, we will look at the voltage of the *rectified* signal and we find that any of the +ve-going parts of the waveform are preserved and any −ve going parts of the signal have been reflected about the $V = 0$ axis. Figure 13.1 shows this for a sinusoid (lower trace), along with the rectified version (upper trace); mathematically, this is $|\sin(\omega t)|$.

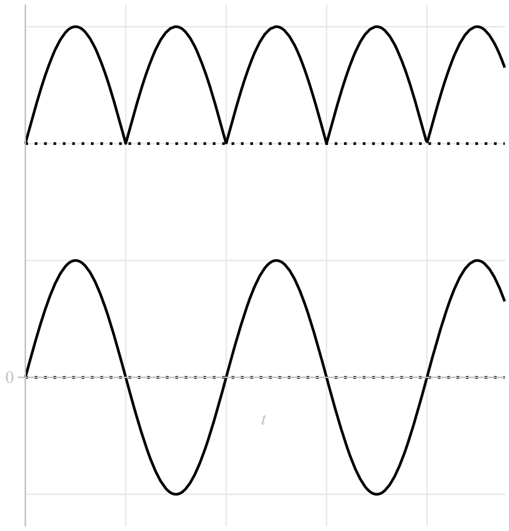


Figure 13.1: Sinusoid (lower trace) and rectified sinusoid (upper trace). In both cases the zero-level is shown by the dotted line.

### 13.1. Diode rectifier

Since the diode only conducts current in one direction we can make a simple *full-wave rectifier* (which produces an output as per Figure 13.1) using a simple circuit of four diodes, as shown in Figure 13.2.[19]

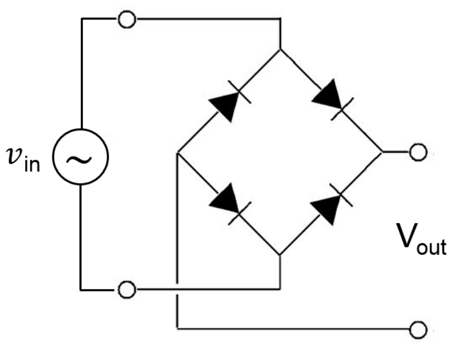


Figure 13.2: Diode bridge full-wave rectifier.

[19] See Horowitz & Hill section 1.26.

**Example: Power Supply**
We might want to build a power supply to charge-up a battery-operated device. This means converting 240 V AC mains to 5 V DC. This is done in three steps. Firstly, a transformer steps the voltage down from 240 V AC to 5 V AC. Secondly, a rectifier gives us a signal per Figure 13.1. Thirdly, the output is *smoothed* using a capacitor. The output is more-or-less a constant voltage, with some *ripple* (see Horowitz & Hill section 1.27); here, we are using the capacitor as a charge-storage device, essentially an integrator.

### 13.2. Synchronous rectifier

There are problems with diode rectifiers; for example, we get a voltage drop across each diode of about 0.6 V, which means that if we are trying to rectify small AC voltages then we can get a significant reduction in the output amplitude.[20] As previously mentioned, op-amps provide a simple, cheap and effective alternative to many traditional techniques, and rectification is no exception. The principle can be seen by looking again at Figure 13.1: the input waveform just needs to be inverted when it swings −ve. That is to say, we need to run it through an amplifier with gain $G = 1$ for +ve voltages and $G = -1$ for −ve voltages (see Figure 13.3). The input signal needs to be multiplied by a square wave at the same frequency (middle trace) to achieve the rectified signal (bottom trace). Op-amps can be used to multiply by a constant gain, so the technique here is to use the square wave to switch the gain between $\pm 1$. A schematic arrangement for achieving this is given in Figure 13.4. We will not cover the actual circuit in these notes (you can see this in Horowitz & Hill figure 15.37) – it is the principle that is important here: that one can use a reference ancillary waveform at the same frequency as our signal to do some useful processing.

In Figure 13.4 the square wave is called the *reference signal* (we shall see why later) and it is used to control a switch, which flicks the signal from inverted to non-inverted and back again each time the reference signal changes sign. We would describe this as an *edge-triggered switch*, as the transition happens on each rising or falling-edge of the reference signal. This kind of arrangement is quite common, for example a mechanical relay would be fine, though commonly there are electronic switches which will do the job much faster. We will take it as read that the switch is a ready-made component and not be concerned any further with it. The main point to take from this is that *if the reference and input are of the same frequency (and zero phase difference) then the output is a rectified version of the input.*

This requirement that the two inputs be at the same frequency seems quite difficult. However, in many real-life

[20] Figure 13.1 shows idealised rectification; however, if you look in detail at Horowitz & Hill figure 1.71 you will see that they have taken the small diode voltage drop into account.

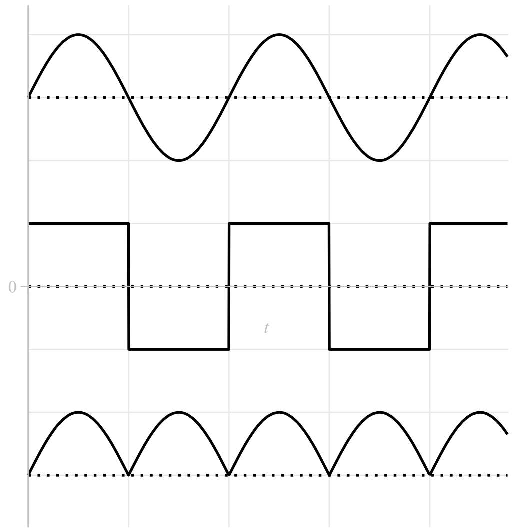

Figure 13.3: Sinusoid multiplied by square wave of same frequency.

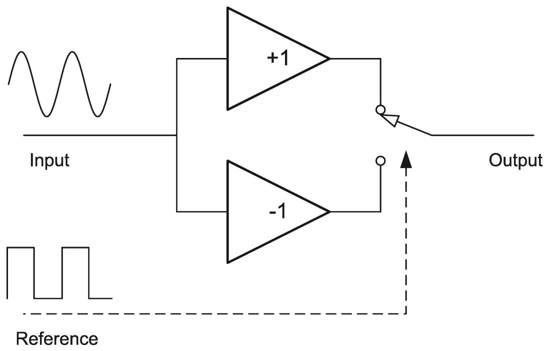


Figure 13.4: Synchronous rectifier.

applications the two signals are derived from a common source so you know they are going to be the same. The fact that this sort of rectifier takes two inputs leads to an application called the *phase detector*, which is a key technique in instrumentation. The phase detector is a synchronous rectifier with a low-pass filter at the output. The importance of the phase detector will be seen when we consider what happens if the two input signals are either out-of-phase, or at slightly different frequencies. First, we need to take a look at the function of the low-pass filter.

*13.3. Averaging circuit*

We will use a low-pass filter to perform an *average* of the rectified signal – see, e.g. the simple RC circuit in Figure 7.1. We have seen how, for repetitive input signals of high frequency (compared to the cut-off, i.e. $\omega \gg 1/RC$), the low-pass filter *approximates* to an integrator (Table 7.1). If we integrate over one time constant ($\tau = RC$) then

$$V_{out} = \frac{1}{\tau}\int_0^\tau V_{in}\,\mathrm{d}t = \langle V_{in}\rangle\,.$$

This makes sense from a physical point of view.[21] When the input voltage is greater than the voltage on the capacitor, the capacitor is charging up; when the input is lower then it is discharging. This charge/discharge cycle will be seen as ripple on the capacitor voltage, but for high frequency inputs ($\omega \gg 1/RC$) the ripple is very small and the capacitor voltage will approximate the average input, i.e. $V_{out} \approx \langle V_{in}\rangle$.

*13.4. Phase detection*

To quote Horowitz and Hill: *"This is a method of considerable subtlety"*, which is their typically understated way of introducing a rather complicated concept. Because of its importance, though, we will go through the topic in some detail. The *phase detector* simply consists of *a synchronous rectifier with a low-pass filter to average the output.* The importance of the phase detector will be seen when we consider what happens if the two input signals are either out-of-phase, or have different frequencies. We will investigate these in turn.

*13.5. Behaviour with a phase difference*

This is best understood graphically, with the help of Figure 13.5. In all the cases presented here, the two input signals have *exactly the same frequency.* In the first panel, there is zero phase difference and we get a constant positive output which turns out to be $2/\pi$ times the amplitude of the input signal. Inverting the input signal ($\phi = \pi$) gives us a negative output of the same size. For the $\phi = \pi/2$ case we get a zero output and for the "in-between" case of $\phi = \pi/4$ we get an in-between output. Clearly then, the phase detector is highly sensitive to differences in phase between the two signals. For an input signal $V_{in} = A\sin(\omega t + \phi)$ the output of the phase detector is

$$V_{out} = \frac{2A}{\pi}\cos\phi\,. \qquad (13.1)$$

This can be checked by performing the rectification/integration directly as

$$V_{out} = \int_0^{\pi/\omega} V_{in}\mathrm{d}t - \int_{\pi/\omega}^{2\pi/\omega} V_{in}\mathrm{d}t\,.$$

In summary then, when the two inputs have the same frequency, the phase detector gives a voltage output which is a function of the phase difference between the two. This in itself is very important and has applications which we will not go into here, because the most useful function of the phase detector becomes apparent when we consider how the circuit behaves as we vary the *relative frequency* of the two inputs.

[21]Consider the Fourier components of the input signal and recall that the filter only lets through the very low frequency components. The rectified signal has a DC component (non-zero value of $A_0$, cf. Section 5) which is the long-term average of the signal.

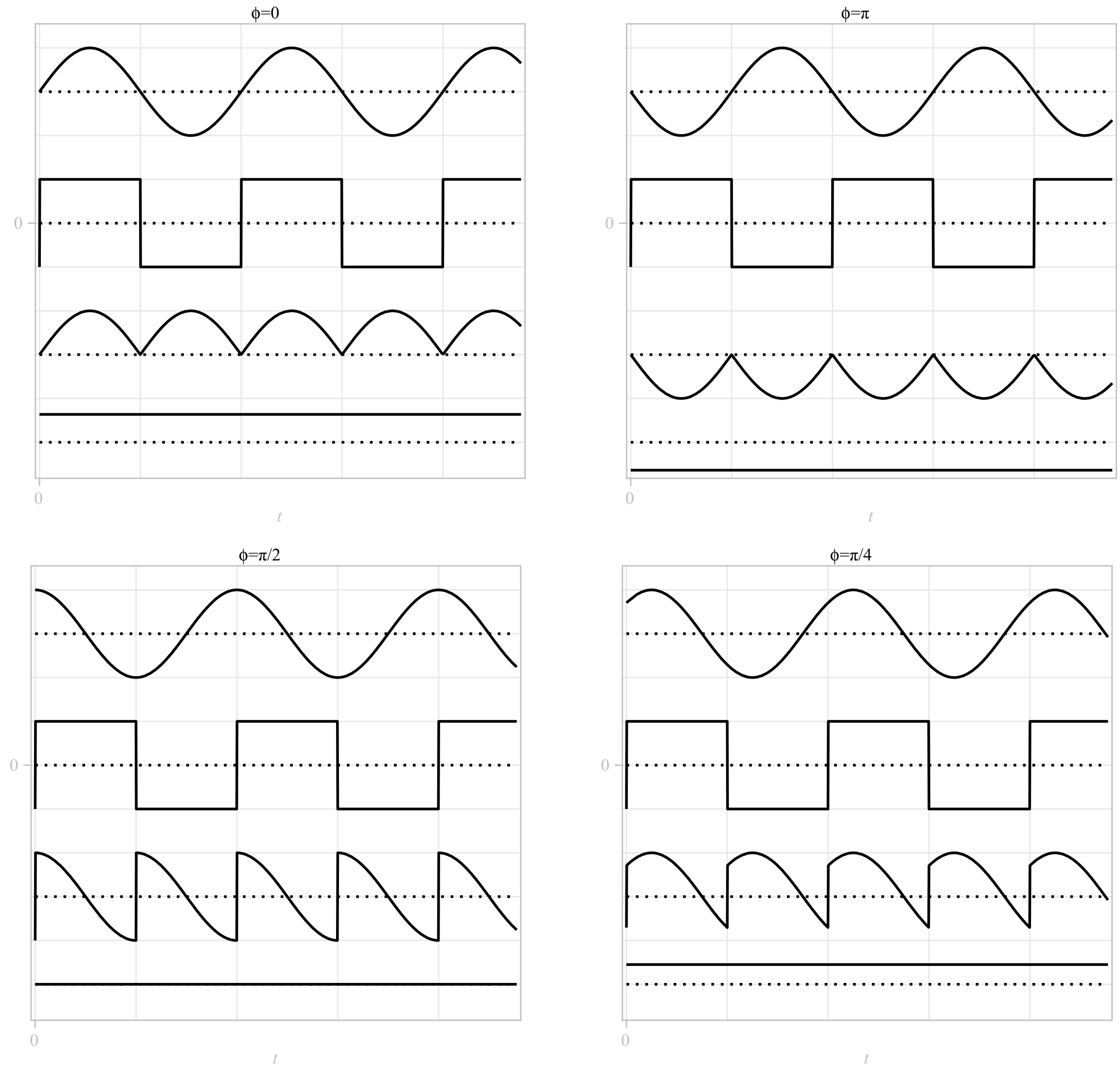


Figure 13.5: Phase detector behaviour with four different values of phase difference $\phi$ between input and reference signal. Note that in all these figures, the waveforms from top to bottom are: input signal, reference signal, rectified signal and averaged (output) signal.

*13.6. Behaviour with a frequency difference*

In this situation we define two different frequencies $\omega_{input}$ for the input signal and $\omega_{ref}$ for the reference. This is illustrated with the four plots in Figure 13.6. For $\omega_{input} = 2\omega_{ref}$ we can see that the output will always be zero. If we choose a ratio such as $\omega_{input} = 1.3\omega_{ref}$ then as long as we average over a sufficiently large number of cycles we always get zero output.[22] We need to reduce the ratio to $\omega_{input} = 1.01\omega_{ref}$ before we start to see any kind of output from the detector. It looks like a constant output, but if we plot the signal over a longer time period (bottom-right panel) we can see that it is sinusoidally varying. In fact the output varies as $\cos(t\Delta\omega)$, where $\Delta\omega$ is the difference between the two input signals, *as long as* $\Delta\omega$ *is a low-enough frequency to pass through the filter.*

We can understand this behaviour by looking at the frequency content of the signals. If the reference frequency is $\omega$ and the input frequency is $\omega + \Delta\omega$,

$$
\begin{aligned}
V_{in} &= A\sin(\omega + \Delta\omega)t \\
V_{ref} &= \frac{4}{\pi}\left[\sin\omega t + \frac{\sin 3\omega t}{3} + \frac{\sin 5\omega t}{5}\right].
\end{aligned}
$$

Here we used a Fourier expansion to represent the square wave, and we just need the first three terms of the Fourier series to understand the behaviour. Multiplying the two together, we find the rectified signal (i.e. *before* the filter)

[22] For these plots, the averaging period is 10 cycles.

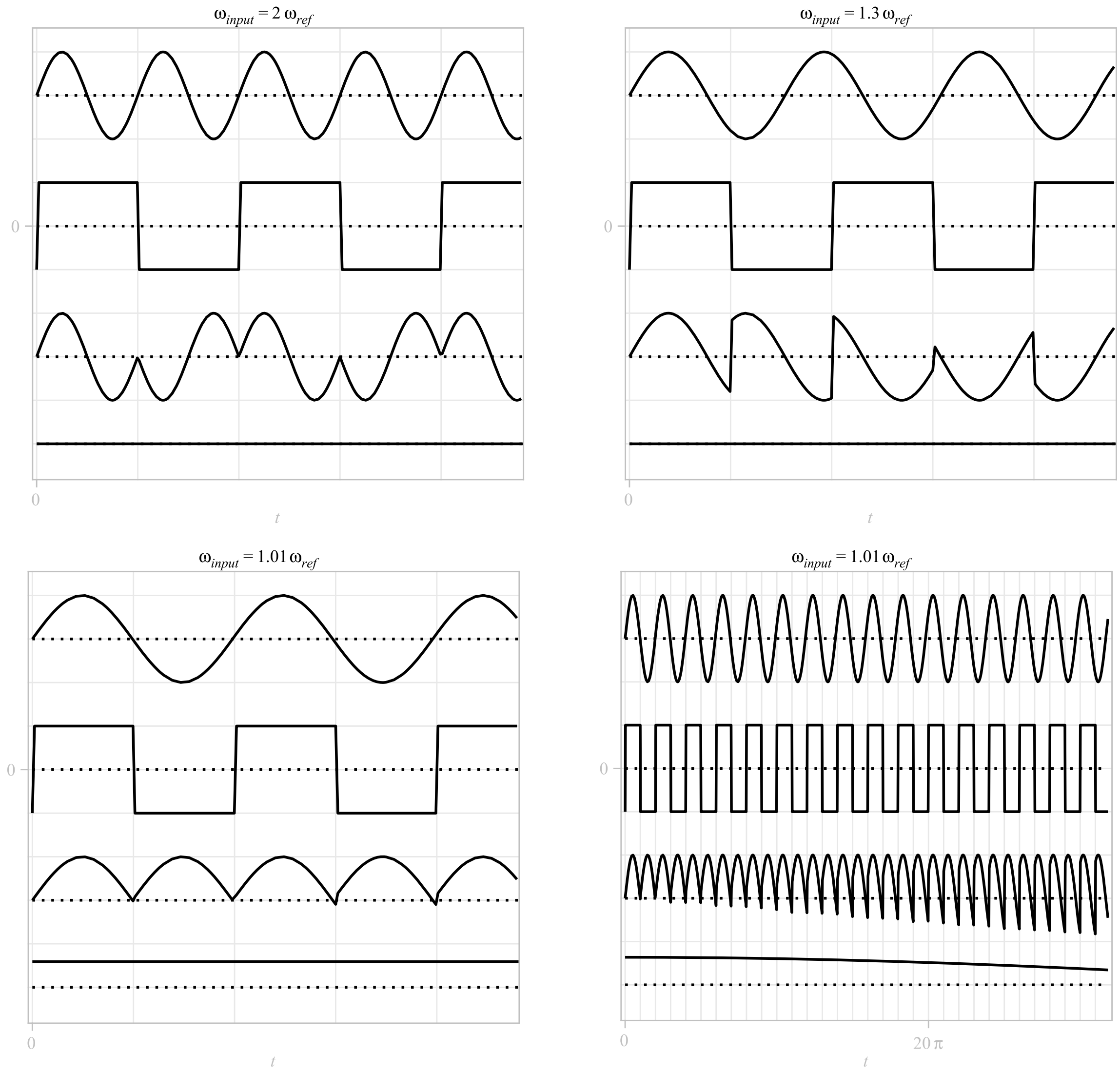


Figure 13.6: Phase detector behaviour where the input signal has a different frequency to the reference frequency. Waveform order in each plot as per Figure 13.5.

is given by[23]

$$
\begin{aligned}
V_{rect} &= \frac{2A}{\pi}[\cos\Delta\omega t - \cos(2\omega + \Delta\omega)t \qquad (13.2)\\
&\quad + \frac{1}{3}\cos(\Delta\omega - 2\omega)t - \frac{1}{3}\cos(4\omega + \Delta\omega)t\\
&\quad + \frac{1}{5}\cos(\Delta\omega - 4\omega)t - \frac{1}{5}\cos(6\omega + \Delta\omega)t]\,.
\end{aligned}
$$

We have implicitly assumed above that $\omega \gg \omega_c$, and so frequencies $\omega$ and above will *not* pass through the filter. But for $\Delta\omega \ll \omega_c$ the first term in equation (13.2) will pass through the filter un-attenuated – i.e. although the filter removes the fundamental frequency of our input signal plus any higher frequency terms, the *difference frequency* relative to the reference signal is slow enough and hence perfectly able to pass through. Thus, we can define the output as follows:[24]

$$
V_{out} = \begin{cases} \frac{2A}{\pi}\cos(t\Delta\omega) & \Delta\omega \ll \omega_c \\ 0 & \Delta\omega \gg \omega_c \end{cases}\,. \qquad (13.3)
$$

We will return to these phase-sensitive techniques in Section 21, the final section in these notes, after a discussion of ‘noise. The ability to narrow the measurement bandwidth to extremely small values is critical when faint signals of known frequency need to be extracted from a large noise, for example.

[23]We use $\sin A \sin B = 1/2[\cos(A-B) - \cos(A+B)]$: the product of two sinusoids is (co-)sinusoids at the difference and sum frequencies.

[24]Note that we can use a similar argument to demonstrate equation (13.1) too.

## 14. Linear Systems

We can characterise a physical system by its ability to accept an input parameter and produce an output in response. Our understanding of a sensor, or transducer, is a perfect example. We will now examine this relationship between input and output in more detail.

Systems can be linear or non-linear. A *linear system* is one for which a linear relation exists between cause and effect. Obvious examples include a spring, for which force is directly proportional to elongation, or a resistor, for which voltage and current are proportional through Ohm's law. A counter example might be a diode, which has a very non-linear relationship between voltage and current. However, for many systems such a casual inspection will not tell us whether it is linear or non-linear. To determine whether a system is linear or not we need to examine the mathematical formulation of the input/output relationship. The reason we are interested in this is because, for linear systems, we can use frequency-domain techniques (Fourier and Laplace) to understand and predict the behaviour of really very complex systems.

### *14.1. Linear time-invariant systems*

Any linear time-invariant system, device or component can ultimately be described by a differential equation of the form

$$a_0 x(t) + a_1 \frac{\mathrm{d}x(t)}{\mathrm{d}t} + a_2 \frac{\mathrm{d}^2 x(t)}{\mathrm{d}t^2} + \ldots + a_n \frac{\mathrm{d}^n x(t)}{\mathrm{d}^n t} = \\ = b_0\, y(t)\,, \tag{14.1}$$

where $a_n$ and $b_0$ are constants, $x(t)$ is the *output* of the system and $y(t)$ is the *input*. Systems which are described by such a differential equation are often termed *Linear Time Invariant* or *LTI* systems. The *time invariance* comes from the fact that the behaviour of the system (its differential equations) does not vary with time, that is to say the behaviour, under a given input, is the same today as it was yesterday, and it will be the same tomorrow.

The input $y(t)$ is also known as the *forcing function* and may be any arbitrary function, for example, it may be a constant (recall the static response), it may be a sinusoid at a given frequency (recall the frequency response), or it may be a transient function such as the unit step or the delta function. LTI systems have two important properties which we explore in turn.

*Frequency preservation.* If the input to an LTI system is a sinusoid then the output is also a sinusoid *of the same frequency*, but it may have its amplitude and phase modified (Figure 14.1). Note that this follows directly from equation (14.1). If the input $y(t)$ is a sinusoid, then $x(t)$ must also be a sinusoid at the same frequency – given that all the derivatives of $x$ are themselves sinusoids *at the same frequency.*

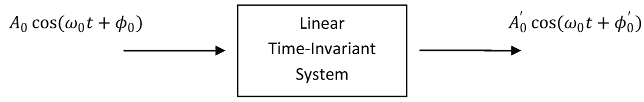


Figure 14.1: Frequency preservation of LTI systems.

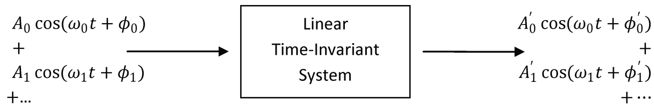


Figure 14.2: Superposition property of LTI systems.

*Superposition.* If the input to an LTI system is the sum of several sinusoids then the output of the system is the sum of the system's response to each individual sinusoid in isolation (Figure 14.2). Note that, for each frequency component, we can read-off the change in amplitude and phase from the Bode plot. A more sophisticated expression of this concept is given below.

### *14.2. Use of Fourier techniques*

The properties of superposition and frequency preservation allow us to apply useful Fourier techniques to real-world problems involving LTI systems. Any real-world signal can be Fourier-decomposed into a sum of sinusoids of suitable frequency, phase and amplitude, and we can follow the propagation of each sinusoid through an LTI system (each component will have its phase and amplitude modified) and then sum the output sinusoids to recover the response of the system to the original signal. We can see from these properties that *if we add several LTI systems together in series, then the resultant system is also LTI.* We will see how to apply these ideas later on in the course.

The most common systems we come across in instrumentation are quite well described by rather simplified versions of equation (14.1). It is worth briefly revising how simple zero-, first- and second-order systems respond to a simple forcing function such as a step $u(t)$ or an impulse $\delta(t)$. The aim of this section is to briefly run through some of the background mathematics, particularly for the second-order system. Note that the second-order system is the damped harmonic oscillator which you will already have covered a number of times before in other courses. What we typically find is that, whilst the differential equations describing isolated zero, first- and second-order systems are generally quite tractable mathematically, it can be extremely difficult to both formulate and solve a differential equation describing the behaviour of a more complex instrument composed of several distinct elements. However, as suggested above there are solutions to this based on Fourier techniques, which we will come on to later.

*14.3. Zero-order systems*

Zero-order systems are described by an equation of the form

$$a_0 x = b_0 y\,,$$

where $a_0$ and $b_0$ are constants, $y(t)$ is the input or forcing function applied to the system, and $x(t)$ its output. They thus represent the most trivial of LTI system. This type of system responds immediately to a changing input, and has no possibility for delay, oscillation, or dependency on past behaviour. Zero-order systems do not care what has happened to them in the past, only what the state of the input is now. This type of behaviour would be ideal for a sensor, we would get infinitely fast time response, and no problems with oscillation, overshoot, settling times and so forth. Unfortunately, such systems are extremely rare in real life instruments, but we will look briefly at them to provide us with some scene-setting examples. A good example of a zero-order system is a variable resistor (Figure 14.3). The input to the system $y(t)$ is the position of a

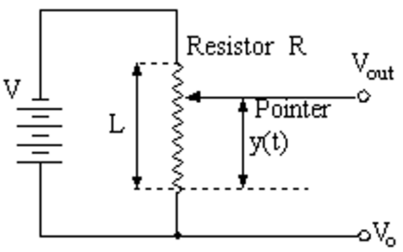


Figure 14.3: Variable resistor (or potential divider) used to measure displacement.

pointer along the resistor, and the output $x(t)$ is a voltage $V_{out}$ that appears at the end of the pointer arm. For the zero-order case $V_{out}$ exactly follows $y(t)$, and at any time,

$$x(t) = \frac{V_0}{L} y(t)\,,$$

where $V_0$ is the voltage across the full length $L$ of the resistor. Any change in $y(t)$ gives an immediate corresponding change in $x$. As soon as $y$ stops changing, so does $x$. This, however, is unphysical for a real-world system, and in our example effects such as the slight flexing of a mechanical pointing arm as it moves or stray inductive or capacitive effects would stop $x$ exactly tracking $y$. More generally, we will have to formulate and solve a differential equation to find the time response of a system to any given input. However, under certain conditions we can approximate the behaviour of some components as zero order, for example a resistor ($v = iR$) or an idealised amplifier (the output is simply an amplified version of the input).

The zero-order system is characterised by the *gain*, or more generally the *static sensitivity* $K_s$; in our example,

$$K_s = \frac{b_0}{a_0} = \frac{V_0}{L} \quad [\mathrm{V/m}]\,.$$

*14.4. First-order systems*

First-order systems are described by a differential equation of the form

$$a_1 \frac{\mathrm{d}x}{\mathrm{d}t} + a_0 x = b_0 y\,. \tag{14.2}$$

The rate of change of the output $x(t)$ is thus proportional to the difference between the system's current state (at a given point in time) and the value of the forcing function. To solve a differential equation of this type we generally try a test solution of the form $x(t) = Ae^{st} + C$, where $A$, $s$ and $C$ are constants. We then apply boundary conditions to find the values of these constants. An example of this method, as applied to the RC circuit, was given in Section 7.

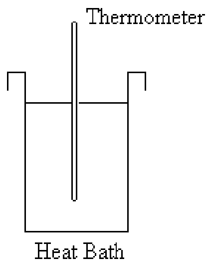


Figure 14.4: Thermometer in heat bath.

By way of example, consider the thermometer in Figure 14.4, which is initially at temperature $y = \theta_0$ and plunged at time $t = 0$ into a heat-bath of temperature $\theta_R$. The rate of heat flow into the thermometer is proportional to the *difference* between the temperature of the thermometer and the heat reservoir $\theta_R$. Our boundary conditions are that at $t = 0$ the thermometer is at its starting temperature $\theta_0$, and for $t \to \infty$ it reaches the same temperature as the heat reservoir. The solution is

$$x(t) = K_s \left[\theta_0 + (\theta_R - \theta_0)(1 - e^{-t/\tau})\right]\,.$$

If $x(t)$ is the length of the mercury column in mm and the $\theta$ parameters are temperatures in °C, then $K_s$ is the static sensitivity in mm/°C. In terms of the original parameters of equation (14.2), we again find that

$$K_s = \frac{b_0}{a_0}\,.$$

Note that $\tau$ is a *time constant* dependent on the conductivity and thermal mass of the thermometer; $\tau$ is the time taken for the system to reach 63.2% of its steady state response, after the application of a step input. This type of time response is very common in instrumentation, see the handout on the AD950 temperature sensor for an example. We often refer to the sort of plot shown in Figure 14.5 as an "RC curve", as it follows the form of the voltage across a capacitor as we charge it up from a DC source through a resistor (see Section 7). The system initially responds rapidly to a step input, as the difference between its starting point and the steady state value is large. As time progresses, the difference between steady state and actual value becomes smaller, and the rate of change falls off correspondingly. After a sufficiently long time the system reaches its steady state value, though for a real sensor, the time taken to reach 95% of its steady state value, or just the time constant $\tau$ will be given on the datasheet.

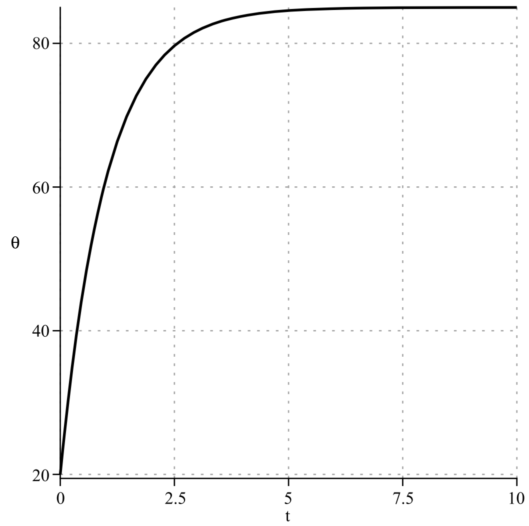


Figure 14.5: Room temperature thermometer plunged into a nice hot cup of tea.

Expressing the time constant $\tau$ in terms of the parameters in equation (14.2), we find

$$\tau = \frac{a_1}{a_0}\,.$$

Referring back to Sections 1 and 2, recall that when we need to calibrate a first-order instrument such as a thermometer we need to hold the input constant and wait a sufficient number of time-constants for the output to settle before taking a reading. As a quick exercise: how many time constants must elapse before the output reaches 99.5% of the steady-state? After this time, we can assume $\mathrm{d}x/\mathrm{d}t = 0$ and so the output:input ratio gives us $K_s$.

### *14.5. Second-order systems*

Second-order systems are described by a differential equation of the form

$$a_2\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + a_1\frac{\mathrm{d}x}{\mathrm{d}t} + a_0x = b_0y\,. \tag{14.3}$$

This is essentially the defining equation of a damped simple harmonic oscillator. A good example of this type of system in instrumentation is a spring balance (Figure 14.6, although there are many non-mechanical physical analogues). Second-order systems can show the type of delayed time-response to a step input that we saw in first-order systems, but they can also oscillate after a step $u(t)$ or impulse $\delta(t)$ input. We can begin by summing all the forces that act on the mass $m$. There is a gravitational force $mg$, a force due to the extension of the spring $kx$, a velocity dependent damping term $c\,\mathrm{d}x/\mathrm{d}t$ and an acceleration term $m\,\mathrm{d}^2x/\mathrm{d}t^2$. Summing all the forces on the mass gives

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + c\frac{\mathrm{d}x}{\mathrm{d}t} + kx = mg\,. \tag{14.4}$$

There are several limiting cases of the behaviour of this system that are easy to find. For the steady state case there are no changes as a function of time and $\mathrm{d}^2x/\mathrm{d}t^2 = \mathrm{d}x/\mathrm{d}t = 0$, the response is the same as a zero-order system. The static response of the system is thus $mg = kx$; therefore, the static sensitivity is

$$K_s = \frac{b_0}{a_0} = \frac{g}{k}\,.$$

For the spring balance example, we intuitively know that we have to wait for any oscillations to be damped out before we take a reading, and therefore in the steady-state case we find $x = mK_s$ where $x$ is the extension of the spring.[25]

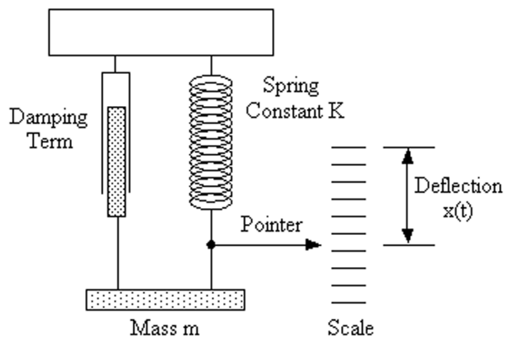


Figure 14.6: Spring balance.

#### *14.5.1. Solution to the second-order differential equation*

To understand the behaviour of the second order system it is convenient to re-write equation (14.5) as

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2\xi\omega_0\frac{\mathrm{d}x}{\mathrm{d}t} + \omega_0^2x = g\,, \tag{14.5}$$

where $\omega_0$ is the *natural frequency* of the system and $\xi$ is the *damping ratio*:

$$\begin{aligned}\omega_0^2 &= \frac{a_0}{a_2} = \frac{k}{m}\\ \xi &= \frac{a_1}{2\sqrt{a_0a_2}} = \frac{c}{2\sqrt{mk}}\,.\end{aligned}$$

[25] Note that for the typical balance this linear displacement $x$ is translated into an angle on a dial by some arrangement of gears.

The exact form of the solution[26] to equation (14.5) depends on the numerical value of the damping ratio; however, in general

- For $\xi = 0$ equation (14.5) becomes that for a simple harmonic oscillator, i.e. the solution is sinusoidal;
- For $\xi < 1$ the solution is an exponentially-decaying sinusoid;
- For $\xi > 1$ the solution is exponential-decay with no oscillation (similar to the first order system).

*14.5.2. Behaviour of the damped harmonic oscillator*

Returning to our spring balance example, we find that for the zero damping case ($c = 0$, thus $\xi = 0$) any disturbance of the system will result in free running oscillations at a frequency $\omega_0$. This type of free running behaviour is almost always a bad thing for an instrument. After all, we want to use it to measure the forcing function. We will, however, see that allowing a little oscillation to occur can actually be a good thing for an instrument, improving its time response at the cost of some small and rapidly damped oscillations after an abrupt change to the input.

The behaviour is summarised in Figure 14.7, which gives the response of the balance to a step change input, i.e. suddenly placing the mass $m$ on the balance at time $t = 0$. The plots are normalised to show a unit-response and give 6 different values of $\xi =$ 0, 0.01, 0.1, 0.5, 1 and 3. For $\xi = 0$ we have free running at $\omega_0$. This would be disastrous in a real balance since we want to use the *static response* to determine the mass, i.e. we want the balance to settle. For $\xi = 0.01$ and 0.1 the system is *underdamped*. At $\xi = 0.6$ is the kind of behaviour we might expect from a real spring balance as it is probably the optimum: some oscillation but rapidly reaching equilibrium. A car's suspension has this sort of behaviour. For $\xi = 1$ the oscillator is *critically damped* – there is no oscillation but the system takes quite a time to reach the equilibrium. The spring closers for doors might be designed for this sort of behaviour. $\xi = 3$ is the *over-damped* case; the behaviour is reminiscent of the first-order system.

[26]The general method consists of finding the solution for both forced and unforced cases and adding them together (this works because the system is linear). This method of solution is not part of the course; however, you can follow it in Poularikas and Seely.

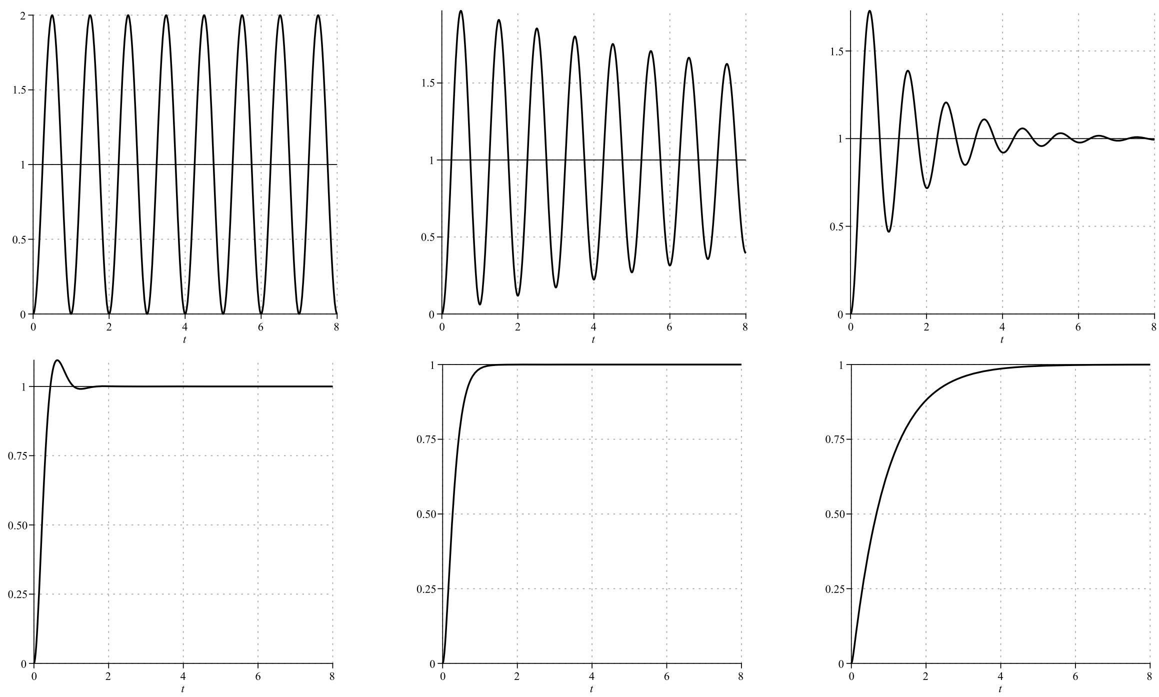


Figure 14.7: Response of the spring balance to a unit step-change input for $\xi =$ 0, 0.01, 0.1, 0.6, 1 and 3. The natural frequency is 1 Hz.

## 15. Laplace transform and the transfer function

The Fourier and Laplace transforms look rather similar at first glance. Both involve an integral of a function multiplied by an exponential term over time. However, the Laplace transform is *single-sided*: it only requires integration over values of time greater than zero, and this is an important difference.

*Fourier transform.* The integral definition is:

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-j\omega t}\mathrm{d}t\,. \tag{15.1}$$

We have transformed a function of time $f(t)$ into a function $F(\omega)$ in the frequency domain. Note that the integral is symmetric in time and involves both positive and negative time.

*Laplace transform.* The integral definition is:

$$\mathcal{L}\{f(t)\} = F(s) = \int_{0}^{\infty} f(t)\, e^{-st}\mathrm{d}t\,. \tag{15.2}$$

Here we say that we have transformed a function of time $f(t)$ into a function $F(s)$ in the $s$-domain, where $s$ is a *complex* variable:

$$s = \sigma + j\omega\,. \tag{15.3}$$

The integral is *single-sided*: it only involves values of time greater than zero. It is important to note that the lower integration limit really means '$0^-$', which is our way of indicating that any discontinuities or impulses at $t = 0$ (e.g. a unit step or a delta function) should be captured by the algebra in the integral.[27] Note also that we can recover a Fourier transform by making the substitution $s \to j\omega$; this is a consequence of the fact that the Fourier transform is a particular case of the Laplace transform, which is the more general treatment.

*Physical interpretation of the transforms.* In general, when we Fourier-transform a real physical observable, the resulting function also represents a physical observable. For example, when we Fourier-transform a pulse shape $f(t)$, a function of time, we find the spectrum function $F(\omega)$, which we can interpret as the amplitude and phase of all the frequencies in the pulse. In contrast, the Laplace transform does not represent any kind of useful physical observable (although $s$ has units of rad/s). We will use it simply as a tool for solving differential equations. There is one exception to this: $s$ can be used to test for stability in a system. We looked at this briefly in the context of feedback amplifiers, and we will cover this more thoroughly again later in the course. Initially, though, we will see how to use the Laplace transform as a highly effective tool for solving differential equations.

[27] This is an area of significant confusion in the literature; a good discussion can be found in this note.

### *15.1. Fourier analysis of systems*

The use of the *Fourier transform* in systems analysis requires the decomposition of an input ("forcing") function $y(t)$ into its spectrum function $Y(\omega)$.[28] This requires an integral over all time. This means that, in practice, we need to know the value of the forcing function over the entire past history of the system and for all future time. If we do know this, then, given that we know the transfer function of the system $G(\omega)$, we can obtain the spectrum of the system output $X(\omega)$ from the relation

$$G(\omega) = \frac{X(\omega)}{Y(\omega)}\,.$$

We can then obtain the system response in the time domain by inverse-transforming:

$$x(t) = \mathcal{F}^{-1}\{X(\omega)\}\,.$$

The *transfer function* $G(\omega)$ contains all the information we need to determine the output. Since it is, in general, a complex function it is more useful to split it into an amplitude $|G(\omega)|$ (called simply the *frequency spectrum*) and a phase term. Although in these lecture notes we focus more on the equivalent transfer function for Laplace transform (as these allow us to solve transient problems), in lab you will work with the Fourier frequency spectrum which is what we tend to measure for periodic signals.

The Fourier approach works perfectly well for systems where the input signal is a continuous repetitive input such as a sinusoid, which for all practical purposes we can take to be infinite in time. This is not just theoretical: in the lab sessions we will see how we can model the transfer functions of the electret microphone, the piezo-sounder, the filter and an amplifier in terms of their frequency response $G(\omega)$. This works because for practical applications we can assume that the input signal is infinite. We know that it is not, but it has existed for long enough that any transient or "start-up" response of the system has long since ceased to be apparent in the output of the system. That is to say, the system output is the same as it would have been had the input existed for all time.

### *15.2. Advantages of the Laplace transform*

When we analyse a complex device in instrumentation, we break the device down into a set of 'blocks', each of which can be understood as a linear system and which has its input/output relationship described by a differential equation of the form described in Section 14. When our device is composed of lots of such blocks connected together, and subject to an arbitrary input, we will find that the overall input/output relationship requires us to solve a

[28] In these notes and in some textbooks the input to a system is labelled $y(t)$ and the output is $x(t)$, which is a little confusing; in other sources this is reversed, so you must be careful... the transfer function is the output divided by the input, essentially a 'gain'.

series of linked differential equations of some considerable complexity. *The usefulness of the Laplace transform lies in its ability to change a differential equation (DE) in the time domain into an algebraic expression in the s-domain.* This method is generally useful when we wish to find the time-domain behaviour of a system subject to an arbitrary (usually transient) input signal. The general method is as follows:

1. Understand the system behaviour in terms of its governing differential equation, its input (forcing function) and any initial conditions;
2. Transform the DE into the $s$-domain;
3. Re-arrange algebraically in order to separate the desired output variable;
4. Transform back into the time-domain to discover the resulting behaviour.

For example, we can use the Laplace transform to examine how a system behaves after the application of an impulse input $\delta(t)$ or a step $u(t)$. Recall the thermometer and spring balance examples of Section 14: the Laplace method allows us to determine how the instrument will respond to the input forcing-function. We could of course use general methods for solving ODEs which you have come across before; however, the Laplace method is significantly more powerful.

### *15.3. Transient analysis*

In systems analysis we often want to understand the response of the system to a transient applied at the input at a given time (usually taken to be $t = 0$ for simplicity). This is known as *transient analysis* and is highly applicable for instrumentation applications, since generally we know the state of our instrument now (i.e. the initial conditions), we know the governing differential equations, and we know what the input is. We do not generally know (or care) about the past history of the system. The Laplace transform is single-sided, so only requires integration over positive (future) time, and is generally rather effective at solving these kind of transient problems. For example, we may have a circuit which is either *initially relaxed* (has no charge stored or currents flowing) or it may have *initial conditions* (due to charge stored on capacitors or current flowing in inductors). If at time $t = 0$ we throw a switch to apply some new input signal (forcing function) or even to change the configuration of the circuit by adding capacitance or inductance, then the Laplace transform allows us to solve these sorts of problems. We can evaluate how the system will evolve for all time $t > 0$. This is a physically realistic situation: we know the state of the system now and we apply an input now; we want to observe how the system will evolve. Note that many of these sort of problems can be solved with the Fourier transform too; however, the Laplace method is generally easier and more flexible.

### *15.4. Properties*

The most important properties of the Laplace transform are its linearity and behaviour under differentiation and integration. These properties are described below.

*Linearity.* Like the Fourier transform, the Laplace transform is a linear operation:

$$\mathcal{L}\{k_1 f_1(t) + k_2 f_2(t)\} = k_1 F_1(s) + k_2 F_2(s)\,. \tag{15.4}$$

*First time derivative.*

$$\mathcal{L}\left\{\frac{\mathrm{d}}{\mathrm{d}t} f(t)\right\} = sF(s) - f(0^+)\,, \tag{15.5}$$

where $f(0^+)$ is the initial condition of the system: it is the value of $f(t)$ as $t \to 0$ from positive $t$. Recall that we do not need to know (or care) about $f$ for $t < 0$.[29]

*Second time derivative.*

$$\mathcal{L}\left\{\frac{\mathrm{d}^2}{\mathrm{d}t^2} f(t)\right\} = s^2 F(s) - sf(0^+) - f^{(1)}(0^+)\,, \tag{15.6}$$

where $f^{(1)}(0^+)$ is the first derivative of $f(t)$ evaluated as $t \to 0^+$.

*Integral with zero initial conditions.*

$$\mathcal{L}\left\{\int_0^t f(\xi)\mathrm{d}\xi\right\} = \frac{F(s)}{s}\,. \tag{15.7}$$

In the presence of initial conditions $y(0^-){=}y_0$, we must add an extra term $y_0/s$ to the Laplace transform, to remain consistent with our use of $0^+$ in the derivative case.

*Time shifting.*

$$\mathcal{L}\{f(t-\lambda)u(t-\lambda)\} = e^{-s\lambda}\,F(s)\,. \tag{15.8}$$

*Frequency shifting.*

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a)\,. \tag{15.9}$$

[29] Initial conditions are defined in the literature in a variety of ways: $f(0^-)$, $f(0)$ and $f(0^+)$. If $f(t)$ is continuous at $t = 0$, they all coincide. If you derive the 'time derivative property' by integrating Eq. 15.2 by parts, you will appreciate that $0^-$ is the most sensible definition mathematically (e.g. try integrating the step function $u(t)$). The $0^+$ definition is common in electronics to highlight that we care about the circuit configuration after some component "turns on", but here we must be careful with how we treat discontinuities at the origin – we usually need to add additional currents or voltages to the circuit equations to capture those initial conditions!

*15.5. Initial conditions*

Perhaps the most delicate aspect of Laplace transforms is how to deal with initial-value problems (e.g. what happens when we close a switch) – due to the various definitions that can be found in the literature.

First, we need to remember that, from the laws of physics: *i*) the *voltage across a capacitor must be continuous*; a capacitor stores charge and, since current is the flow of charge, any finite current can only lead to a finite charge – and hence voltage; and *ii*) the *current through an inductor must be continuous*; an inductor stores energy in the magnetic flux and, by Faraday's law, an infinite voltage would be needed to support a step change in current. In both cases the key quantity remains finite also if we apply an impulse stimulus, since the delta function has a well-defined finite integral. These considerations usually allow us to write equations for the continuity of voltage or current across a discontinuity and hence relate those variables before and after $t = 0$.

If we use the transform definition in Eq. 15 with lower limit $0^-$ (sometimes called the "pre-switch" initial conditions), these will automatically be taken care of by the algebra, e.g. the voltage across an inductor in $s$-domain becomes $V(s) = sLI(s) - Li(0^-)$. In our course, we tend to use the $0^+$ convention, which means that that we need to *explicitly add an extra term* to represent the voltage or current initial conditions that existed before the switch was closed (see Problem Sheet 8). We can do this because of the linearity property: the response of the system should equal the sum of the response with no charges or currents, plus any preexisting charges or currents.

*15.6. Elementary Laplace transform pairs*

The forward transform is relatively straightforward to derive for many common functions. The inverse transform, by contrast, is given by

$$\mathcal{L}^{-1}\{F(s)\} = f(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} F(s)\, e^{st}\mathrm{d}s\,, \qquad (15.10)$$

and therefore requires integration in the complex plane. Don't try this at home! In order to perform most inverse transforms (and many forward transforms too), it is usually sufficient to use tables of elementary transform pairs[30] such as that given in Figure 15.2 at the end of this section.

*15.7. Transfer function*

Also with Laplace transforms, the *transfer function* is by definition the ratio of the output of a system to its input, where now both are $s$-domain representations – see Figure 15.1. So, if we have a system where the input is represented by some function $y(t)$ and the output is $x(t)$, then the transfer function is

$$G(s) = \frac{X(s)}{Y(s)} = \frac{\mathcal{L}\{x(t)\}}{\mathcal{L}\{y(t)\}}\,. \qquad (15.11)$$

There are some conditions attached to this definition:

1. The input and output signals $y(t)$ and $x(t)$ must be continuous functions of time;
2. The system in question must be linear time-invariant;
3. The system must be *initially relaxed* or in a quiescent state. Generally, for the kind of electrical systems we are looking at, this means the system shall not be energised (no energy stored on capacitors or in inductors) at time $t = 0$ (the definition of transfer function would be less useful otherwise, as we want it to characterise the system rather than the initial conditions).

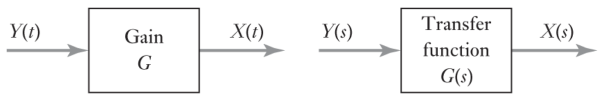


Figure 15.1: The transfer function.

For many 'complex systems' – such as an instrument – we can break the design down into component building-blocks such as:

- Sensors;
- Filters;
- Amplifiers (including feedback loops);
- Output transducers.

Typically, each block can be represented as a zero-, first- or second-order LTI system, for which we can individually determine the transfer function according to the definition of equation (15.11). The real power of the transfer function comes when we build, according to simple rules, the transfer function for the whole system. This allows us to find the output of the whole system $x(t)$ in response to any arbitrary input $y(t)$. The rules for this *block diagram algebra* are simple. If we know the transfer functions $G_1$ and $G_2$ of two functional blocks, then:

- The transfer function of the *series* (or *cascade*) connection of $G_1$ followed by $G_2$ is $G = G_2G_1$, i.e. the product of the individual transfer functions (note: the ordering can be important!);
- The transfer function for the *parallel* connection is $G = G_1 + G_2$;
- The transfer function of the *feedback* connection, with (open-loop) amplifier block transform $G_1$ and (negative) feedback block transform $-G_2$ is equal to $G = G_1/(1 + G_1G_2)$.

We will see how to apply this method in Section 16.

[30] As a general guide, you should know by heart some of the basic transforms such as entries 1, 2 and 4, while any others you might need would be given on an exam paper.

| Entry No. | $f(t) = \frac{1}{2\pi j}\int_{\sigma - j\infty}^{\sigma + j\infty} F(s)e^{st}\, ds$ | $F(s) = \int_0^{\infty} f(t)\bar{e}^{-st}\, dt$ |
|---|---|---|
| 1 | $\delta(t)$ | $1$ |
| 2 | $u(t)$ | $\frac{1}{s}$ |
| 3 | $t^n \quad \text{for} \quad n > 0$ | $\frac{n!}{s^{n+1}}$ |
| 4 | $e^{-at}$ | $\frac{1}{s+a}$ |
| 5 | $te^{-at}$ | $\frac{1}{(s+a)^2}$ |
| 6 | $\frac{t^{n-1}e^{-at}}{(n-1)!}$ | $\frac{1}{(s+a)^n}$ |
| 7 | $\frac{1}{b-a}(e^{-at} - e^{-bt})\, a \neq b$ | $\frac{1}{(s+a)(s+b)}$ |
| 8 | $-\frac{1}{b-a}(ae^{-at} - be^{-bt})\, a \neq b$ | $\frac{s}{(s+a)(s+b)}$ |
| 9 | $\sin \omega t$ | $\frac{\omega}{s^2 + \omega^2}$ |
| 10 | $\cos \omega t$ | $\frac{s}{s^2 + \omega^2}$ |
| 11 | $e^{-at} \sin \omega t$ | $\frac{\omega}{(s+a)^2 + \omega^2}$ |
| 12 | $e^{-at} \cos \omega t$ | $\frac{s+a}{(s+a)^2 + \omega^2}$ |
| 13 | $\sinh \omega t$ | $\frac{\omega}{s^2 - \omega^2}$ |
| 14 | $\cosh \omega t$ | $\frac{s}{s^2 - \omega^2}$ |
| 15 | $\frac{\sqrt{a^2 + \omega^2}}{\omega} \sin(\omega t + \phi),\ \phi = \tan^{-1} \frac{\omega}{a}$ | $\frac{s+a}{s^2 + \omega^2}$ |
| 16 | $\frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2})t \qquad \lvert\zeta\rvert < 1$ | $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ |
| 17 | $\frac{1}{a^2 + \omega^2} + \frac{1}{\omega\sqrt{a^2 + \omega^2}} e^{-at} \sin(\omega t - \phi),$ $\phi = \tan^{-1}\left(\frac{\omega}{-a}\right)$ | $\frac{1}{s[(s+a)^2 + \omega^2]}$ |
| 18 | $1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2}\, t + \phi),$ $\phi = \cos^{-1} \zeta,\ \lvert\zeta\rvert < 1$ | $\frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$ |

Figure 15.2: Elementary Laplace transform pairs (from Poularikas and Seeley).

### 16. Solving problems with the transfer function

Laplace transforms allow us to find the time response of complex systems driven by an arbitrary forcing function, often without having to directly formulate and solve the differential equation. The method works well for problems which can be formulated to start at some given time, usually taken to be $t = 0$ for simplicity. This is what we actually deal with on most experiments. We set up some apparatus and then throw a switch at a starting time $t = 0$. We then want to know what happens for positive time. To formulate and solve Laplace transforms for simple input functions we can look up the relevant time domain functions and their $s$-domain transforms in a table (see Figure 15.2). For a more complex system we will probably need to simplify the Laplace transform representation somewhat before we can transform back into the time-domain. The technique we quite commonly use to do this involves partial fractions (see last page in this section).

Note that the Laplace transform of the delta function $\delta(t)$ is 1. This means that the time response of a system subject to an impulse at its input is given by the inverse Laplace transform of the transfer function, i.e. $x(t) = \mathcal{L}^{-1}\{G(s)\}$ for a $\delta$-function input. That is to say, *the impulse response – which we may call $g(t)$ – is the time-domain representation of the transfer function.* This is one of the reasons why the delta function is so commonly used in analysis of systems. Another way of understanding this is to recall that the Fourier transform of a $\delta$-function is an infinite flat spectrum, so a perfect impulse contains all frequencies in equal amounts. In other words, when we apply an impulse to a system, we stimulate it with all frequencies *simultaneously* and hence we can measure its frequency response – rather than some property of the input pulse. In practical applications, such as the Elvis Bode Analyser, it is inefficient to stimulate the system with all frequencies at once (and indeed impossible to generate a mathematically *true* impulse, so the Bode Analyser sweeps the frequency input in order to buildup the frequency response.

#### *16.1. Step-input response for a series system*

Take an instrumentation system constructed from several stages as shown in Figure 16.1; we wish to determine its response to a unit-step applied to the input.

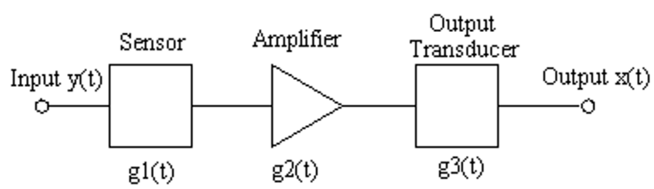


Figure 16.1: System consisting of three functional blocks in series.

We begin by determining the individual responses of the sensor, amplifier and transducer to the $\delta(t)$ function. This is the all-important *impulse response function* which can be determined either analytically or experimentally.

For the instrument in the figure let us say we determine the impulse responses in the time domain, that is to say we have measured the input and output responses of each transducer to impulse inputs. As mentioned previously, this is not always the most efficient method, but it is sometimes used in reality, especially for certain mechanical/acoustic transducers or for photomultiplier tubes (the single photoelectron response is essentially the impulse response function of a PMT). In this case let us say we measure the impulse responses to be:

- Sensor: $g_1(t) = e^{-t}$ ;
- Output transducer: $g_3(t) = e^{-2t}$.

Let us further say that we have tested the amplifier and found it to have a voltage gain of 4. If we assume that it is an ideal amplifier then (as a zero-order system) it will have an impulse response given by:

- Amplifier: $g_2(t) = 4\delta(t)$.

What is the output of the system as a function of time for, say, an input step $y(t) = u(t)$? First we need to formulate a transfer function for the complete system. We know the time domain responses of the individual components and from these we can find their transfer functions. From our table of Laplace transforms in Figure 15.2,

$$\begin{aligned} G_1(s) &= \frac{1}{s+1}\,; \\ G_2(s) &= 4\,; \\ G_3(s) &= \frac{1}{s+2}\,. \end{aligned}$$

Now we can find $G(s)$, the transfer function for the whole system:[31]

$$\begin{aligned} G(s) &= G_1(s)G_2(s)G_3(s) \\ &= \frac{4}{(s+1)(s+2)}\,. \end{aligned}$$

The input to the system is a unit step function $y(t) = u(t)$, and the Laplace transform of this is simply $1/s$. To find the time response of the system we recall that $X(s) = G(s)Y(s)$, where $X(s)$ and $Y(s)$ are the Laplace transforms of the output and input of the system and $G(s)$ is the transfer function; so,

$$\begin{aligned} x(t) &= \mathcal{L}^{-1}\{X(s)\} \\ &= \mathcal{L}^{-1}\{G(s)Y(s)\} \\ &= \mathcal{L}^{-1}\left\{\frac{4}{s(s+1)(s+2)}\right\}. \end{aligned}$$

[31] Note that the order for this cascade system should be more correctly $G = G_3G_2G_1$, but for most LTI systems with scalar transform functions the terms clearly commute. We need to be careful, for example, if one block *loads* the next...

Our task now is to evaluate the inverse Laplace transform of this function, and we do this for our example using partial fractions. We write our $s$-domain function as a sum of partial fractions (see the final page in this section):

$$X(s) = \frac{4}{s(s+1)(s+2)} = \frac{A_0}{s} + \frac{A_1}{s+1} + \frac{A_2}{s+2}\,.$$

This is an identity and valid for all values of $s$, therefore we can choose values of $s$ which make some of the constants disappear. Multiplying through by $s$ and evaluating for $s = 0$ gives

$$\left|\frac{4}{(s+1)(s+2)} = A_0 + \frac{A_1 s}{s+1} + \frac{A_2 s}{s+2}\right|_{s=0},$$

$$A_0 = 2\,.$$

Similarly, for $A_1$ we can multiply through by $s+1$ and evaluate for $s = -1$:

$$\left|\frac{4}{s(s+2)} = \frac{A_0(s+1)}{s} + A_1 + \frac{A_2(s+1)}{s+2}\right|_{s=-1},$$

$$A_1 = -4\,.$$

And for $A_2$ we can multiply through by $s+2$ and evaluate for $s = -2$:

$$A_2 = 2\,.$$

Putting these constants back into our identity for $X(s)$ we obtain

$$X(s) = \frac{2}{s} - \frac{4}{s+1} + \frac{2}{s+2}\,.$$

This is now in a form that we can simply write down a time response for, with reference to our Laplace transform tables. If we do this for the individual components of $X(s)$ we find, for positive times,

$$x(t) = 2 - 4e^{-t} + 2e^{-2t}\,.$$

This function is plotted in Figure 16.2 for $0 < t < 8$ seconds. It shows (nearly) the sort of time behaviour we expect from first order systems subject to a step input (or a heavily-damped second order system). The response to the step is an initial rise, which then slows down, and tends to some steady state value for large $t$. Note that transfer functions of the form $G_1(s)$ and $G_3(s)$ are characteristic of first-order systems while the ideal amplifier (zero-order) has no time-dependence and hence its transfer function is a constant.

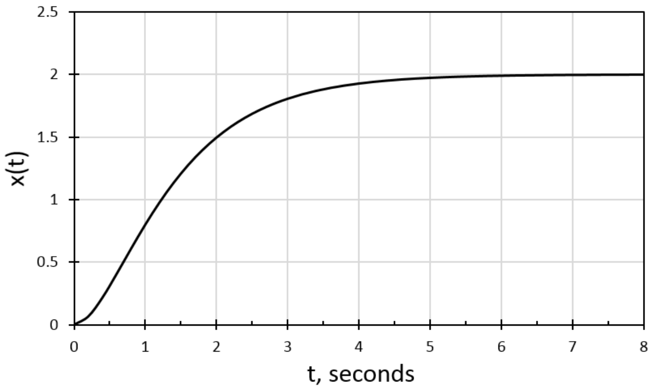


Figure 16.2: System response to a step-input $y(t) = u(t)$.

*16.2. Impulse response with known transfer function*

We now attempt to find the time-domain impulse response of a system with a (known) transfer function

$$G(s) = \frac{s+3}{(s+2)^2}\,.$$

To find the time-domain response of the system we note that $g(t) = \mathcal{L}^{-1}\{G(s)\} = \mathcal{L}^{-1}\{X(s)\}$ (since $y(t) = \delta(t)$ and, hence, $Y(s) = 1$) and so

$$g(t) = \mathcal{L}^{-1}\left\{\frac{s+3}{(s+2)^2}\right\}.$$

As before, we need a partial fraction expression to simplify the inverse transform. For denominators of order $> 1$ we must write the partial fraction as

$$\frac{s+3}{(s+2)^2} = \frac{A_2}{(s+2)^2} + \frac{A_1}{(s+2)}\,.$$

To find $A_2$ we multiply $G(s)$ by $(s+2)^2$ and set $s = -2$ to obtain $A_2 = 1$. We now need to find $A_1$; however, there is a problem with continuing the method. To find this coefficient we cannot use $s = -2$ again as we have already used this to find $A_2$. Instead, we can *differentiate* $G(s)(s+2)^2$:

$$\frac{\mathrm{d}}{\mathrm{d}s}G(s)(s+2)^2 = \frac{\mathrm{d}}{\mathrm{d}s}(s+3) = \frac{\mathrm{d}}{\mathrm{d}s}(A_2 + A_1(s+2))\,.$$

$$1 = A_1\,.$$

Therefore,

$$G(s) = \frac{1}{(s+2)^2} + \frac{1}{s+2}\,,$$

which can be easily inverse-transformed into

$$g(t) = t\,e^{-2t} + e^{-2t}\,.$$

*16.3. Other applications*

We have seen here how Laplace methods allow complex systems to be solved by transforming differential equations in the time-domain into algebraic expressions in the $s$-domain. The fact that the Laplace transform is single-sided and takes into account initial conditions means that it is very well suited for solving transient analysis problems in mechanical and electrical engineering. These methods are applicable across the physical and engineering disciplines, and biological systems (including complex feedback loops) can be also modelled. Other applications are in economics, where the parameter to be determined could be price and the constants relate to supply and demand. In general, any continuous-time system represented by ordinary differential equations can be studied, especially where we want to set the system up in some starting configuration and observe how it evolves with time.

As a final note, you should be aware that there is an equivalent to the Laplace transform for discrete (quantised) signals called the $Z$-transform. This is of central importance in the field of digital signal processing, though we will not be able to cover this in the course.

*16.4. Partial fractions*

Consider the identity

$$\frac{2}{5} - \frac{3}{4} + \frac{1}{2} = \frac{8 - 15 + 10}{20} ,$$

where 20 is the lowest common multiple. *Partial fraction decomposition* is the *reverse* of this process. In general, any rational function

$$\frac{P(s)}{Q(s)}$$

can be re-written using partial fractions. First, it is essential to check that the degree of the numerator is less than the degree of the denominator. If not, it is necessary to first divide-out by long division. This is tedious so we will avoid it in this course, and in general the sort of real-world problems we will encounter naturally lead a higher degree of $s$ in the denominator. We proceed as follows.

For each factor of $Q(s)$ in the form $(as + b)^m$ introduce terms

$$\frac{A_1}{as + b} + \frac{A_2}{(as + b)^2} + ... + \frac{A_m}{(as + b)^m} .$$

For factors of the form $(cs^2 + ds + e)^n$ introduce terms

$$\frac{C_1 s + D_1}{cs^2 + ds + e} + \frac{C_2 s + D_2}{(cs^2 + ds + e)^2} + ... + \frac{C_n s + D_n}{(cs^2 + ds + e)^n} .$$

There are various methods to solve for the constants, but remember that

$$\frac{P(s)}{Q(s)} = \frac{A_1}{as + b} + \frac{C_1 s + D_1}{cs^2 + ds + e} + ...$$

is an identity valid for all values of $s$. We can rearrange and substitute values of $s$ which cause some unknowns to disappear. For example, in

$$X(s) = \frac{s + 3}{(s + 2)^2} = \frac{A_1}{s + 2} + \frac{A_2}{(s + 2)^2} ,$$

we can multiply $X(s)$ by $(s+2)^2$ and set $s = -2$ to get $A_2$. To get $A_1$ we can differentiate $X(s)(s+2)^2$ with respect to $s$ and then use $s = -2$ a second time. Alternatively, we can gather terms in powers of $s$ to obtain a set of simultaneous equations

$$X(s)(s + 2)^2 = s + 3 = A_1(s + 2) + A_2 .$$

Gather terms:

$$0 = s(A_1 - 1) + (A_2 + 2A_1 - 3) ,$$

from which

$$\begin{aligned} A_1 - 1 &= 0 \\ A_2 + 2A_1 - 3 &= 0 . \end{aligned}$$

Whichever method is best depends on the form and complexity of the functions.

## 17. Stability in LTI systems

If we have a mathematical model of our system, then we can test for stability by exciting all frequencies at the input using a delta function. We generally find two cases:

*Stable.* When the impulse is applied to the system, the output is characterised by transient behaviour (sometimes with oscillation) which reduces over time leaving the system back in a quiescent state.

*Unstable.* The transients do not reduce but actually increase with time, such that the quiescent condition is never returned to.

There is a third case on the limit between these two: the output oscillates infinitely (similar to the un-damped harmonic oscillator in Section 14), or moves to a new constant output. As long as the output does not grow, this is still considered stable.

### *17.1. Stability and feedback*

We briefly looked at stability in the context of feedback amplifiers. We saw how a negative feedback system can go into unstable positive feedback when the phase change at the output tends to 180°. In an amplifier this is very bad, since such feedback then tends to increase the gain rather than decrease it. The output tends to grow until it reaches some physical limit, usually $V_{ss}$ (the supply voltage). We will now investigate this in more detail. Consider again the equation for an amplifier wired for negative feedback:

$$G = \frac{A}{1+\beta A}\,,$$

where $A$ and $G$ are variables which we saw were a function of frequency and $\beta$ is the feedback fraction. We stated that we must have $A \ll 1$ as the output phase $\phi \to -180°$. For the purposes of stability analysis we assume the voltage follower configuration which means that *all* the output is fed back to the input (this is the worst-case situation for stability analyses) so $\beta = 1$. For the case $\phi = -180°$ negative feedback becomes positive feedback and we can rewrite $G$ as

$$G = \frac{A}{1-A}\,.$$

We must avoid $A = 1$ and $\phi = -180°$ since then $G \to \infty$. We will now do this more formally for the general case of a transfer function $G(s)$.

The model we adopted for the amplifier in the last section – that of a static, zero-order system with constant gain – is over-simplified, since we know there is a significant dynamic behaviour in any feedback amplifier; we therefore propose a more realistic dynamic model with $s$-domain representation given by:[32]

$$G(s) = \frac{A(s)}{1+H(s)A(s)}\,.$$

According to the argument above, we must avoid the condition $1 + H(s)A(s) = 0$.

### *17.2. Zeros and poles*

Writing the transfer function as a general fraction

$$G(s) = \frac{P(s)}{Q(s)}\,,$$

we can make the following associations:

- Values of $s$ such that $P(s) = 0$ are *zeros*, i.e. $G(s) \to 0$ as $P(s) \to 0$;
- Values of $s$ such that $Q(s) = 0$ are *poles*, i.e. $G(s) \to \infty$ as $Q(s) \to 0$.

In real (physical) systems the transfer function $G(s)$ is *always* a ratio of two polynomials with real coefficients. This results in the poles and zeros being either real *or* complex-conjugate pairs (recall the previous definition $s = \sigma + j\omega$). We can check for stability by seeing where the roots of $Q(s)$ lie on the complex plane. Generally,

- Roots in the $\sigma < 0$ side of the plane are stable;
- Roots in the $\sigma > 0$ side of the plane are unstable.

### *17.3. Roots of the transfer function*

Sometimes roots will be complex and, as stated above, in real physical systems these always appear as complex conjugate pairs. Generally, as we have seen, we can expand the transfer function as

$$G(s) = \sum_{k=1}^{n} \frac{A_k}{s - s_k}\,.$$

This is a simplification in order to illustrate the point, though in practice many real-world transfer functions are actually expandable in this form. The complex-conjugate roots will be of the form

$$s_k = \sigma_k \pm j\omega_k\,,$$

so we obtain a set of solutions of the form

$$g_k(t) = A_k\, e^{s_k t} + A_k^*\, e^{s_k^* t}\,.$$

Since $A_k = a + jb$ and $s_k = \sigma_k + j\omega_k$ we can simplify to

$$\begin{aligned} g_k(t) &= 2\sqrt{a^2+b^2}\, e^{\sigma_k t} \cos(\omega_k t + \beta_k) \\ \beta_k &= \tan^{-1}\frac{b}{a}\,. \end{aligned}$$

These results are illustrated in Figure 17.1. In the top panel, a single, real negative root exists and the system is stable. In the second panel, the system is unstable as the output grows without limit. In the third panel complex-conjugate roots exist to the left of the $\sigma = 0$ axis; the result is damped oscillation and the system is stable. With complex roots on the RHS of the axis we have growing

[32] Following the third rule for block diagram algebra given at the end of Section 15.

oscillation and the system is unstable. The last two panels show some special cases. With purely imaginary roots we have sustained oscillation, which is nonetheless stable. If the root is at $s = 0$ then the response is a constant, which is stable.

Note that *all* roots must be on the $\sigma < 0$ side of the plane for stability. *Any one root* on the right of the plane will give instability.

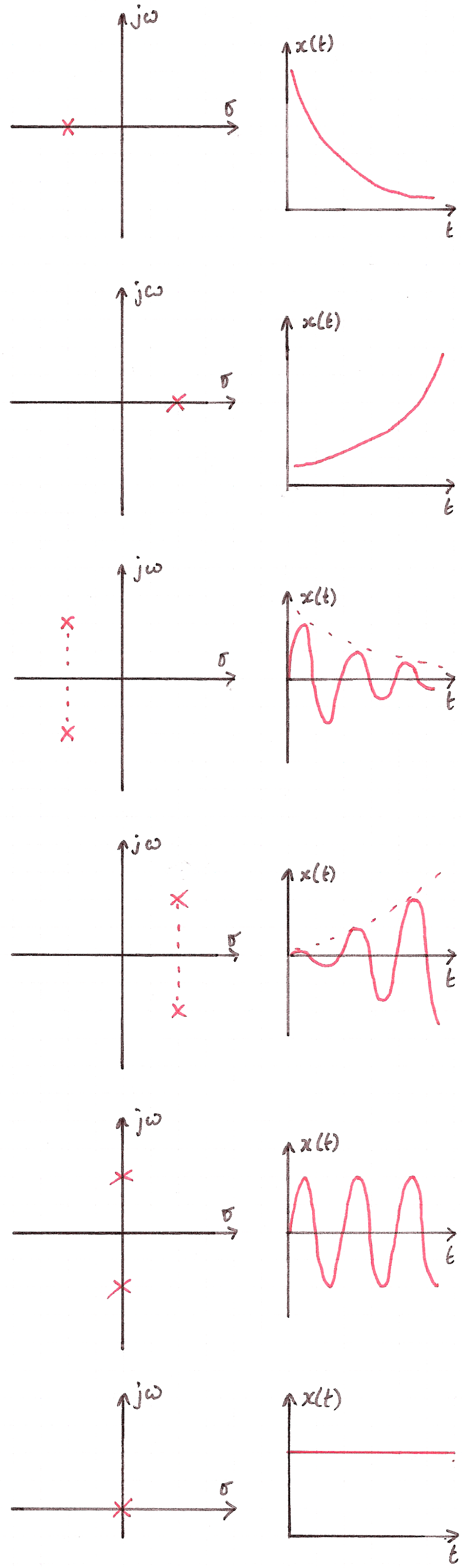


Figure 17.1: Graphical illustration of roots in the complex plane.

## 18. Systems with initial conditions

We will now use the Laplace transform to analyse a simple electrical circuit. One of the major uses of the Laplace transform in electrical engineering is for initial value problems where we throw a switch, or turn on a circuit. What we then want to know is the behaviour of the circuit for all times after we throw the switch.

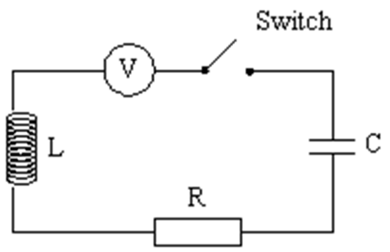


Figure 18.1: Switched RLC circuit.

Figure 18.1 shows the simple RLC series circuit with a voltage source $V$ and a switch. If the voltage source were a sine wave generator, and the switch had been closed for a long time, then we could use our usual expressions for the complex impedance (essentially a Fourier approach) to find the current and voltage for various components in the circuit. However, here the applied voltage is a constant $V$ and is applied once at time $t = 0$. AC circuit analysis is not appropriate for this sort of transient input problem where we want to know "what happens immediately after we throw the switch?" We can use the Laplace transform to answer this.

Consider the following problem: for a DC voltage source $V = 300$ V, inductance $L = 2$ H, capacitance $C = 0.02$ F and resistance $R = 16\ \Omega$, we want to find an expression for the charge on the capacitor $q(t)$ and the current $i(t)$ in the circuit for all times $t > 0$ after the switch is closed. Assume that the circuit is initially relaxed.

We begin by determining the differential equation which governs the circuit. We can write the voltage across each of the components as

- Voltage across the inductor $= L\frac{\mathrm{d}i}{\mathrm{d}t}$;
- Voltage across the resistor $= iR$;
- Voltage across the capacitor $= \frac{q}{C}$.

Using Kirchhoff's voltage law to sum the voltages around the series circuit we can write (for time $t > 0$, i.e. *after* the switch is closed):

$$L\frac{\mathrm{d}i}{\mathrm{d}t} + iR + \frac{q}{C} = V\,. \tag{18.1}$$

Since $i = \mathrm{d}q/\mathrm{d}t$ (we know that the instantaneous current is the same everywhere in the circuit, due to conservation of charge, or Kirchhoff's Current Law) and also $V = 0$ for all time $t < 0$, we can write this as

$$Vu(t) = L\frac{\mathrm{d}^2q}{\mathrm{d}t^2} + R\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{C} \tag{18.2}$$

where $u(t)$ is the unit step function. The differential equation is second-order in time, therefore we may expect that the system may be capable of oscillation as well as tending to some steady-state condition at some time $t$ much after we close the switch. To solve the DE we take the Laplace transform:

$$\begin{aligned}\frac{V}{s} &= L\left(s^2Q(s) - sq(0^+) - \frac{\mathrm{d}q(0^+)}{\mathrm{d}t}\right)\\ &\quad + R(sQ(s) - q(0^+))\\ &\quad + \frac{Q(s)}{C}\,.\end{aligned} \tag{18.3}$$

Here we see how the initial conditions of the system are automatically taken care of by the Laplace transform. Since the circuit is initially relaxed, we start with $q = 0$ and $\dot{q} = \ddot{q} = 0$, so we can further simplify the algebraic expression

$$\frac{300}{s} = 2s^2Q(s) + 16sQ(s) + 50Q(s)\,,$$

$$Q(s) = \frac{150}{s(s^2 + 8s + 25)}\,.$$

Again, using partial fractions, here we must write

$$\frac{150}{s(s^2 + 8s + 25)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 8s + 25}\,.$$

Multiplying by $s$ and setting $s = 0$ yields $A = 6$. Multiplying by $s(s^2 + 8s + 25)$, taking $\mathrm{d}/\mathrm{d}s$ and setting $s = 0$ yields $B = -6$. Again, taking $\mathrm{d}/\mathrm{d}s$ and setting $s = 0$ yields $C = -48$:

$$Q(s) = \frac{150}{s(s^2 + 8s + 25)} = \frac{6}{s} - \frac{6s + 48}{s^2 + 8s + 25}\,.$$

This requires further simplification before we can use the standard transforms:

$$\begin{aligned}Q(s) &= \frac{6}{s} - \frac{6(s+4) + 24}{(s+4)^2 + 9}\\ &= \frac{6}{s} - \frac{6(s+4)}{(s+4)^2 + 9} - \frac{24}{(s+4)^2 + 9}\,.\end{aligned}$$

We can now use entries 2, 11 and 12 in Figure 15.2 to get:

$$q(t) = 6 - 6e^{-4t}\cos(3t) - 8e^{-4t}\sin(3t)\,.$$

Now we have $q(t)$ we can differentiate to find the current:

$$i(t) = \frac{\mathrm{d}q}{\mathrm{d}t} = 50e^{-4t}\sin(3t)\,.$$

This is illustrated in Figure 18.2. Note that this is the kind of response we expect from a damped harmonic oscillator. The result is physically meaningful: at the instant at which the switch is closed, the rate of change of voltage across the capacitor is very large, so a large current will flow, with the rate of change of current limited by the inductor. Electrical energy is stored in the electric field across the capacitor and in the magnetic field of the inductor. The two exchange energy in an oscillatory fashion like a mechanical oscillator exchanging kinetic and potential energy. The $\sin 3t$ term reflects this. As the system evolves, electrical energy is converted to heat energy in the resistor, so the amplitude of the response decays (the exponential term) – and, in fact, it overshoots into negative territory a little. For large $t$ the applied voltage (the forcing function $V$) is a constant, and the voltage across the capacitor tends to a constant – hence the impedance of the capacitor tends to $\infty$ and the current tends to zero.

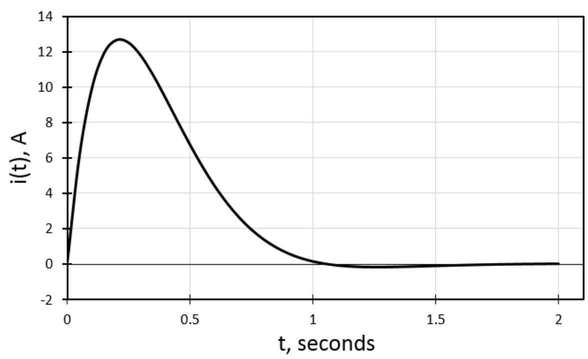


Figure 18.2: Time response of the switched RLC circuit.

In this example we started from the understanding that the system was "initially relaxed". This is usually the starting point for most systems. When we switch-on the Elvis prototyping board, we assume that the voltages and currents are zero, and this is usually right; however, consider in this case that someone has snuck into the lab overnight and charged the capacitor up to 100 V. The charge on the capacitor will stay put until the switch is closed, and we can imagine that the $i(t)$ response will be very different! The defining differential equation for the circuit is still valid (see equations (18.1) and (18.2) and we can see that the transformation into the $s$-domain (equation (18.3)) automatically includes an initial value for $q$.

## 19. Noise in electronics

A signal will always be contaminated by noise, either from some physical principle fundamental to the measurement process or the instrument (usually referred to at *intrinsic noise*), or from outside sources (*extrinsic noise* or, more generally, *interference*). Here, we will look at three of the most common intrinsic noise processes: *thermal*, *shot*, and *flicker*. You will encounter practical questions on noise in the last problem sheet of the course.

### *19.1. Thermal noise*

A resistor sitting on the bench in the lab is a source of noise due to the random thermal motion of the electrons in the resistive material. If we notionally divide the resistor into two halves, then at any instant in time the left half and the right half will tend to have different numbers of electrons. Since the material has some resistance, it is able to sustain a potential difference across its ends and hence we will observe a voltage which is a function of the resistance.[33] This type of noise was first observed by Johnson at Bell labs in the 1920s, and the effect was subsequently explained by Nyquist, hence thermal noise is sometimes referred to as *Johnson* or *Johnson-Nyquist* noise. To characterise this or any other type of noise we need to consider both its amplitude and frequency characteristics.

*Amplitude characteristics.* The resistor on the bench has an average noise voltage of zero. Since the voltage is caused by an ensemble of random processes, we will not be surprised to find that the amplitude distribution is Gaussian and, in this case, with a standard deviation given by

$$v_n = \sqrt{4k_B TRB}\,, \tag{19.1}$$

where $T$ is the temperature in kelvin, $R$ the resistance in ohm, $k_B$ is Boltzmann's constant and $B$ is the bandwidth of the measurement. The noise voltage $v_n$ is the RMS voltage which will be measured with an instrument of bandwidth $B$ (recall that for a Gaussian centered at zero RMS $= \sigma$). A $10\,\mathrm{k\Omega}$ resistor measured with a really good true-RMS voltmeter with 10 kHz bandwidth will be seen to generate an RMS noise voltage of about 1.3 $\mu$V.

*Frequency characteristics.* The resistor appears to generate power. The noise power generated per unit frequency is simply

$$\frac{v_n^2}{RB} = 4k_B T\,. \tag{19.2}$$

Equation (19.2) is the *noise power density*, i.e. the power per Hz of frequency, and since this is a constant we know that the noise has a flat spectrum, i.e. it is *white noise*.[34] This presents a conceptual problem since integrating over all frequencies implies that noise power is infinite. In practice, thermal noise is observed to be white up to tens of GHz, above which the capacitance between the two leads (tiny, but ultimately non-zero) serves to short out the very high frequencies. For practical purposes, consider the noise to be white, therefore we get a very important result: *the RMS noise voltage measured is a function of the measurement bandwidth; if we reduce the bandwidth we see less noise.*

*Thermal noise model.* It is helpful to think about thermal noise in terms of an ideal (noiseless) resistor $R$ in series with a noise voltage source $v_n$, i.e. a Thevenin equivalent. We can also think of some stray capacitance across the whole, as above. If the resistor generates power, how can we extract this? The maximum power transfer theorem tells us that we need to connect $R$ to another resistor $R_1$ of the same value. Then the voltage $v_n$ is halved across both resistors and the power transferred to $R_1$ is $k_B T$ per Hz. We could imagine trying to measure this with an ideal band-pass filter of width 1 Hz and a voltmeter. However. the problem is that $R_1$ is also at temperature $T$ and so it generates power $k_B T$ back into $R$. The *total* power in the system is still $4k_B T$ per Hz, the same as with one resistor, i.e. independent of $R$. Despite the impossibility of extracting the power to do useful work, the noise voltage is real and measurable, particularly when our signal is small and comes from a source with some resistance (as all real sources do). The resistive part of a source impedance generates noise, as do the resistances we use in our op-amp and filter circuits. Thermal noise usually sets the lower-limit for voltage measurements.

### *19.2. Signal-to-noise ratio and noise figure*

This is a useful figure of merit for any measurement but we need to be sure we are comparing like with like. Generally,

$$\mathrm{SNR_{dB}} = 10\log_{10}\frac{P_s}{P_n} = 20\log_{10}\frac{v_s}{v_n}\,.$$

$P_s$ is the signal power ($\propto v_s^2$) which must be measured in the same bandwidth $B$ as the noise measurement, and correspondingly the signal voltage $v_s$ must also be an RMS measurement. For example, a $10\,\mathrm{k\Omega}$ resistor carrying a *narrow-band* signal such as a 10 mV amplitude sine-wave will be measured by a voltmeter with 10 kHz bandwidth to have $v_s = 7.1$ mV, hence the signal-to-noise ratio is about 75 dB.

### *19.3. Shot noise*

An electric current is not a smooth fluid flow but a flow of discrete electronic charges with some statistical variation. A DC current $I$ will see an RMS noise current (superimposed on the DC current) given by

$$i_n = \sqrt{2eIB}\,, \tag{19.3}$$

[33] Note that, in an ideal conductor, while the electrons are in thermal motion, the conductor is unable to sustain an electric field and therefore there is no observable potential difference.

[34] Formally, we should start from thermodynamics: the equipartition theorem is used to derive the noise power spectrum which ultimately results in equation (19.1).

where $B$ is the bandwidth and $e$ is the electronic charge. An analogy is rain on a tin-roof. When it is raining hard, we hear a continuous sound level (the signal). As the rain eases off, the natural statistical fluctuation in the number of drops per second (charge carriers) becomes more apparent. Hence, the variation (noise) is more apparent when the signal is small, as we can see from the signal-to-noise ratio:

$$\mathrm{SNR} = \frac{I^2}{i_n^2} \propto I\,.$$

*Amplitude and frequency characteristics.* For a signal with a very small number of charge carriers, the statistics of the number passing a given point as a function of time follows a Poisson distribution since we assume the charge carriers are independent and random. For realistic currents, the distribution tends to the normal distribution (i.e. the amplitude distribution is Gaussian, as for thermal noise), and similarly the noise power per Hz into a load $R$ is

$$\frac{i_n^2 R}{B} = 2eIR\,,$$

which is constant with frequency: shot noise is also white.

*Sources.* It is tempting to think that any DC current $I$ will exhibit shot noise according to equation (19.3); in reality, in a good conductor there are long-range Coulomb interactions which act to smooth-out any variations so, to a good approximation, *there is zero shot-noise in a macroscopic current flowing in a wire or a purely passive circuit* (e.g. a battery connected to a resistor). However, where we have a semiconductor junction then charge carriers cross the junction only according to random thermal excitation and hence a diode is a source of shot noise. Shot noise was discovered in thermionic diodes by Schottky; in these vacuum devices the electrons "boil off" from a hot cathode according to Poisson statistics, then cross the vacuum to the anode. Since all modern electronics contains large numbers of semiconductor devices, shot noise is present generally, and is particularly apparent when the signal current is very small. In summary, a DC current flowing in a resistor will not generally have shot noise, however if there is already shot noise in the signal (e.g. it came from a semiconductor source) then we will see this as an RMS noise voltage $v_n = i_n R$ across the resistor.

### *19.4. Noise in amplifiers*

The ideal amplifier is, of course, noiseless, but the real device will exhibit thermal, shot and $1/f$ noise – all of these are associated with semiconductors. In any given frequency band, the amplifier will amplify the signal presented at its input but it will also add some amount of noise. It is common to "refer this noise to the input", as this helps us select a suitable amplifier: we compare the noise already present in the signal we want to amplify to the noise we can expect from the amplifier as it that were also present at the input: this way we are comparing apples with apples.

It is common to use a *noise factor* when quantifying the amount of noise added by an amplifier or some other component of the circuit:

$$F = \frac{\mathrm{SNR_{in}}}{\mathrm{SNR_{out}}}\,.$$

Here the SNR is a simple ratio, not expressed in dB.[35] The ideal, noiseless amplifier has $F = 1$, since it will amplify the signal and existing noise by the same factor. A real one has $F > 1$, since the SNR at the output will be a little worse due to the noise added by the amplifier itself.

Noise can be reduced in various ways. For example, one may decrease the bandwidth of the system by selecting carefully the pass-band of the amplifier or through additional filtering (e.g. a narrow-band filter around our frequency of interest). In some applications (e.g. astronomy, but not only), it is common to operate the amplifier at very low temperatures to decrease thermal noise.

[35] One may instead present a 'noise figure', which is $F$ given in dB: $NF = 10\log_{10} F$; there is also a 'noise temperature', which is a similar way to quantify the noise added by an amplifier – but we will tend to use the simpler *noise factor* in this course.

## 20. Flicker noise

Flicker noise gets its name from the "flickering", or randomly varying current, seen in vacuum tubes. Its *power spectrum* is inversely proportional to frequency:

$$\text{power} \propto \frac{1}{f}\,, \tag{20.1}$$

hence, it is often referred to as $1/f$*-noise* or *"pink noise"* (due to the fact that lower frequencies have a higher power content). Flicker noise results in equal noise power per decade of frequency, i.e. there will be the same amount of noise power in the band 1–10 Hz as there will be in the band 10–100 Hz, and so on. So:

- For high frequencies, the amount of power in the flicker noise rapidly tends to zero. Typically, the flicker noise is below other sources of noise for frequencies above a few hundred Hz;
- For low to DC frequencies, the amount of flicker noise power tends to infinity, which is problematic, but we can deal with this, as we shall see later.

Figure 20.1 shows the typical noise landscape which plagues most instrumentation systems. Flicker noise dominates up to some 'knee' frequency typically in the range 100–1,000 Hz. Making measurements in the kHz–MHz range is often desirable.

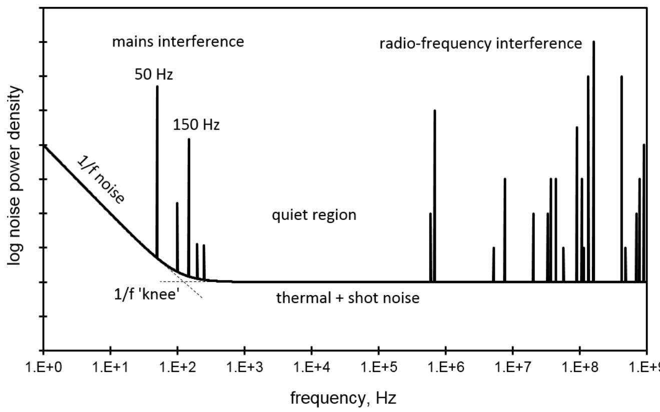


Figure 20.1: Typical noise power spectrum, showing $1/f$ noise dominating at low frequencies up to ~100–1,000 Hz, mains interference at 50 Hz and its harmonics, and RF interference at high frequencies (analogue and digital radio and TV, wireless and mobile networks, etc.), leaving a relatively quiet region in the kHz–MHz region where the white noise sources (thermal and shot noise) dominate.

Thermal and shot noise are irreducible forms of noise which are generated according to well-determined physical processes. Equations (19.1) and (19.3) can be derived from well-understood physics. The thermal noise voltage on a $10\,\mathrm{k\Omega}$ resistor is given by fundamental physical constants and quantities; it does not depend on the type of resistor. For example, the cheapest carbon-compound resistor has the same amount of thermal noise as the most-expensive high-precision wire-wound resistor of the same value. This is not true of flicker noise, which is dependent on the type of technology in use. While flicker noise seems to be a universal phenomenon associated with all measurements, we will use the example of the resistor to illustrate. Imagine a $1\,\mathrm{k\Omega}$ resistor carrying a constant of 1 mA applied from a really good, steady source. Assume for the moment that there is no thermal or shot noise. Then. if we measure the the RMS flicker noise voltage across the resistor in, say, any decade of frequency, we will find different results for different resistor types – see Table 20.1. This is extraordinary, given that we assume the current supplied to the resistor is constant, so the voltage should be too (assuming no shot/thermal noise). There is no satisfactory universal physical explanation for this effect, which gives rise to another name: *excess noise*; it is likely that there are multiple phenomena which cause this type of noise in any system (e.g. the nature of the electrical contacts), rather than a universal explanation.

Table 20.1: Flicker noise according to resistor type.

| Resistor technology | Flicker noise voltage |
|---|---|
| Carbon composition | $1\ \mu\mathrm{V_{RMS}}$ |
| Carbon film | $0.2\ \mu\mathrm{V_{RMS}}$ |
| Metal film | $0.1\ \mu\mathrm{V_{RMS}}$ |
| Wire-wound | $<0.1\ \mu\mathrm{V_{RMS}}$ |

Flicker noise can be understood by imagining that the *resistance* of the resistor fluctuates, hence generating the noise voltage. This "understanding" (as it is not physically-based) is useful as it allows us to reason why different resistor technologies show different levels of noise. It seems that details of the construction of the resistor influence the overall amount of noise, such as the type of resistive materials used and how the leads are connected to it.

The flicker noise power in a bandwidth $B = f_1 ... f_2$ is

$$P = k \int_{f_1}^{f_2} \frac{\mathrm{d}f}{f}\,, \tag{20.2}$$

where $k$ is some constant which depends on the technology.

Flicker noise is seen in almost all electronics applications, and is particularly apparent in any kind of semiconductor device, where the idea that a junction has some resistance is also useful for understanding the effect. However, flicker noise seems to be a more fundamental property of any measurement, and indeed a fundamental property of nature itself. If we record the speed of ocean currents (for example) over a long period of time and then take a frequency spectrum from our time-series data, then the result tends to have a $1/f$ power spectrum. What this tells us is that the longer-period variations carry relatively more power in the signal than the faster variations. This pattern is repeated across nature. Another example is that the loudness of a reasonably long and complex piece of music (say a symphony) tends to have $1/f$ characteristics. Since the power increases as frequency decreases, the question becomes: how low can we go in measurement frequency? According to equation (20.2), the total integrated power between DC (zero Hz) and 1 Hz tends to infinity, which seems to imply that any truly DC measurement is impossible. This is generally true, since any true DC measurement of any quantity would take an infinite amount of time to complete. In this sense, there is no such thing as a truly constant parameter (such as voltage, current, temperature,...), we can just constrain the lower frequency of our measurement. Consider another often-cited example of $1/f$ behaviour: the level of water in the Nile. This is probably the single quantity with the longest history of recording: there are measurements of the water height every day going back about 4,000 years. The lowest frequency in the data is therefore about $4 \times 10^{-12}$ Hz. This is a very low frequency indeed, but not constant or DC. Thus, measurement-time is an effective lower-limit on frequency, which serves to keep the flicker noise finite.

On a log-log plot of flicker noise *power* vs. frequency such as that in Figure 20.1, the flicker noise falls-off at $-10$ dB/decade, while we have seen that thermal and shot noise are constant with frequency. Therefore, a typical overall noise plot for any measurement will show $1/f$ dominating at low frequencies, and thermal/shot at higher frequencies. An important consequence of flicker noise is that low-frequency measurements are harder to do than higher frequency measurements.

## 21. The lock-in amplifier

The *lock-in amplifier* is an application of the *phase detector* which we looked at in Section 13. Lock-in techniques can be used to achieve extraordinary levels of noise rejection through simultaneous application of two techniques:

1. *Bandwidth narrowing*, which as we have seen is the best way to minimise noise in our measurement;
2. *Modulation* of a low frequency, or essentially 'DC', up to a higher frequency where it will be less affected by flicker noise.

A very important implication of equation (13.3) is that the difference between the two input frequencies has to be very small if we are to get any output from the phase detector. We can set the value of $\omega_c$ by choosing the values of $R$ and $C$ in the output filter, and therefore we can make the range of frequencies that the circuit responds to very narrow indeed (values as low as 1 Hz are possible). Since the reference frequency may be kHz or MHz, this means the circuit is very good at "picking-out" frequency components of the input signal with very high *spectral (frequency) resolution*. Generally, our input signal will consist of many frequencies (and noise) so the phase detector is good for asking questions such as *"What is the amplitude of the input signal at 1 kHz measured in a bandwidth of 1 Hz?"*.

A typical commercially-available implementation of the phase detector is known as the *lock-in amplifier* and in its simplest implementation this is a box with an input, two control knobs and an output. The first control is used to set the reference frequency (an internally generated square wave) and the second allows the phase of the reference to be adjusted. The procedure to be used goes as follows: the input is connected to the signal we wish to study and the output $V_{out}$ is usually sent to an oscilloscope. The reference frequency is set to 1 kHz and if there is any component of the input signal at this frequency then we will see an output on the scope. According to equation (13.3) we may find that the output varies a little so we fine-adjust the frequency knob to get a fix (or 'lock') with a steady output. At this point $\Delta\omega = 0$, so now equation (13.1) comes into play. We now adjust the phase knob to get a maximum output, and from this we can determine the value of the amplitude $A$ at the given frequency.

Practical lock-in amplifiers also have complicated extra features but the only one we need to be concerned with is the ability to take the reference frequency as an external input so we can generate this signal ourselves.

### *21.1. Signal analyser*

The lock-in amplifier gives us the capability to measure the amplitude of a signal at a specific frequency and with a very narrow bandwidth. This is the basic requirement of a *signal analyser*, a device which gives as output the amplitude of a signal as a function of frequency. We can use a signal generator to sweep the reference frequency over the required range and we can measure the amplitude at the output as a function of frequency. This will give us the amplitude spectrum of the signal. Note that, in practice, signal analysers – such as the one you have used with the ELVIS equipment – use a different technique; however, it is important to understand the concept, and in fact some high-end *spectrum analysers* do use a variation on the method described above.

### *21.2. Extracting signal from noise*

One of the most powerful uses of this technique is when we have a signal we wish to measure at a known frequency, but heavily contaminated with noise. For example, we may have a sensor which we know produces a signal at 1 kHz but with a very small amplitude, say tens of $\mu$V. We may find that our signal has noise of some few volts RMS (measured in a bandwidth of say 10 kHz). The signal to noise ratio is then some $-100$ dB. The situation seems hopeless: were we to plot the data in the time-domain we would just see the noise; the signal at 1 kHz would be pretty well buried. However, recalling that noise is usually broad-band, we understand that we can reduce the amount of noise in a measurement by reducing the bandwidth of the measurement. For example, for thermally-generated noise we found that $v_{n,RMS} \propto \sqrt{B}$. The lock-in amplifier can recover the signal because:

1. We expect a well-defined signal frequency which is repetitive and can be synchronously detected (in comparison with the noise which is random and ultimately averages to zero);
2. The lock-in technique is highly selective in frequency, therefore it reduces the bandwidth of the measurement.

In practice, a SNR of $-100$ dB or worse is manageable with this technique.

**Bandwidth limiting**
Consider a measurement where the observed noise is $10^6$ times larger than the signal. If the noise is principally thermal in origin, then the noise per unit Hz is a constant so we can reduce the total noise measured by reducing the bandwidth of the measurement. Measuring with an oscilloscope (typical bandwidth 100 MHz or more) we would be unable to see the signal. However, if we first filter the signal with a band-pass filter of width 100 Hz, then the SNR improves by a factor of a million. Note that the part of the signal we are interested in must of course be within the filter range.

In summary, we should think of this as a frequency-domain technique. If the signal of interest has a definite and fixed frequency (i.e. it is *narrow-band*), and if the noise is *broad-band* (i.e. across the whole frequency range of the measurement), then we can see that in the frequency-domain the two sources look quite different, and can be separated by the lock-in technique.

*21.3. Low-frequency or DC signals*

This approach works well for AC signals with a frequency that it is sufficiently high that it is above the regime where the $1/f$ noise dominates. Under these circumstances, the noise – measured in the same band as the signal – is usually tolerable. However, as the signal frequency reduces, and ultimately as we go towards a 'DC' measurement, then the $1/f$ noise dominates and can overwhelm the small signal.

We may have a detector which gives a constant voltage of a few $\mu$V with some volts RMS of noise. Another example would be measuring the intensity of light from a distant source, when the detector output is swamped by ambient light. Or we could be trying to detect faint radio-emission from a star. These measurements are all small, constant values subject to noise. The solution here is to shift the frequency of the signal to be measured from near DC to a higher frequency (well above the regime where the $1/f$ dominates).

This can be done by a variety of methods which we will look at next, though they are all essentially *modulation techniques* or, as it is more typically known for this application, "*chopping*".

*21.4. Modulation techniques*

We can move a DC or very low-frequency signal to a higher frequency by modulating or chopping the signal. The name derives from optical experiments where a 'chopper' wheel (a rotating blade) physically interrupts the light beam in front of the detector. This modulates the light beam from a constant to an on/off signal at a well defined frequency (given by the rotation rate of the chopper wheel). In Fourier terms, the signal changes from a delta function at zero frequency to a delta function at the modulation frequency. This has the added advantage that we can use the same frequency source for the modulation and the reference of the lock-in amplifier, so we are guaranteed to have the precise same frequency. This is best illustrated with an example.

*Demonstration experiment using light intensity.* Imagine we have an LED powered by a battery, and on the other side of the room we have a photodetector. The question is: can the detector tell whether the light is on or not? In a very dark room, with not too large a distance between the two, it is possible. With the LED off, the trace on the oscilloscope is a noisy straight line, and when we switch the LED on then the line goes up just a little. Note here that the detector is acting as an intensity meter. The LED light is a constant output 'DC' signal. The detector responds to light of all visible wavelengths, so when we open the curtains the detector is flooded with ambient light millions of times brighter than the LED. We have to drop down the scale of the scope by several orders of magnitude to get a new reading of the intensity. Now, if we switch the LED off/on we see no difference in the trace because it is simply buried by the ambient light.

Now imagine we discard the battery and instead power the LED with a square wave from a signal generator. This modulates the light intensity from DC to the frequency of the signal generator. Further, we also connect the signal generator output to the reference input of a lock-in amplifier, and the output of the detector to the signal input of the lock-in. Now we can synchronously detect the LED intensity at the modulation frequency. Because modulation and reference frequencies are the same, we just need to adjust the phase knob to get a maximum output. We will see a nice strong constant signal out of the lock-in. If we disconnect the LED it goes to zero. This works the same both in the dark and in bright ambient light.

This is a striking example of extracting a tiny signal from noise. To be clear about what is happening here, we need to recall that

1. The modulated signal has a very narrow, well-defined (fundamental) frequency;
2. The noise has lots of sources: mostly the more-or-less constant ambient light, plus some 50 Hz from mains lights, plus $1/f$ and thermal noise contributions from the detector system;
3. Modulation moves the signal above the low-frequency noise, to where we have only broad-band thermal noise, etc.;
4. Consequently, the noise in the 1 Hz bandwidth of the lock-in is actually quite small.

*Chopping.* The demonstration experiment above is not very realistic, since it is usually impractical to electrically modulate the signal without introducing significant systematic error. Usually, the quantity to be measured needs to be mechanically or optically 'chopped'. The name derives from optical experiments where a rotating wheel or blade physically interrupts a light beam and turns it from a constant to a modulated quantity. Since we know the frequency of the 'chopper' wheel we can use this as the reference for the lock-in. Typically the arrangement is similar to that described above except here the signal generator drives the chopper.[36]

*21.5. An example from radioastronomy*

A pleasing example of the chopping technique comes from the world of radioastronomy. A radio-telescope is a dish antenna used to measure radio-frequency emission from distant stars. The signal is very faint, while noise sources abound: we have the general cosmological background, terrestrial (man-made) radio-sources, plus all the usual electrical noise sources in the receiver. To make this worse, fluctuations in the atmosphere and ionosphere add extra low-frequency noise to the signal. Radioastronomers use the chopping/synchronous-detection scheme as

[36] You can watch a good explanation of the chopper-plus-lock-in amplifier here: https://www.youtube.com/watch?v=SwPt9grlsME.

described previously. This can be done by physically rocking the dish side-to-side thus sweeping it across the target star. This directly modulates the signal intensity, while all the noise sources remain the same. With large dishes this is impractical, so often there is a secondary reflector at the focus of the dish (Cassegrain type), and it suffices to rock the secondary. A further refinement, which avoids any mechanical action (and can be run much faster) is a setup which uses a single dish with two signal-receiving waveguide/antennae. One is exactly on-axis and receives the target signal, the other is slightly off-axis so it receives all the noise sources but none of the signal from the star. The radiometer (powermeter for radio-frequency signals[37] is electrically switched between the two sources at the modulation frequency.

[37]Ultra-sensitive radiometers use another noise minimisation technique we have already discussed: the receiver/amplifier stages are cooled to cryogenic temperatures in order to minimise thermal noise.

## A. Circuit theory (revision)

In this appendix we will review some of the basic concepts in circuit theory: we will go back to basics. This should be:

- A refresher, in that you will have seen all of this before in the first year electronics course and labs;
- An *aide-memoire*, summarising the basic concepts and equations which we will need for this course.

There are two main types of circuits, DC and AC, which we will discuss in turn. DC is short for *direct current*, which refers to a static or constant current flowing in a circuit. AC denotes *alternating current*, and refers to a (usually sinusoidal) time-varying current in a circuit. A DC current is what you get from a battery, while the mains provides AC at 50 Hz. In physics, and certainly in engineering, these terms are used rather loosely to mean any signal which is either constant or varying. For example, we might talk about a DC voltage of 1.5 V (from an AA battery), or 240 $V_{AC}$ (mains voltage). We might even describe the intensity of radiation from a *pulsar* as an "AC signal", or the weight of a kilogram mass as a "DC signal".

### 1) DC circuits

*Current.* Current is the rate of flow of charge:

$$i = \frac{\mathrm{d}q}{\mathrm{d}t}\,.$$

The SI unit is the ampere (or 'amp'), with 1 ampere (A) being equivalent to 1 coulomb (C) per second: 1 A ≡ 1 C/s.

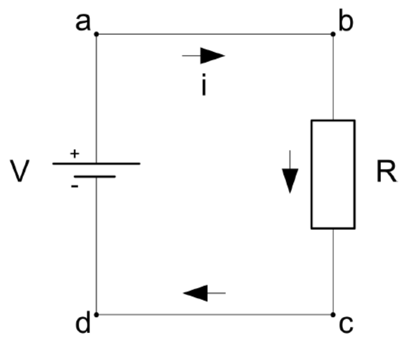


Figure A.1: Conventional current.

Note that, by convention, we use the letter $i$ to represent current, but it can appear as either lower or uppercase.[38] Our charge carriers are of course electrons, and since there are plenty of free electrons in metals it makes sense to use metal wires (usually copper) to connect together the parts of our circuits, but generally the wires play no part in the circuit other than as conduits for electrons – or, more specifically, to connect different parts of the circuit to the same potential. Refer to Figure A.2 for the simplest circuit possible, which may represent a torch. An old-fashioned bulb is effectively a resistor: when current flows it gets so hot it glows incandescently. The battery provides a potential difference across its terminals. By convention, the positive terminal is the larger "plate".

[38] Typically, an upper case $I$ will denote a DC current, and we reserve $i$ for an AC current, but this is not universally true – even in these notes!

*Conventional current.* Electrons are attracted to the positive terminal and hence flow anti-clockwise in the figure; however, conventional current is taken to flow in the opposite direction to actual electron current, and hence is *defined as flowing from positive to negative potential.* We might think of conventional current as flow of positive charge or "holes". We – like most textbooks – will always use conventional current, even when we are talking about beams of electrons flowing in a vacuum tube.

*Conservation of charge.* The total number of electrons (or holes) in the circuit is constant. This is a statement of conservation of charge. Consequently, the current $i$ at point $a$ is the same as that at $b$, the same as that flowing through the resistor, to $c$, $d$ and back through the battery. This is an important concept, always true, and it holds for AC circuits too.

*Resistance.* The sheer abundance of conduction electrons in the copper wires means that appreciable currents will flow with a net electron flow speed of the order $10^{-4}$ m/s. This is the *drift velocity* of the electrons. Contrast this with the average thermal speed of the electrons at room temperature which is about $10^6$ m/s. The overall behaviour of the electrons is therefore rather random and collision-dominated, and we must see current as a statistical average drift of electrons rather than an orderly procession. Because the electrons are in constant collision with each other and the atomic lattice of the conductor, the mobility of the electrons is key to their behaviour. Mobility is high in metals, but in carbon it is about a thousand times lower. Clearly, then, if we make a resistor from an insulating tube covered with a thin film of carbon then mobility will be much reduced and current will be severely restricted. This is captured by *Ohm's law*:

$$i = \frac{v}{R}\,. \tag{A.1}$$

*Potential.* A difference in electrical potential causes current to flow, and of course when a charge moves through a potential difference (PD) work is done. The SI unit of potential difference is the volt, which equals one joule per coulomb: 1 V ≡ 1 J/C.

Here, the battery does the work, converting stored chemical energy into current. When measuring PD it is important to understand the sense of the measurement. If our battery is a single AA then $v_{ad}$= 1.5 V. We would measure this with a meter by placing the red probe at $a$ and the black probe at $d$. If we reversed the probe we would measure $v_{da} = -1.5$ V. Since (ideally) the wires have zero resistance, $v_{ab} = v_{cd} = 0$. Therefore, $v_{bc} = v_{ad}$. Since we know the PD across the resistor we can calculate the current through it using equation (A.1).

*Kirchhoff's voltage law*

Kirchhoff's Voltage Law (KVL) states that *the sum of the PDs around any loop is zero*:

$$\sum v = 0\,. \tag{A.2}$$

Imagine sitting in the centre of the circuit with the meter probes and measuring each of the four sides in turn, going round clockwise and taking care to keep the sense of the measurement the same as you go:

$$v_{ad} + v_{ba} + v_{cb} + v_{dc} = 0\,.$$

This is, of course trivial, but there could be any number of components filling up the top and bottom sides. Since we know the current is the same in each, we can calculate the PD across each. This works for all components, i.e. including inductors, capacitors, etc, and is suitable for AC circuits too.

*Ground.* As the name indicates, potential difference gives us the *difference* in potential between two points. What is the actual potential at $a$ or $d$?. The answer is that we do not know, as there is no reference potential in this circuit.[39] It is said to be *floating*. Any battery-powered kit from a multimeter to a laptop is floating. We could connect the circuit to ground as in Figure A.2, then we would know that the potential at $d$ (and $c$) is zero volts. Ground (the Earth pin on a plug) is genuinely connected physically to a large copper spike in the basement of the building driven into the earth. This is taken by convention as the zero reference for electrical potential. Now we can say that

$$v_a = 1.5 \text{ V}\,.$$

Scientists and engineers use voltage and potential difference interchangeably. This is wrong but in such common use as to be unavoidable. We might say "the voltage across $R$ is 1.5 V" when we mean "the potential difference...".

*Series resistors and the voltage divider*

Resistors in series add, i.e. $R = R_1 + R_2$. In Figure A.3 we can use Ohm's law to state $v_{ab} = iR_1$. Note the sense of the subscript. Ohm's law gives us the voltage drop from $a$ to $b$ when $a \to b$ is the direction of conventional current. So, $a$ is at a higher potential than $b$ (which Kirchhoff confirms) and hence

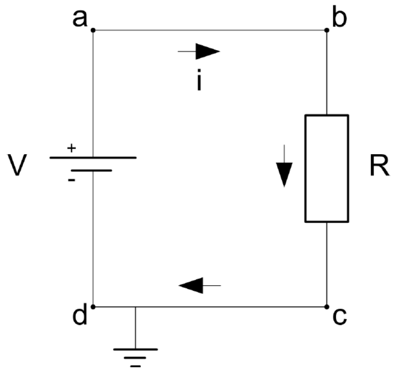


Figure A.2: Grounded circuit.

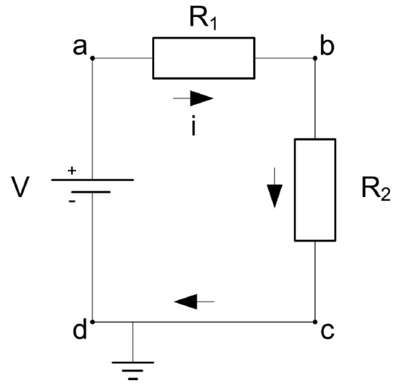


Figure A.3: Resistors in series.

$$v_b = v_a - iR_1\,.$$

We find the current flowing in both resistors as

$$i = \frac{v_{ac}}{R_1 + R_2}\,.$$

Combining the two equations and noting that $v_{ac} = v_a$ we get the *voltage divider* equation:

$$v_b = v_a \frac{R_2}{R_1 + R_2}\,. \tag{A.3}$$

*Parallel resistors and the current divider*

For resistors in parallel – such as Figure A.4 – we use:

$$\frac{1}{R_{||}} = \frac{1}{R_2} + \frac{1}{R_3}\,.$$

We can calculate the current as

$$i = \frac{v_a}{R_1 + R_{||}}\,.$$

We can calculate $v_b$ since $R_1$ and $R_{||}$ are a voltage divider.

[39]Note that there is has a deeper physical question here: only differences in potential can be measured for conservative vector fields... What we are discussing here it more trivial: can we measure potential differences relative to some reference, e.g. the potential of the environment around us.

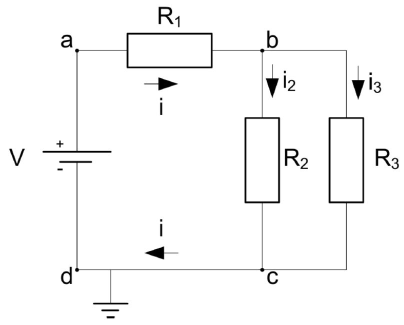


Figure A.4: Resistors in parallel.

Then,

$$i_2 = \frac{v_b}{R_2}, \qquad i_3 = \frac{v_b}{R_3}\,. \tag{A.4}$$

We would find that the *current divider* rule is

$$i_2 = i\frac{R_3}{R_2 + R_3}\,.$$

Note the similarities and differences with the voltage divider rule. Generally, this can get confusing and it is easier to just remember the voltage divider rule and then use the method of equations (A.4) to obtain the currents in the branches.

*Kirchhoff's current law*

Conservation of charge tells us that the current going into point $b$ must equal the current coming out, so $i = i_1 + i_2$. This is enshrined in Kirchhoff's Current Law (KCL) which states that *the sum of all the currents into a node (junction) is zero*:

$$\sum i = 0\,.$$

To draw this we would change the direction of the $i_2$ and $i_3$ arrows *which also changes their algebraic sign* and then indeed the sum *into* $b$ is zero.

*General method for circuit analysis*

The above rules give us everything we need for the vast majority of circuit analysis. Usually, we are given as a starting point some knowledge of the current or some potentials. As a general approach:

- If you know the current in any branch of a circuit then calculate the PD across any components and then determine the potentials (using Ohm's law and KVL) of any branching points in the circuit;
- Conversely, if you are given the potential relative to ground at any point then try to calculate the current in that branch and use a combination of Ohm's Law and KCL to deduce any currents.

*Voltage and current sources*

An ideal voltage source provides a fixed voltage across its terminals which is independent of the load resistance to which it is connected (and hence the amount of current drawn). It appears to have zero resistance, in that it dissipates no power. In reality, there is a limit to how much current a voltage source can supply, and there is always some internal resistance. A battery is a good voltage source example: a 9 V battery connected to a 3 ohm load will supply a full 3 A (for a few minutes, until the chemical energy source is exhausted). The battery will get hot due to its non-zero internal resistance (about 1 ohm). The load resistor will get very hot! The battery manufacturer has put a lot of effort into making the battery as close to an ideal voltage source as possible. This means keeping the internal resistance as low as possible. In traditional zinc-carbon batteries, this is done by mixing carbon into the electrolyte chemicals in order to improve the overall conductivity. However, there always remains some internal resistance.

On circuit diagrams you will often see the battery symbol as used here or sometimes just a circle labelled 'V'. Either of these should be taken to represent an ideal voltage source.

An ideal current source provides a fixed current which is independent of the load resistance to which it is connected (and the amount of voltage it must generate). In reality, there is a limit to how much voltage a current source can generate. There is no simple physical example of a current source; when you need a powerful current source for a lab experiment, often they are quite complicated and expensive pieces of equipment. There are two sensor examples which are reasonably good current sources, though they can only produce quite small current signals in the micro- to milli-ampere range: the AD590 temperature sensor and the photomultiplier tube.

The standard circuit symbol is two part-overlapping circles, though quite often you will see just a single circle marked 'I', so take care to distinguish these from voltage sources.

*Thevenin equivalent circuit*

As a final remark on DC circuits, *Thevenin's theorem* is frequently used in instrumentation since the *Thevenin equivalent circuit* is a convenient way of encapsulating the behaviour of more complex circuits which may represent, for example, a sensor. Thevenin states that

*Any network of many sources and impedances can be replaced by a single source in series with a single impedance.*

Sources are voltage sources such as in Figure A.5, or current sources. Here we consider pure resistances but the method works also for generalised impedances, as we shall see later.

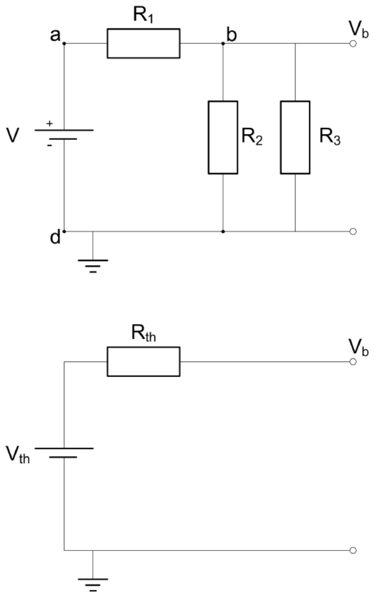


Figure A.5: Thevenin equivalent circuit.

Thevenin's theorem tells us that the top and bottom circuits in Figure A.5 are exactly equivalent. If the two circuits were inside a "black box" with just the two terminals on the outside, then nothing we could do from the outside could tell the difference. To get the Thevenin equivalent circuit we just need to determine – by calculation or measurement – the values of $v_{th}$ and $R_{th}$.

*Thevenin voltage.* $v_{th}$ is the *open-circuit* PD across the terminals. "Open circuit" means that no current is drawn down the terminal branches, i.e. when there is no "load" attached to the circuit. For example, if we measure the voltage at $b$ with a multimeter then this has an input impedance of M$\Omega$ so, and as long as $R_n \sim$k$\Omega$, we can confidently assert that the measured voltage is very close to the open-circuit value. The meter hardly "loads" the circuit at all. In this case, we have already calculated $v_{th}$ since it is equal to $v_b$.

*Thevenin resistance.* $R_{th}$ is the combined impedance (resistance) when:

1. Any voltage sources are short-circuited;
2. Any current sources are open-circuited.

We short-circuit the battery by removing it and connecting across points $a$ and $d$. Then, all three resistors are in parallel so

$$\frac{1}{R_{th}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\,.$$

This is the resistance we would measure if we connected a resistance meter across the terminals of the circuit.

The point of the Thevenin circuit is that it behaves *exactly* like the original, but it is easier to see how it behaves under different load conditions. Connecting a high-impedance meter across the terminals we measure $v_{th}$, since no current flows in $R_{th}$; however, were we to connect a low-impedance load (say a light bulb) then it is easy to see how the current flowing means there is a voltage drop across $R_{th}$. Issues of input and output impedance are very important and we will come back to this later.

In the Thevenin equivalent circuit, $v_{th}$ is an idealised voltage source, and $R_{th}$ is the apparent series (internal) resistance of the circuit. As an example, the Thevenin equivalent of the 9 V battery circuit mentioned earlier would be a voltage source of 9 V in series with a resistance of $\sim$1.5 $\Omega$.

### 2) AC circuits

As previously mentioned, electronic circuits are good analogues for other physical systems. For example, a first-order system such as the thermometer can be very effectively modelled by a Resistor-Capacitor (or RC) circuit. The mechanical damped-oscillator has an exact physical equivalent in the Resistor-Inductor-Capacitor (or RLC) circuit. In these cases, and many others, the dynamical systems are represented by circuits processing time-varying signals. Before the advent of computer-modeling, this was a very important way of modeling physical systems. For our studies, we need to understand how AC circuits can manipulate electrical signals from sensors and more complex instrumentation systems.

*Capacitor*

Essentially two parallel metal plates with an insulating dielectric in-between, the capacitor has ideally infinite ohmic resistance to DC current. When a PD is applied it stores energy in the electric field. By definition, the capacitance, which is measured in farad (1 F≡1 C/V), is

$$C = \frac{q}{v}\,,$$

from which we obtain the key relations

$$v(t) = \frac{1}{C}\int_{\infty}^{t} i(\tau)\mathrm{d}\tau\,,$$

$$i(t) = C\frac{\mathrm{d}v(t)}{\mathrm{d}t}\,.$$

Note that, for AC signals, the current integrated over one cycle leaves no net charge on a capacitor, and hence $\langle v(t)\rangle = 0$ (see Poularikas Example 2.1). Capacitors in parallel add together (i.e. opposite to resistor behaviour) which is logical when we consider that 2 capacitor plates in parallel are the same as a single capacitor with plates twice as large.[40] As a consequence of the above equations, when we apply a sinusoidal voltage to a capacitor, the current through the capacitor *leads* the voltage across it by 90°.

[40] For further information see Diefenderfer 2.2 & 2.3.

*Inductor*

An inductor is essentially a coil of wire (or a solenoid), usually wound around a core of high magnetic permeability such as iron or ferrite to increase the inductance (in henry, with 1 H≡1 weber/amp = 1 V s/A):

$$L = \frac{\psi}{i}\,.$$

Using Faraday's law $v(t) = \mathrm{d}\psi/\mathrm{d}t$ we obtain the relations

$$i(t) = \frac{1}{L}\int_{\infty}^{t} v(\tau)\mathrm{d}\tau\,, \tag{A.5}$$

$$v(t) = L\frac{\mathrm{d}i(t)}{\mathrm{d}t}\,. \tag{A.6}$$

As a consequence, when we apply a sinusoidal voltage to an inductor, the current through the inductor *lags* the voltage across it by 90°.

*Complex representation of signals*

Consider a voltage $v(t) = V_0 \cos \omega t$ driving a capacitor C. The current which flows in the capacitor is

$$i(t) = C\frac{\mathrm{d}v(t)}{\mathrm{d}t} = -C\omega V_0 \sin \omega t\,.$$

Clearly, there is a voltage/current phase difference of $\pi/2$. It is frequently more convenient to use a complex representation. A voltage $v(t) = V_0 \cos(\omega t + \phi)$ is represented by the complex number $\mathbf{v} = V_0\, e^{j\phi}$. This represents the voltage as a vector in the complex plane. There are a few points to note:

1. The signal (voltage) is of course a real quantity and is represented by the real part of the complex quantity;
2. We will often represent complex voltages, currents, impedances, etc, by bold-face letters to stress that they are complex quantities: $\mathbf{v}$, $\mathbf{i}$, ...;
3. Engineers typically use $j$ instead of $i$ to avoid confusion with current;
4. In most situations we are just concerned with the relative amplitude and phase of the various voltages in the circuit so the time-varying part of the representation $e^{j\omega t}$ is implied (as is the case above).

To get the actual voltage signal, multiply by $e^{j\omega t}$ and take the real part:

$$v(t) = \mathcal{R}\{\mathbf{v}\, e^{j\omega t}\} = V_0 \cos(\omega t + \phi)\,.$$

*Phasor representation*

Consider the vector $\mathbf{P} = (a + jb)$ plotted on the Argand diagram in Figure A.6: we have $a = P\cos\theta$ and $b = P\sin\theta$, with

$$P = \sqrt{a^2 + b^2} = \sqrt{\mathbf{PP}^*}\,.$$

Using the Euler relations we see that the vector is represented as $Pe^{j\theta}$. If we advance the vector by a phase of $\pi/2$ we get $Pe^{j\theta+\pi/2}$, and if we advance by $\pi$ then we get $Pe^{j\theta+\pi} = -Pe^{j\theta}$, from which we can see that $\mathrm{e}^{j\pi} = -1$. Taking the square-root of both sides, $e^{j\pi/2} = \sqrt{-1}$. Consequently, multiplication by $j$ represents an anticlockwise rotation by $\pi/2$. The relevance of this to AC circuits comes from the fact that a multiplication by $e^{j\theta}$ represents a rotation $\theta$ in the complex plane. In general, then, any sinusoidal physical quantity (voltage, current etc.) can be thought of as a vector rotating anti-clockwise. The Argand diagram is a "snapshot", e.g. at time $t = 0$.

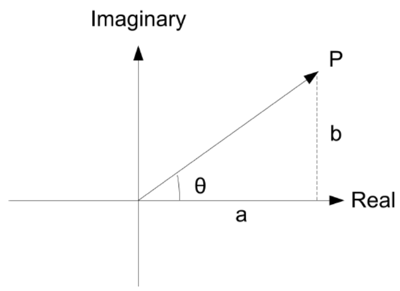


Figure A.6: Phasor representation in the Argand diagram.

*Reactance*

Returning to the capacitor example, we can see that

$$v(t) = V_0 \cos(\omega t + \phi) = \mathcal{R}\{\mathbf{v}\, e^{j\omega t}\}\,,$$

$$\begin{aligned} i(t) &= C\frac{\mathrm{d}v(t)}{\mathrm{d}t} = -C\omega V_0 \sin(\omega t + \phi) \\ &= \mathcal{R}\left\{\frac{\mathbf{v}\, e^{j\omega t}}{\frac{-j}{\omega C}}\right\} \\ &= \mathcal{R}\left\{\frac{\mathbf{v}\, e^{j\omega t}}{-jX_C}\right\}\,, \end{aligned}$$

where $X_C$ is the *reactance* of the capacitor. Similarly, we define the reactance of the inductor $X_L$. The reactance is the *AC impedance* of the circuit element:

$$X_C = \frac{-1}{\omega C}\,, \tag{A.7}$$

$$X_L = \omega L\,. \tag{A.8}$$

*Complex impedance*

Clearly, inductors and capacitors have purely imaginary impedance, which gives them the quality that they induce a $\pi/2$ phase shift between voltage and current *and hence dissipate no power.* The resistor is the opposite: it has purely a real resistance, hence it dissipates power, and the voltage is always in phase with the current. The *complex impedance* of a circuit element is thus:

$$\begin{aligned} \mathbf{Z_R} &= R \\ \mathbf{Z_C} &= \frac{-j}{\omega C} \\ \mathbf{Z_L} &= j\omega L\,. \end{aligned}$$

The complex impedance $\mathbf{Z}$ of any circuit element or combination of elements is the vector combination of the resistive and reactive components according to the same rules as pure resistances:

$$\mathbf{Z}_{\text{series}} = \mathbf{Z_1} + \mathbf{Z_1}\,; \tag{A.9}$$

$$\frac{1}{\mathbf{Z}_{||}} = \frac{1}{\mathbf{Z_1}} + \frac{1}{\mathbf{Z_2}}\,. \tag{A.10}$$

We can visualise this with the Argand diagram – see Figure A.7. If we have a resistor in series with an inductor then $\mathbf{Z} = R + j\omega L$, where $|\mathbf{Z}| = \sqrt{R^2 + \omega^2 L^2}$ and $\phi = \tan^{-1}(\omega L/R)$.

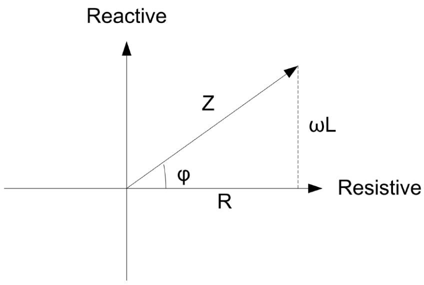


Figure A.7: Complex impedance on the Argand diagram.

Figure A.7 requires some interpretation. The Argand diagram gives us the relative phases of the voltages in our circuit. The voltage across the inductor (positive imaginary axis) leads the voltage across the resistor (real axis) by $\pi/2$ (Note: this is consistent with the "$V$ leads $I$" rule for inductors; $V$ is in phase with $I$ for the resistor). Let us say that we apply a voltage $v(t) = V_0\cos(\omega t + \phi) = \mathcal{R}\{\mathbf{v}\,e^{j\omega t}\}$. Then, the Argand diagram gives us a "snapshot" of the voltages at time $t = 0$. Actual voltages (as would be measured with the oscilloscope) are projections onto the real axis. The voltage across the resistor is at its peak value (phase $= 0$). The voltage across the inductor is zero (phase $= \pi/2$), and the applied voltage (across both) is $V_0\cos\phi$.

Note that at all times Kirchhoff's Voltage Law applies: the instantaneous applied voltage always equals the sum of the instantaneous voltages on the resistor and inductor.

*Ohm's law generalised*

The capacitive and inductive reactances have units of ohms ($\Omega$). Ohm's law is applicable to impedances generally:

$$\mathbf{v} = \mathbf{i}\,\mathbf{Z}\,, \tag{A.11}$$

where all the quantities are complex. Note, however, that only $\mathbf{v}$ and $\mathbf{i}$ can be represented as *phasors* (time-varying complex quantities). The impedance $\mathbf{Z}$ is a complex *constant*. This simultaneously illustrates the usefulness and the limitation of the phasor representation: if we know the phasor current $\mathbf{i}$, we can multiply it by the complex impedance $\mathbf{Z}$ to get the phasor voltage $\mathbf{v}$:

$$V_0\,e^{j\phi} = I_0\,e^{j\psi}|\mathbf{Z}|\,e^{j\theta}\,,$$

where the amplitudes obey $V_0 = I_0|\mathbf{Z}|$ and the phases $\phi = \psi + \theta$. This is much easier than solving problems using the trigonometric forms. However, *we can never multiply two phasors together* – that would represent the product of two sinusoids which would give us new frequency components! In practice, the use of phasors is limited to situations like equation (A.11) where a complex constant modifies the amplitude and phase of a phasor. This does turn out to be very useful and will be the basis later on for much of our work on linear systems. This also means that phasor notation can only represent systems at a given frequency (that is to say, while $\mathbf{Z}$ is a function of frequency, for any $i$ we get a $v$ of the *same* frequency).

Reactive components have two important properties which we will see the use of in future sections:

1. Phase shifting (as discussed above);
2. Frequency dependence. At DC (zero Hz) the reactance of the inductor is zero. It is a short-circuit. At high-frequencies, the reactance tends to $\infty$ (open circuit). The capacitor is the opposite way round. As we have seen before, it is open-circuit at DC.

Note that if we just want to work with the magnitudes of the voltages and currents (i.e. we do not care about the detailed phase relationships) then we can use $|\mathbf{Z}| = |\mathbf{v}|/|\mathbf{i}|$, so for a capacitor we can calculate the amplitude of the current as $I_0 = V_0\omega C$. For most purposes this is the most useful result from the preceding discussion: *where we do not care about the relative phases of the signals in the circuit, we can just work with the amplitudes of the signal and the magnitude of the impedance.*

*Power in AC circuits*

If we have a sinusoidal voltage $v = V_0\sin\omega t$, then, even though the average value of $v$ is zero, the signal has the capacity to dissipate power when it is applied across a resistor $R$ according to

$$P = vI = \frac{v^2}{R} = \frac{V_0^2\sin^2\omega t}{R}\,. \tag{A.12}$$

This is the *instantaneous power* at any time $t$. More useful (see boxed example) is the *average power*:

$$\langle P\rangle = \frac{V_0^2}{R}\langle\sin^2\omega t\rangle = \frac{V_0^2}{2R}\,; \tag{A.13}$$

the average[41] is taken over a cycle of $\omega$.

[41] To see why the average value of a squared-sine function is 1/2 consider the identity $2\sin^2\theta = 1 - \cos 2\theta$. Over a cycle, $\langle\cos 2\theta\rangle = 0$.

*Root mean square*

The *root mean square (RMS)* of a signal is a measure of the power in the signal. For our voltage $v$, it is

$$v_{\mathrm{RMS}} = \sqrt{\langle v^2 \rangle}\,; \tag{A.14}$$

(it does exactly what the name says!) Hence, for a sinusoidal signal

$$v_{\mathrm{RMS}} = \frac{V_0}{\sqrt{2}}\,. \tag{A.15}$$

The average power dissipated in a resistor is

$$\langle P \rangle = \frac{v_{\mathrm{RMS}}^2}{R} = v_{\mathrm{RMS}}\, i_{\mathrm{RMS}}\,. \tag{A.16}$$

The above is always true, regardless of the shape of the waveform. RMS is a useful measure since it gives the equivalent power (or heating effect) as the constant 'DC' voltage of the same value.

**Example**
The UK mains voltage has $V_{\mathrm{RMS}}$=240 V (often written as $v$=240 $V_{\mathrm{RMS}}$), the amplitude of the mains signal is approximately $V_0$= 340 V, and the peak-to-peak voltage is thus $V_{\mathrm{pp}}$=680 V. A 60 W light bulb (old-fashioned tungsten-type) will have a resistance of 960 Ω.

*Power in reactive circuits*

For circuits with inductance and/or capacitance, the average power becomes

$$\langle P \rangle = v_{\mathrm{RMS}}\, i_{\mathrm{RMS}} \cos\phi\,, \tag{A.17}$$

where $\phi$ is the phase difference between $v$ and $i$ and $\cos\phi$ is often called the 'power factor'. To understand this consider again the capacitor. The voltage is $\pi/2$ out of phase with the current. The instantaneous power delivered to the capacitor is $P = vi$ (sketch this to visualise it). Positive power is when the capacitor is charging, and negative power is when discharging. Clearly, over a cycle the average power is zero. This is true for any circuit with purely imaginary impedance. Circuits with some real (resistive) component of impedance will dissipate power. In general, we could determine average power by integrating over a cycle and dividing by the time for one cycle. This gives

$$\begin{aligned} \langle P \rangle &= \frac{\mathcal{R}\{\mathbf{v}\,\mathbf{i}^*\}}{2} \\ &= \frac{\mathcal{R}\{\mathbf{v}^*\,\mathbf{i}\}}{2} \\ &= v_{\mathrm{RMS}}\, i_{\mathrm{RMS}} \cos\phi\,. \end{aligned}$$

## Further reading

There are very readable sections on DC and AC circuits in Diefenderfer (Chapter 1) and Horowitz & Hill (Chapter 1).